

Partitions into Catalan Numbers

All-Orders Asymptotics, Periodic Corrections, and Inverse Growth

A proof-focused research report for the ProveIt project

October 1, 2026

Abstract

Let $p(n)$ count ordinary integer partitions whose allowed part sizes are $1, 2, 5, 14, 42, \dots$, the distinct Catalan numbers. This is OEIS A033552. We develop a complete expansion in inverse powers of a logarithmic size parameter, including a nonconstant periodic amplitude, an explicit first correction, and an inverse expansion for the threshold at which $p(n)$ reaches a prescribed value. The argument combines an all-orders coefficient saddle estimate, a Barnes G product identity, and an exponentially convergent Fourier series involving $\Gamma(s)\zeta(1+s)$. The contour estimate controls all noncentral arcs; the expansion is therefore more than a formal saddle calculation. A further theorem gives a phase-dependent limiting distribution for the largest allowed-part index in a uniformly chosen partition. Exact computations through $n = 10^6$ and independent high-precision checks accompany the proofs. The inspected OEIS entry contains no asymptotic formula; we do not claim that it records an explicit asymptotic conjecture, or that an exhaustive literature-priority search has been completed. The sequence-specific formulas and proofs below are the substantive output, not a claim that the classical saddle or periodic-partition methods themselves are new.

Principal formula. Put $\alpha = \log 4$ and let μ be the large solution of $n = 4^\mu / \sqrt{\pi\mu}$. Then

$$p(n) = e^{C + \Psi_\alpha(\mu)} \mu^{23/8} \exp\left(\frac{\alpha}{2}\mu^2 - \frac{1 + 3\alpha}{2}\mu\right) \left(1 + \frac{D_\alpha(\mu)}{\mu} + O(\mu^{-2})\right).$$

Here C is explicit and Ψ_α, D_α are smooth, one-periodic functions given below. Every subsequent inverse-power coefficient is constructively determined. Replacing Ψ_α by a constant does *not* give an asymptotic equivalent.

[Write note, batch 7301, 1 October 2026.] This is the collection edition of the report, written into ProveIt's research-report collection from manuscript 39 of batch 7301 (provenance in [appendix C](#)). Every label carries the prefix `ctp:`; `requirements.txt` now lives in `data/` and `build.sh` in `code/`. Text added in the write is marked like this note; everything else is the delivered text. As the abstract says, the OEIS entry states *no* asymptotic conjecture: this report answers no posed question, it supplies the asymptotics the entry lacks.

[Write note, batch 98, 5 October 2026.] Since batch 98 the report has two Parts. Part I is manuscript 39 of batch 7301, as described above. Part II ([section 12](#)) merges manuscripts 06 and 10 of batch 98, which answer Part I's questions [10.4](#) and [10.5](#) and supply the growing-offset and moment results that Theorem [7.1](#) does not assert: the exact boundary of independence for the upper multiplicity process, corrected joint edge laws, uniform tails and all moments of the largest index, the limit law of the number of parts and its joint behaviour with the largest index, and the explicit law of that limit. Its labels carry the prefixes `ctp:cl:` and `ctp:te:`. Part I is unchanged except for notes like this one and its Part heading.

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Part I

All-orders asymptotics, periodic corrections and inverse growth

1 The problem, its status, and the contribution

1.1 The sequence and the normalization

For $k \geq 1$ define

$$c_k = \frac{1}{k+1} \binom{2k}{k}, \quad F(z) = \prod_{k \geq 1} (1 - z^{c_k})^{-1}, \quad p(n) = [z^n]F(z). \quad (1.1)$$

Thus the allowed part sizes are $c_1 = 1, c_2 = 2, c_3 = 5, c_4 = 14, \dots$. The initial terms are

$$1, 1, 2, 2, 3, 4, 5, 6, 7, 8, 10, 11, 13, 14, 17, 19, 22, 24, 27, 30, \dots$$

The index starts at $n = 0$. It is important not to include both c_0 and c_1 : doing so would create two colors of the unit part and a different sequence. The definition and generating function agree with OEIS A033552 [1].

The inspected entry supplies a generating function, a recursive computation, and a reference to Pak's enumerative-complexity survey, but no asymptotic formula. Pak explicitly discusses partitions into Catalan numbers among restricted-partition counting problems that may be less tractable than some classical examples [2, §2.5]. That discussion concerns exact counting complexity. An asymptotic expansion, however accurate to fixed order, does not by itself settle that complexity question.

All logarithms in this article are natural. A periodic function is evaluated on a real argument by periodic extension. All O -estimates concern the limit of that argument, or of n , to positive infinity; their constants may depend on a fixed truncation order but not on a fractional phase.

1.2 What is established here, and what is not

The principal contribution is a self-contained chain from the exact product (1.1) to the following results: an all-orders saddle expansion with a contour proof; a completely normalized, phase-resolved explicit expansion; a first correction containing derivatives of the periodic amplitude; an inverse threshold expansion with error transport; and a phase-dependent law for the largest part index. A general proposition explains which coefficients depend only on the exponential and polynomial growth of the allowed part sizes.

Periodicity in asymptotics of geometrically restricted partitions is a classical phenomenon. Erdős and Richmond discuss the de Bruijn–Mahler phenomenon and its extensions [3]. Mellin analysis of harmonic sums is developed systematically by Flajolet, Gourdon, and Dumas [4]. We use that context, but prove the concrete estimates and normalizations needed here rather than importing a vaguely matched theorem. The explicit Catalan-part formulas below were derived for this report. A limited search found no source giving this complete collection of formulas; that observation is not a proof of publication priority.

[Write note, batch 7301, 1 October 2026.] The phenomenon Erdős and Richmond name goes back to de Bruijn's treatment of Mahler's partition problem [9], which the delivered bibliography did not list: partitions into powers of a fixed integer, whose logarithm carries a periodic function of a logarithmic variable built from $\Gamma(s)\zeta(1+s)$ on the Mellin lattice. The Catalan parts $c_k \sim 4^k/(\sqrt{\pi}k^{3/2})$ are a polynomially perturbed geometric sequence of that kind, and lemma 4.1 is the corresponding Mellin computation; see also the note after its proof.

The manuscript contains conventional mathematical proofs and executable checks. It is not a Lean formalization. The numerical checks corroborate, but do not replace, the arguments. No computer search is used as a substitute for a missing contour estimate or for a missing asymptotic remainder.

1.3 Relation to the ProveIt repository

The inspected repository snapshot is commit `e6190a94552f119e9a2140e5fea8450a045a0837`. Its transseries documentation separates asymptotic scales from inversion, distinguishes a continuous inverse from an integer staircase inverse, and records qualifications on formalized and unformalized results [7]. Those distinctions are useful here. The companion discusses asymptotics of Catalan numbers themselves, whereas our object is the partition sequence whose *allowed parts* are Catalan numbers. An initial repository code search for A033552 returned no match; that is a duplication check, not a mathematical premise.

The present article uses an exactly invertible dominant size coordinate and then transports residual errors through inversion. Its periodic coefficients also make a useful boundary case for a purely polynomial–logarithmic transseries calculus: the coefficient algebra must retain a periodic phase.

[Write note, batch 7301, 1 October 2026.] At the repository’s current state the search result above still holds: no report or formal development treats A033552 or partitions into Catalan numbers. The pin quoted above is a commit; the delivered `SOURCE_NOTES.md` also names `1f1981f682b2878bde51a6ad40c22777f362fc05`, the commit its first repository request returned, and calls it “a tree SHA, not a commit SHA”. That is mistaken: it is a ProveIt commit of 1 October 2026 (the root tree of `e6190a945` is the separate object `8e77886c`). Neighbouring material is listed in [appendix C](#). No statement of this report is formalized.

2 Main asymptotic and inverse formulas

2.1 Constants and the periodic amplitude

Set

$$\alpha = \log 4, \quad A = \frac{1+3\alpha}{2}, \quad \eta = \frac{23}{8}, \quad C = \frac{1}{4} \log 2 + \frac{3}{4} \log \pi + \log G(3/2), \quad (2.1)$$

where G is the Barnes function, normalized by $G(1) = 1$ and $G(z+1) = \Gamma(z)G(z)$ [5]. For the moment allow $\alpha > 0$ to vary, and define

$$\kappa = \frac{\pi^2}{12} - \frac{\gamma^2}{2} - \gamma_1, \quad (2.2)$$

$$\Psi_\alpha(\theta) = \frac{\alpha}{12} + \frac{\kappa}{\alpha} + \frac{1}{\alpha} \sum_{\ell \in \mathbb{Z} \setminus \{0\}} \Gamma\left(\frac{2\pi i \ell}{\alpha}\right) \zeta\left(1 + \frac{2\pi i \ell}{\alpha}\right) e^{2\pi i \ell \theta}. \quad (2.3)$$

Here γ is Euler’s constant and γ_1 is the first Stieltjes constant, with convention $\zeta(1+s) = s^{-1} + \gamma - \gamma_1 s + O(s^2)$. The Fourier series and all of its derivatives converge absolutely. Conjugate terms pair to make Ψ_α real. A prime denotes differentiation with respect to θ ; a partial α -derivative holds θ fixed. The correction is

$$D_\alpha(\theta) = \frac{1}{6} - \frac{1}{2\alpha} - \frac{3}{2} \partial_\alpha \Psi_\alpha(\theta) + \frac{17}{8\alpha} \Psi'_\alpha(\theta) - \frac{\Psi'_\alpha(\theta)^2 + \Psi''_\alpha(\theta)}{2\alpha^2}. \quad (2.4)$$

Only after taking the partial derivative do we set $\alpha = \log 4$.

Define the large real size coordinate

$$n = \frac{4^{\mu(n)}}{\sqrt{\pi \mu(n)}}, \quad \mu(n) = -\frac{1}{2\alpha} W_{-1}\left(-\frac{2\alpha}{\pi n^2}\right). \quad (2.5)$$

The negative real branch W_{-1} is intended. For sufficiently large n this is the unique large solution and $\mu(n) \sim (\log n)/\alpha$.

Theorem 2.1 (Explicit expansion, including the first correction). *For the sequence (1.1), as $n \rightarrow \infty$ through integers,*

$$\log p(n) = \frac{\alpha}{2}\mu^2 - A\mu + \eta \log \mu + C + \Psi_\alpha(\mu) + \frac{D_\alpha(\mu)}{\mu} + O(\mu^{-2}). \quad (2.6)$$

Equivalently,

$$p(n) = e^{C+\Psi_\alpha(\mu)} \mu^\eta e^{\alpha\mu^2/2-A\mu} \left(1 + \frac{D_\alpha(\mu)}{\mu} + O(\mu^{-2})\right). \quad (2.7)$$

For every fixed $J \geq 0$ there are smooth, one-periodic functions $L_1, \dots, L_J$, with $L_1 = D_\alpha$, such that

$$\log p(n) = \frac{\alpha}{2}\mu^2 - A\mu + \eta \log \mu + C + \Psi_\alpha(\mu) + \sum_{j=1}^J \frac{L_j(\mu)}{\mu^j} + O_J(\mu^{-J-1}). \quad (2.8)$$

Sections 3–5 prove this statement and give a finite coefficient-construction procedure at each order.

The phrase “all orders” here means a remainder theorem for every fixed truncation order. It does not assert convergence of the resulting infinite inverse-power series, nor does it resolve exponentially smaller sectors.

Corollary 2.2 (An unavoidable periodic factor). *There is no constant B for which*

$$p(n) \sim B\mu(n)^\eta e^{\alpha\mu(n)^2/2-A\mu(n)}.$$

More precisely, the limit points of the normalized sequence are exactly

$$\left[\exp\{C + \min_\theta \Psi_\alpha(\theta)\}, \exp\{C + \max_\theta \Psi_\alpha(\theta)\} \right],$$

an interval with distinct endpoints.

Proof. The gamma function has no zeros, and $\zeta(1+it) \neq 0$ for real $t \neq 0$ [6]; hence the first Fourier coefficient in (2.3) is nonzero. Thus Ψ_α is nonconstant. For any fixed phase θ , round $4^{j+\theta}/\sqrt{\pi(j+\theta)}$ to the nearest integer n_j . Implicit differentiation of (2.5) shows $\mu(n_j) = j + \theta + o(1)$. Every phase therefore occurs as a subsequential limit. Apply (2.7) and continuity of Ψ_α . $\square$

The first Fourier harmonic has amplitude approximately 0.00100279457386. The modulation is small but does not tend to zero. Its mean is approximately 0.64116608028709. These are reproducible floating-point evaluations, not interval-certified bounds.

Corollary 2.3 (The first three logarithmic growth terms). *With $X = \log n$,*

$$\log p(n) = \frac{X^2}{2\alpha} + \frac{X \log X}{2\alpha} + \left(\frac{\log(\pi/\alpha) - 1}{2\alpha} - \frac{3}{2} \right) X + O((\log X)^2). \quad (2.9)$$

Proof. The defining equation is $X = \alpha\mu - \frac{1}{2} \log \mu - \frac{1}{2} \log \pi$. Hence

$$\alpha\mu = X + \frac{1}{2} \log X + \frac{1}{2} \log(\pi/\alpha) + O((\log X)/X).$$

Substitute this into (2.6) and collect terms of orders X^2 , $X \log X$, and X . $\square$

The positive sign of $X \log X$ is a concrete effect of the factor $k^{-3/2}$ in the growth of c_k . Replacing c_k by 4^k too early loses that information.

2.2 The inverse threshold

For real $y > 1$ define the integer staircase

$$N(y) = \min\{n \geq 0 : p(n) \geq y\}. \quad (2.10)$$

Put

$$Y = \log y, \quad R = \sqrt{2\alpha Y}, \quad s = R/\alpha. \quad (2.11)$$

Theorem 2.4 (Inverse growth and its first correction). *As $y \rightarrow \infty$,*

$$\begin{aligned} \log N(y) = & R + A - \frac{1}{2} \log s - \frac{1}{2} \log \pi \\ & + \frac{\frac{1}{2}(A^2 - A) - \alpha \eta \log s - \alpha \{C + \Psi_\alpha(s + A/\alpha)\}}{R} + O\left(\frac{(\log R)^2}{R^2}\right). \end{aligned} \quad (2.12)$$

In particular,

$$N(y) \sim 8 \sqrt{\frac{e\alpha}{\pi}} \frac{\exp(\sqrt{2\alpha \log y})}{(2\alpha \log y)^{1/4}}. \quad (2.13)$$

The inverse has an expansion to arbitrary fixed order in R^{-1} , with coefficients polynomial in $\log R$ and smooth periodic functions of $s + A/\alpha$. Section 6 proves the passage from a smooth asymptotic inverse to the integer threshold.

3 An all-orders coefficient saddle estimate

3.1 Positive cumulants and the saddle

For $t > 0$ write

$$H(t) = \log F(e^{-t}), \quad K_j(t) = (-1)^j H^{(j)}(t) = \sum_{k \geq 1} c_k^j \sum_{\ell \geq 1} \ell^{j-1} e^{-\ell t c_k} \quad (j \geq 1). \quad (3.1)$$

Absolute convergence on each half-plane $\operatorname{Re} t \geq t_0 > 0$ justifies termwise differentiation. Since $K'_1(t) = -K_2(t) < 0$, K_1 decreases strictly from infinity to zero. Thus, for each $n > 0$, there is a unique $t_n > 0$ satisfying

$$K_1(t_n) = n. \quad (3.2)$$

The real logarithmic interpolation of the part sizes is

$$r(x) = \alpha x - \frac{1}{2} \log \pi + \log \Gamma(x + \frac{1}{2}) - \log \Gamma(x + 2), \quad x \geq 1. \quad (3.3)$$

It obeys $r(k) = \log c_k$. Since $r'(1) = 1/2$ and $r''(x) = \psi'(x + 1/2) - \psi'(x + 2) > 0$, it is strictly increasing, with $r'(x) \rightarrow \alpha$. Define

$$u_n = \log(1/t_n), \quad m_n = r^{-1}(u_n). \quad (3.4)$$

The parameters m_n and $\mu(n)$ are not identical; their small difference is responsible for several cancellations in the explicit formula.

Lemma 3.1 (Size of the scaled cumulants). *Let $t = e^{-r(m)}$ and $m \rightarrow \infty$. For each fixed integer $j \geq 1$,*

$$B_j(t) := t^j K_j(t) = (j-1)! m + O_j(1). \quad (3.5)$$

In particular $nt_n \sim m_n$ and $m_n \sim (\log n)/\alpha$.

Proof. Set $f_j(v) = v^j \sum_{\ell \geq 1} \ell^{j-1} e^{-\ell v}$. For $0 < v \leq 1$, $f_j(v) = (j-1)! + O_j(v)$, by the elementary Laurent expansion of the geometric sum and its derivatives. For $v \geq 1$, $f_j(v)$ decays exponentially up to a fixed polynomial factor. The exact ratio

$$\frac{c_{k+1}}{c_k} = \frac{2(2k+1)}{k+2} \in [2, 4) \quad (3.6)$$

implies $\sum_{tc_k \leq 1} tc_k = O(1)$ and a bounded sum of the large- v tails. There are exactly $\lfloor m \rfloor$ parts with $tc_k \leq 1$. Summing $f_j(tc_k)$ proves (3.5). At the saddle, $nt = B_1 = m + O(1)$, and $\log n = r(m) + \log(m + O(1)) = \alpha m + O(\log m)$. $\square$

Theorem 3.2 (All-orders saddle expansion). *For every fixed $J \geq 0$, with all cumulants evaluated at $t = t_n$,*

$$p(n) = \frac{\exp(H(t) + nt)}{\sqrt{2\pi K_2(t)}} \left(1 + \sum_{j=1}^J E_j(t) + O_J(m_n^{-J-1}) \right). \quad (3.7)$$

The correction E_j is the finite sum

$$E_j = \sum_{\substack{k_3, \dots, k_{2j+2} \geq 0 \\ \sum_{r=3}^{2j+2} (r-2)k_r = 2j}} (-1)^{R/2} (R-1)!! \prod_{r=3}^{2j+2} \frac{1}{k_r!} \left(\frac{K_r}{r! K_2^{r/2}} \right)^{k_r}, \quad R = \sum_{r=3}^{2j+2} r k_r. \quad (3.8)$$

Here R is even, and $E_j = O_j(m_n^{-j})$. In particular,

$$E_1 = \frac{K_4}{8K_2^2} - \frac{5K_3^2}{24K_2^3}, \quad (3.9)$$

$$E_2 = -\frac{K_6}{48K_2^3} + \frac{7K_3K_5}{48K_2^4} + \frac{35K_4^2}{384K_2^4} - \frac{35K_3^2K_4}{64K_2^5} + \frac{385K_3^4}{1152K_2^6}. \quad (3.10)$$

3.2 The noncentral arcs: why the coefficient estimate is valid

The Cauchy coefficient integral is

$$p(n) = \frac{e^{H(t)+nt}}{2\pi} \int_{-\pi}^{\pi} \exp\{H(t-i\theta) - H(t) - in\theta\} d\theta. \quad (3.11)$$

Its normalized absolute value satisfies the exact identity

$$\left| \frac{F(e^{-t+i\theta})}{F(e^{-t})} \right| = \prod_{k \geq 1} \left(1 + \frac{4e^{-tc_k} \sin^2(c_k \theta/2)}{(1 - e^{-tc_k})^2} \right)^{-1/2}. \quad (3.12)$$

All factors on the right are at most one. Fix a requested error order. In the estimates below, $B > 0$, the number of fixed factors K , and a small $\delta > 0$ are chosen in that order and then held fixed as $m \rightarrow \infty$.

For $|\theta| \leq tm^B$, use every part with $c_k \leq 1/(tm^B)$. There are $m - O(\log m)$ such parts. For them, $|c_k \theta| \leq 1$, and elementary sine and exponential estimates give

$$\left| \frac{F(e^{-t+i\theta})}{F(e^{-t})} \right| \leq \left(1 + c(\theta/t)^2 \right)^{-c'm} \quad (3.13)$$

for fixed positive c, c' . Between $t \log m / \sqrt{m}$ and t , integration of this bound gives $O(t \exp\{-c''(\log m)^2\})$, after changing constants. Between t and tm^B , it gives $O(tm^B e^{-c'm})$. Both are smaller than $(t/\sqrt{m})m^{-L}$ for every fixed L .

Next, for $tm^B \leq |\theta| \leq \delta$, select the first K factors and choose $\delta c_K \leq 1$. Each selected factor in (3.12) is at most $C_K t/|\theta|$. Hence

$$\int_{tm^B \leq |\theta| \leq \delta} \left| \frac{F(e^{-t+i\theta})}{F(e^{-t})} \right| d\theta = O_K \left(t m^{-B(K-1)} \right). \quad (3.14)$$

Taking K sufficiently large makes this negligible to any specified inverse-power order relative to the main width $t/\sqrt{m}$.

Finally consider $\delta \leq |\theta| \leq \pi$. The unit-part factor is $O_\delta(t)$. The functions $\sin(2\theta/2)$ and $\sin(5\theta/2)$ do not vanish simultaneously on this compact set: a simultaneous zero would require $\theta \in 2\pi\mathbb{Z}$ because $\gcd(2, 5) = 1$. Therefore at least one of the part-2 and part-5 factors is uniformly $O_\delta(t)$. Thus

$$\int_{\delta \leq |\theta| \leq \pi} \left| \frac{F(e^{-t+i\theta})}{F(e^{-t})} \right| d\theta = O_\delta(t^2). \quad (3.15)$$

Since $t = \exp(-\alpha m + O(\log m))$, this is again negligible to every inverse power of m relative to $t/\sqrt{m}$.

These estimates eliminate possible competing root-of-unity arcs at the precision claimed in [theorem 3.2](#). They do not assert that all those arcs vanish, or identify their exponentially smaller contributions.

3.3 The central expansion and its remainder

At the saddle the linear term in (3.11) cancels. With $v = \theta\sqrt{K_2}$, the exponent becomes

$$-\frac{v^2}{2} + \sum_{r \geq 3} \frac{i^r K_r}{r! K_2^{r/2}} v^r. \quad (3.16)$$

For each fixed r , [lemma 3.1](#) gives $K_r/K_2^{r/2} = O_r(m^{-(r-2)/2})$. The central interval $|\theta| \leq t \log m/\sqrt{m}$ corresponds to $|v| \leq C \log m$. Taylor estimates for $H(t - i\theta)$ follow directly from its absolutely convergent sums: the absolute value of a derivative along this vertical segment is bounded by the corresponding positive derivative sum at t .

Expand the exponential in (3.16), assigning weight $r - 2$ to the term of degree r . Retain total weight through $2J + 1$. The integrated terms of odd weight vanish because their total degree in v is odd. The weight- $2j$ terms have Gaussian integral $(-1)^{R/2}(R - 1)!!$ after division by $\sqrt{2\pi}$, giving (3.8).

For completeness, the remainder estimate does not require retaining a power of $\log m$. On $|v| \leq C \log m$, the absolute correction to the Gaussian exponent is $O(m^{-1/2}(|v|^3 + 1))$, which is at most $v^2/4 + o(1)$ outside a fixed bounded interval. A Taylor remainder through weight $2J + 1$ is bounded by

$$C_J m^{-J-1} (1 + |v|^{M_J}) e^{-v^2/4}$$

for a fixed integer M_J , after including the Gaussian factor and increasing the Taylor order of H to $2J + 4$ if necessary. Its integral is $O_J(m^{-J-1})$. The omitted Gaussian tails are superpolynomially small. Combining this with (3.13)–(3.15) proves [theorem 3.2](#).

The leading values of the correction polynomials are useful checks:

$$E_1 = -\frac{1}{12m} + O(m^{-2}), \quad E_2 = \frac{1}{288m^2} + O(m^{-3}), \quad E_3 = \frac{139}{51840m^3} + O(m^{-4}). \quad (3.17)$$

They follow by substituting $B_r \sim (r - 1)!m$ in (3.8). At a finite m , using the exact cumulants is preferable to replacing each E_j by only its first universal term.

3.4 A reusable sufficient hypothesis

The same proof applies to an increasing sequence of positive integers $a_1, a_2, \dots$ when $a_1 = 1$, the ratios are eventually between two fixed constants $1 < q \leq Q < \infty$, and a finite subset of $\{a_2, a_3, \dots\}$ has greatest common divisor one. Replace m by the number of active parts $a_k \leq 1/t$, up to an $O(1)$ difference. The finite gcd condition is sufficient for the last compact arc estimate; it is not claimed to be necessary. In particular, the proof as stated should not be applied unchanged to pure powers of two, whose tail has a nontrivial gcd.

4 The Catalan product and its moving periodic phase

4.1 The logarithm of a Catalan number and the Barnes product

The standard gamma quotient in (3.3) gives the complete expansion

$$r(x) = \alpha x - \beta \log x + c + \sum_{j \geq 1} b_j x^{-j}, \quad \beta = \frac{3}{2}, \quad c = -\frac{1}{2} \log \pi, \quad b_1 = -\frac{9}{8}, \quad b_2 = \frac{1}{2}. \quad (4.1)$$

The meaning is asymptotic, to every fixed order with differentiated remainders. A convenient exact coefficient formula is

$$b_j = \frac{(-1)^{j+1}}{j(j+1)} (B_{j+1}(1/2) - B_{j+1}(2)), \quad (4.2)$$

where $B_j(x)$ is a Bernoulli polynomial. For example, $b_3 = -21/64$. This follows by subtracting the shifted Stirling expansions of the two log-gamma functions; it also gives an effective coefficient source for every subsequent step.

For an integer $M \geq 0$, put $S(M) = \sum_{k=1}^M r(k)$. Repeated use of $G(z+1) = \Gamma(z)G(z)$ gives the exact identity

$$S(M) = \frac{\alpha}{2} M(M+1) - \frac{M}{2} \log \pi + \log G(M+3/2) - \log G(M+3) - \log G(3/2). \quad (4.3)$$

The Barnes expansion [5], or Euler–Maclaurin summation followed by the exact identity to fix the constant, yields

$$S(M) = \frac{\alpha}{2} M^2 - \frac{3}{2} M \log M + \left(\frac{\alpha}{2} - \frac{1}{2} \log \pi + \frac{3}{2} \right) M - \frac{15}{8} \log M + K \\ - \frac{19}{16M} + \frac{65}{128M^2} + O(M^{-3}), \quad (4.4)$$

$$K = -\frac{3}{4} \log(2\pi) - \log G(3/2). \quad (4.5)$$

Subtracting the expansion at $M-1$ from the expansion at M recovers (4.1); this is a useful local check, but the Barnes identity is what determines the constant K .

4.2 Separating the bulk from a localized lattice sum

Let

$$g(v) = -\log(1 - e^{-v}) - (-v)_+, \quad (-v)_+ = \max\{-v, 0\}. \quad (4.6)$$

As $v \rightarrow -\infty$, $g(v) = O(e^v)$; as $v \rightarrow +\infty$ it decays superexponentially. It is continuous and piecewise analytic, with a jump in its first derivative at zero. This mild kink will cancel against the fractional-part polynomial.

For $m \geq 1$, let $M = \lfloor m \rfloor$ and $\theta = m - M$. Define

$$\mathcal{H}(m) = H(e^{-r(m)}). \quad (4.7)$$

Then the following identity is exact:

$$\mathcal{H}(m) = Mr(m) - S(M) + \sum_{k \geq 1} g(r(k) - r(m)). \quad (4.8)$$

Indeed, the subtracted linear part $(-v)_+$ contributes $r(m) - r(k)$ precisely for $k \leq M$. If m is an integer, the boundary term is zero, so no convention ambiguity occurs.

Replacing $r(m + d) - r(m)$ by αd in the localized sum suggests

$$\Psi_\alpha(\theta) = \frac{\alpha}{2}\theta(1 - \theta) + \sum_{j \in \mathbb{Z}} g(\alpha(j - \theta)), \quad 0 \leq \theta \leq 1. \quad (4.9)$$

The sum converges absolutely, exponentially fast on the left and faster on the right. The polynomial is forced by expansion of $Mr(m) - S(M)$; it is not an adjustable normalization.

Lemma 4.1 (Fourier representation of the phase). *The lattice formula (4.9) equals (2.3). In particular it extends to a smooth, nonconstant one-periodic function, and termwise derivatives of its Fourier series of every fixed order are legitimate.*

Proof. For $\operatorname{Re} s > 0$, change variables $y = e^v$ and expand $-\log(1 - e^{-y}) = \sum_{\ell \geq 1} e^{-\ell y} / \ell$ to obtain

$$\int_{\mathbb{R}} g(v) e^{sv} dv = \Gamma(s) \zeta(1 + s) - \frac{1}{s^2}.$$

The integral involving g extends analytically to $\operatorname{Re} s > -1$. The apparent singularity on the right at zero is removable: the product of the Laurent expansions of $\Gamma(s)$ and $\zeta(1 + s)$ has leading term s^{-2} , no s^{-1} term, and constant term κ . Consequently the mean of the lattice sum is κ/α .

For $\ell \neq 0$, integrating its Fourier coefficient over $0 \leq \theta \leq 1$ and unfolding the lattice gives

$$\frac{1}{\alpha} \left\{ \Gamma(s_\ell) \zeta(1 + s_\ell) - \frac{1}{s_\ell^2} \right\}, \quad s_\ell = \frac{2\pi i \ell}{\alpha}.$$

The coefficient of $\alpha\theta(1 - \theta)/2$ is $-\alpha/(4\pi^2\ell^2)$, which cancels the contribution of $-1/(\alpha s_\ell^2)$. Its mean is $\alpha/12$. This proves (2.3). The gamma factor has exponential decay on the vertical axis; the zeta factor and its fixed-order derivatives have at most polynomial growth there, as follows for example from Euler–Maclaurin continuation. Hence every differentiated Fourier series converges absolutely. Nonconstancy was proved in [corollary 2.2](#). $\square$

[Write note, batch 7301, 1 October 2026.] The same Mellin kernel and the same constant occur elsewhere in ProveIt, for a different product. The Fabius-tree compilation *Geometric Uniform Convolutions and New Frontiers* [10], in its section “Mellin analysis and the hidden log-periodic phase”, treats the Laplace product $H_q(z) = \prod_{k \geq 0} (1 - e^{-zq^k}) / (zq^k)$, with kernel $g(x) = \log((1 - e^{-x})/x)$, Mellin transform $-\Gamma(1 + s)\zeta(1 + s)/s = -\Gamma(s)\zeta(1 + s)$, and constant $c_1 = \pi^2/12 - \gamma^2/2 - \gamma_1$, which is the κ of (2.2). Neither text proves the other’s theorem: the products, the lattices and the normalizations differ, and the coincidence is the de Bruijn–Mahler mechanism common to both.

4.3 The first two bulk orders, with a uniform remainder

Theorem 4.2 (Moving-lattice expansion). *Uniformly in the fractional part of m , as $m \rightarrow \infty$,*

$$\mathcal{H}(m) = \frac{\alpha}{2}m^2 - \left(\frac{3}{2} + \frac{\alpha}{2}\right)m + \frac{15}{8}\log m - \frac{9}{8} - K + \Psi_\alpha(m) + \frac{P_1(m)}{m} + O(m^{-2}), \quad (4.10)$$

$$P_1(\theta) = \frac{27}{16} - \frac{3}{2}\partial_\alpha \Psi_\alpha(\theta). \quad (4.11)$$

There is a complete extension with $\sum_{j=1}^J P_j(m)m^{-j}$ and remainder $O_J(m^{-J-1})$. The P_j are smooth and one-periodic. The expansion may be differentiated any fixed number of times if sufficiently many terms are retained.

Proof. In (4.8), put $d = k - m$. For a bounded d ,

$$r(m+d) - r(m) = \alpha d - \frac{\beta d}{m} + O((1+|d|)^2 m^{-2}).$$

Taylor expansion of the localized term therefore begins as

$$g(\alpha d) - \frac{\beta d}{m}g'(\alpha d) + O(m^{-2}(1+|d|)^L e^{-c_0|d|})$$

for a fixed L ; on the positive side an even stronger tail is available. One-sided values at $d = 0$ cause no difficulty because the multiplier d vanishes. The missing terms with $k \leq 0$ in the infinite lattice are exponentially small.

Expansion of the exact bulk term using (4.1) and (4.4) gives

$$\begin{aligned} Mr(m) - S(M) &= \frac{\alpha}{2}m^2 - (\beta + \alpha/2)m + (\beta/2 - b_1)\log m + b_1 - K \\ &\quad + \frac{\alpha}{2}\theta(1-\theta) + \frac{27/16 - (\beta/2)\theta(1-\theta)}{m} + O(m^{-2}). \end{aligned}$$

Combining this with the localized sum gives (4.10), since

$$\partial_\alpha \Psi_\alpha(\theta) = \frac{1}{2}\theta(1-\theta) + \sum_{j \in \mathbb{Z}} (j-\theta)g'(\alpha(j-\theta)).$$

Here is a uniform justification of the arbitrary-order procedure. Split the lattice into $|d| \leq m^{1/3}$ and its complement. In the first range, expansions of $r(m+d) - r(m)$ and Taylor expansions of g have remainders bounded by a fixed polynomial in $|d|$ times $m^{-J-1}e^{-c_0|d|}$. The signs of d and of the two arguments of g coincide, so Taylor expansion never crosses the kink. Summing preserves $O_J(m^{-J-1})$. In the complementary range, monotonicity and the positive lower bound for r' give an exponentially small tail, including the initial indices far to the left. The same reasoning applies after any fixed number of derivatives, using more terms at the start. At integral m , the jumps of the one-sided lattice derivatives cancel exactly with those of the fractional-part bulk polynomials. This can be seen either from the exact decomposition, by transferring the boundary index between the two pieces, or coefficient by coefficient in that identity. The resulting coefficient functions therefore match with all fixed-order derivatives across the endpoints of a period. $\square$

4.4 A constructive rule for every coefficient

The preceding proof can be made into a finite symbolic recipe. Write

$$r(m+d) - r(m) = \alpha d + \sum_{j \geq 1} D_j(d)m^{-j}, \quad (4.12)$$

where

$$D_j(d) = \frac{\beta(-1)^j d^j}{j} + \sum_{h=1}^{j-1} b_h (-1)^{j-h} \binom{j-1}{h-1} d^{j-h}. \quad (4.13)$$

For example, $D_1(d) = -\beta d$ and $D_2(d) = \beta d^2/2 - b_1 d$. The localized contribution to the coefficient of m^{-j} is

$$T_j(\theta) = \sum_{d \in \mathbb{Z}-\theta} \sum_{\ell=1}^j \frac{g^{(\ell)}(\alpha d)}{\ell!} \sum_{\substack{r_1+\dots+r_\ell=j \\ r_\nu \geq 1}} \prod_{\nu=1}^{\ell} D_{r_\nu}(d). \quad (4.14)$$

Each coefficient is thus an absolutely convergent lattice sum of explicit functions. At $d = 0$, all $D_j(0)$ vanish, so arbitrary consistent one-sided values of the derivatives give the same term, namely zero.

Expand $Mr(m) - S(M)$ with $M = m - \theta$, then add (4.14). This produces $P_j(\theta)$. Bernoulli polynomials supply the bulk coefficients; finite compositions of the D_j supply the localized coefficients. This is an all-orders construction, not merely a claim that unspecified coefficients exist. Fourier transformation can be used afterward to replace the localized sums by rapidly convergent Fourier series.

5 Deriving the explicit coefficient expansion

5.1 The saddle coordinate and its correction

Let $t = e^{-r(m)}$, and define

$$Q(m) = t K_1(t) = \frac{\mathcal{H}'(m)}{r'(m)}. \quad (5.1)$$

Differentiation of (4.10), followed by series division, gives

$$Q(m) = m + q_0(m) + \frac{q_1(m)}{m} + O(m^{-2}), \quad (5.2)$$

$$q_0 = -\frac{1}{2} + \frac{\Psi'_\alpha}{\alpha}, \quad q_1 = \frac{P'_1}{\alpha} + \frac{\beta \Psi'_\alpha}{\alpha^2}. \quad (5.3)$$

The phase functions on the right are evaluated at m . The scaled variance satisfies the exact identity

$$B_2 = t^2 K_2 = Q + \frac{Q'}{r'}, \quad (5.4)$$

so that

$$B_2 = m + q_0 + \frac{1 + q'_0}{\alpha} + O(m^{-1}). \quad (5.5)$$

More generally, the recurrence

$$B_{j+1} = j B_j + \frac{B'_j}{r'} \quad (5.6)$$

generates every scaled cumulant from Q . This is useful both for the proof and for the all-orders construction.

At the saddle, $n = e^{r(m)} Q(m)$. Consequently

$$\log n = \alpha m - \frac{1}{2} \log m + c + \frac{b_1 + q_0(m)}{m} + \frac{b_2 + q_1(m) - q_0(m)^2/2}{m^2} + O(m^{-3}). \quad (5.7)$$

The core coordinate (2.5) removes precisely the first three terms. Write

$$m = \mu + \frac{a(\mu)}{\mu} + \frac{b(\mu)}{\mu^2} + O(\mu^{-3}). \quad (5.8)$$

Taylor substitution into (5.7) gives the explicit functions

$$a = -\frac{b_1 + q_0}{\alpha}, \quad (5.9)$$

$$b = \frac{(1/2 - q'_0)a - (b_2 + q_1 - q_0^2/2)}{\alpha}. \quad (5.10)$$

All functions here are evaluated at μ . The residual derivative in (5.7) is $\alpha + O(1/m)$, which is bounded away from zero. Thus the formal coefficient comparison also gives the asserted analytic error by the mean value theorem. Repeating the argument proves an all-orders reversion with smooth periodic coefficients.

5.2 The normalization and the first correction

Taking logarithms of (3.7), using (3.17), and writing $K_2 = e^{2r(m)} B_2$ gives

$$\log p(n) = \mathcal{H}(m) + Q(m) - r(m) - \frac{1}{2} \log(2\pi B_2(m)) - \frac{1}{12m} + O(m^{-2}). \quad (5.11)$$

Let $d = b_1 + q_0$. Substituting the preceding expansions into this equation first produces the expression in the *true* saddle coordinate:

$$\log p(n) = \frac{\alpha}{2} m^2 - Am + \eta \log m + C + \Psi_\alpha(m) + d(m) + \frac{U(m)}{m} + O(m^{-2}), \quad (5.12)$$

$$U = P_1 + q_1 - b_1 - \frac{q_0}{2} - \frac{1 + q'_0}{2\alpha} - \frac{1}{12}. \quad (5.13)$$

The constant simplifies because

$$-K - c - \frac{1}{2} \log(2\pi) = \frac{1}{4} \log 2 + \frac{3}{4} \log \pi + \log G(3/2) = C.$$

Now substitute (5.8). At order one, $\alpha a + d = 0$, so the apparently additional phase derivative in d *cancels*. At order $1/\mu$, the coefficient is

$$\alpha b + (-A + \Psi'_\alpha + q'_0)a + U. \quad (5.14)$$

Using (5.9)–(5.10), this simplifies first to

$$P_1 - \frac{7}{12} - \frac{1}{2\alpha} + (1/2 - b_1)q_0 - \frac{1}{2}q_0^2 - \frac{q'_0}{2\alpha},$$

and then, with $b_1 = -9/8$ and (4.11), exactly to (2.4). In particular the P'_1 terms disappear. This explicit cancellation is one reason to distinguish m and μ throughout. Equations (2.6) and (2.7) follow.

5.3 Completion of the all-orders proof

For a fixed requested order J , use the following finite operations. First compute sufficiently many b_j from (4.2), bulk coefficients from (4.3), and P_j from (4.14). Next form Q and $B_2, B_3, \dots$ by (5.1) and (5.6). Substitute these in the finite Gaussian polynomials (3.8) and take the logarithm of the saddle prefactor and correction. Finally revert (5.7) about μ and substitute.

The derivative-controlled estimates in theorem 4.2 justify each finite calculus operation. The saddle theorem supplies the coefficient-extraction remainder. The reversion has derivative bounded away from zero, so it preserves the desired error once one extra order has been retained before substitution into the quadratic term. All coefficient operations are addition, multiplication, differentiation, and division by series with nonzero constant term in an algebra of smooth periodic functions. Thus no new unbounded coefficient, such as $\log \mu$, is generated beyond the single displayed $\eta \log \mu$. This proves (2.8) and completes theorem 2.1.

A safe implementation rule is to retain two orders more than the final requested order during the m -to- μ substitution. This is a bookkeeping rule, not a new analytic assumption.

6 Inverse growth and the integer staircase

6.1 Reversion of the quadratic logarithmic core

Start with $Y = \log y$ and $s = \sqrt{2Y/\alpha}$. The leading quadratic term suggests $\mu = s + A/\alpha + o(1)$. Set

$$\mu = s + \frac{A}{\alpha} + \frac{v(s)}{s} + O\left(\frac{(\log s)^2}{s^2}\right).$$

Substitution into (2.6), at constant order after the quadratic and linear terms cancel, yields

$$v(s) = \frac{A^2}{2\alpha^2} - \frac{\eta \log s + C + \Psi_\alpha(s + A/\alpha)}{\alpha}. \quad (6.1)$$

Since $\log n = \alpha\mu - \frac{1}{2} \log \mu - \frac{1}{2} \log \pi$, (2.12) follows by Taylor expansion. The term $D_\alpha(\mu)/\mu$ in $\log p$ first affects the next inverse order, not the displayed $1/R$ correction.

Higher inverse terms are obtained by the same procedure. The leading inverse derivative is of size s , so a residual term is removed by division by s and a bounded perturbation. Every differentiation of a periodic coefficient leaves it in the same coefficient algebra, while products of the already occurring $\log s$ terms produce polynomials in $\log s$. The phase is always reducible by Taylor expansion to $s + A/\alpha$. This proves the asserted form of the all-orders inverse.

6.2 Why a continuous calculation describes the discrete threshold

Adding a unit part proves that $p(n)$ is nondecreasing. Moreover, $p(n) - p(n-1)$ counts partitions of n without a unit part. Every integer $n \geq 4$ is a nonnegative combination of 2 and 5: even n use only 2, and odd $n \geq 5$ use one 5 and then 2. Therefore $p(n) > p(n-1)$ for all $n \geq 4$, and the threshold (2.10) is unambiguous for large y .

Let $f_J(x)$ denote the smooth right side of (2.8), with n replaced by a large positive real x . Its derivative with respect to $X = \log x$ satisfies

$$\frac{d}{dX} f_J(e^X) = \mu(e^X) + O(1) > 0. \quad (6.2)$$

The asymptotic theorem says, at integers, $\log p(n) = f_J(n) + O(\mu(n)^{-J-1})$ with a uniform constant. If $f_J(x) = Y$, a change of $O(\mu^{-J-2})$ in $\log x$ moves $f_J(x)$ by the entire coefficient error. Bracketing the corresponding real arguments by neighboring integers therefore gives

$$\log N(y) - \log x = O(\mu(x)^{-J-2}) + O(x^{-1}), \quad (6.3)$$

provided x is the inverse of the chosen smooth truncation. The bracket uses only monotonicity, not an assumed canonical interpolation of p .

The $O(x^{-1})$ term comes from rounding an argument to an integer; it is smaller than every fixed inverse power of R . The first term is the analytic truncation uncertainty and is much larger at a fixed order. Accordingly, the theorem gives an asymptotic formula for the threshold, not an exact integer reconstruction algorithm for every y . Equations (6.1) and (6.3) prove [theorem 2.4](#).

6.3 A practical caution

The expansion parameter is approximately $1/\log n$, not $1/n$. Even at $n = 10^6$, the effective number of active parts is only about 11.36. The leading inverse equivalent can therefore have a large finite-argument error. The first correction helps substantially, but a numerical application at moderate size should use the implicit saddle formula and invert it numerically, preferably with a certified remainder if an integer decision is required. [Section 9](#) reports actual errors rather than presenting an asymptotic equivalent as a uniformly accurate approximation.

7 The largest Catalan part in a random partition

Choose uniformly among the $p(n)$ partitions of n . Let J_n be the largest index k for which a part of size c_k occurs. For $n > 0$ this is well-defined. It is natural to center J_n at the saddle cutoff, not at $\log_4 n$ alone.

Theorem 7.1 (A phase-dependent largest-index law). *Let $m = m_n$ be the true saddle coordinate, $M = \lfloor m \rfloor$, and $\theta_n = m - M$. For every fixed integer h ,*

$$\mathbb{P}(J_n \leq M + h) = \prod_{j=h+1}^{\infty} (1 - \exp\{-4^{j-\theta_n}\}) + O_h(m_n^{-1}). \quad (7.1)$$

In particular, along any subsequence with $\theta_n \rightarrow \theta \in [0, 1)$, $J_n - M$ converges in distribution to the integer-valued law with cumulative probabilities

$$G_h(\theta) = \prod_{j=h+1}^{\infty} (1 - e^{-4^{j-\theta}}). \quad (7.2)$$

The estimates are for fixed h ; a growing- h or moment theorem is not being asserted.

Proof. Under the product probability measure at parameter t , let the numbers of parts of sizes c_k be independent geometric random variables:

$$\mathbb{P}(Z_k = z) = (1 - e^{-tc_k})e^{-tc_k z}, \quad z = 0, 1, 2, \dots$$

Conditioning on $\sum_k c_k Z_k = n$ gives the uniform distribution on partitions of n , since every such partition has the same product weight. This provides an exact probabilistic interpretation, not an approximation.

Put $K = M + h$ and split the product at K . Let H_{pre} and H_{tail} be its logarithms. At the full saddle, the removed tail has scaled cumulants of every fixed order $O_h(1)$. To see this, write $k = M + j$ and use

$$tc_{M+j} = \exp\{r(M+j) - r(m)\},$$

which is bounded away from zero for each fixed starting $j = h + 1$ and then grows geometrically. The corresponding sums of the functions f_r from [lemma 3.1](#) are bounded. Thus the removed mean is $O_h(1/t)$, whereas the full standard deviation is $\asymp \sqrt{m}/t$.

Apply the same coefficient saddle argument to the prefix product. Its saddle differs from t by $O_h(t/m)$: the mean deficit is $O_h(1/t)$ and the derivative of the mean is of size m/t^2 . Taylor's theorem then shows that its saddle action differs from the prefix action evaluated at t by $O_h(1/m)$. The variance ratio is $1 + O_h(1/m)$, and the two first saddle correction factors are also $1 + O_h(1/m)$. The arc estimates remain uniform because the prefix still contains $m - O_h(1)$ active parts and all the fixed small factors used in [Section 3](#). Consequently,

$$\frac{[z^n] \prod_{k \leq K} (1 - z^{c_k})^{-1}}{p(n)} = e^{-H_{\text{tail}}(t)} (1 + O_h(m^{-1})) = \prod_{k > K} (1 - e^{-tc_k}) (1 + O_h(m^{-1})).$$

This ratio is exactly the desired conditional probability.

Finally, for each fixed j ,

$$tc_{M+j} = 4^{j-\theta_n} \left(1 + O\left(\frac{1+|j|}{m}\right) \right).$$

For $0 \leq j - \theta_n \leq m/2$, the corresponding logarithmic error is bounded by $C(1 + |j|)/m$. The derivatives of the logarithmic tail factors have a summable, superexponentially decaying majorant throughout this range. The range $j - \theta_n > m/2$ contributes a superexponentially small tail, and the finitely many negative indices are controlled separately. Thus summation replaces the tail product by (7.2) with error $O_h(m^{-1})$. This proves (7.1). For each phase, $G_h \rightarrow 1$ as $h \rightarrow +\infty$ and $G_h \rightarrow 0$ as $h \rightarrow -\infty$, so it is the cumulative distribution of a proper integer-valued law. Pointwise convergence at the integer cutoffs gives the distributional assertion. $\square$

[Write note, batch 98, 5 October 2026.] Part II supplies what this theorem does not assert: the first correction $G_h(1 + C_h/m)$ (corollary 16.5, proved in both batch-98 manuscripts), uniform two-sided tails and convergence of every polynomial and exponential moment (lemma 17.3 and corollary 17.4), the first correction summed over every threshold (theorem 17.5) with $\mathbb{E}J_n$ and $\text{Var } J_n$ to $O(m^{-2})$ (17.11)–(17.12), and a growing-offset estimate for $|h| \leq m^{1/8}$ (theorem 17.1).

The local scale $4^{j-\theta}$ in this law is the same scale that produces the Fourier amplitude in (2.3). Both results describe the bounded-width transition between parts that are typically available in many copies and parts that are rarely used. The phase is not noise: it records where the continuous cutoff falls between two consecutive Catalan indices.

8 A universality statement for polynomially perturbed geometric parts

The preceding method separates universal local behavior from sequence-specific normalization. The following proposition makes that separation precise.

Proposition 8.1 (General exponential–polynomial part growth). *Suppose an increasing integer part sequence a_k satisfies the sufficient saddle hypotheses at the end of Section 3. Suppose its logarithm has a smooth eventual interpolation with a termwise differentiable expansion*

$$r(k) = \alpha k - \beta \log k + c + b_1/k + b_2/k^2 + \cdots, \quad \alpha > 0.$$

Define $K_{\mathcal{A}}$ by

$$\begin{aligned} \sum_{k=1}^M \log a_k &= \frac{\alpha}{2} M^2 - \beta M \log M + (\alpha/2 + c + \beta) M \\ &\quad + (b_1 - \beta/2) \log M + K_{\mathcal{A}} + O(M^{-1}). \end{aligned}$$

Let $p_{\mathcal{A}}(n)$ be the corresponding partition number, and let μ be the large solution of $n = e^c e^{\alpha\mu} \mu^{1-\beta}$. Then

$$\begin{aligned} \log p_{\mathcal{A}}(n) &= \frac{\alpha}{2} \mu^2 + (1 - \beta - 3\alpha/2)\mu + (3\beta/2 - b_1 - 1/2) \log \mu \\ &\quad - K_{\mathcal{A}} - c - \frac{1}{2} \log(2\pi) + \Psi_{\alpha}(\mu) + O(\mu^{-1}). \end{aligned} \tag{8.1}$$

There is an all-orders extension with smooth periodic inverse-power coefficients. The function Ψ_{α} is exactly (2.3); it depends on α , not on β or the finite initial segment.

Proof. The constant $K_{\mathcal{A}}$ exists by Euler–Maclaurin summation of the assumed expansion, or by subtracting enough explicitly summable terms. The bulk and local argument gives

$$\mathcal{H}(m) = \frac{\alpha}{2} m^2 - (\beta + \alpha/2)m + (\beta/2 - b_1) \log m + b_1 - K_{\mathcal{A}} + \Psi_{\alpha}(m) + O(m^{-1}).$$

The local limit $r(m+d) - r(m) \rightarrow \alpha d$ is unchanged. Also $Q = m + q_0 + O(m^{-1})$, where $q_0 = -1/2 + \Psi'_{\alpha}/\alpha$, independent of β . Therefore

$$\log n = \alpha m + (1 - \beta) \log m + c + (b_1 + q_0)/m + O(m^{-2}).$$

Reversion about μ again gives $m = \mu - (b_1 + q_0)/(\alpha\mu) + O(\mu^{-2})$. In the logarithm of the saddle estimate, the constant $b_1 + q_0$ cancels against the quadratic-core displacement just as before. The remaining linear, logarithmic, and constant terms are exactly (8.1). The proof of every higher order is the same finite construction with the assumed coefficients b_j in place of their Catalan values. $\square$

This proposition explains why merely computing the exponential ratio of part sizes is insufficient. The coefficient β changes the linear term and the size coordinate; b_1 changes the power of μ ; and the whole initial sequence contributes to $K_{\mathcal{A}}$. The local periodic shape, on the other hand, is determined by the limiting logarithmic gap.

9 Computation, independent checks, and reproducibility

9.1 What the supplied programs do

The package contains an exact integer dynamic program for $p(0), \dots, p(N)$, a separate logarithmic-derivative recurrence check, arbitrary-precision saddle evaluation, independent Fourier and real-lattice evaluations of the phase, the explicit first-correction formula, and the inverse formula. The numerical program implements E_1 and E_2 directly; a rational multi-index generator lists the Gaussian polynomials through E_4 in the recorded data. The all-orders theorem does not depend on a claim that an unlimited symbolic implementation is already supplied.

Exact dynamic programming starts with $p(0) = 1$ and $p(n) = 0$ for $n > 0$. For each allowed part $c \leq N$, in increasing order, replace $p(n)$ by $p(n) + p(n - c)$ for $n = c, c + 1, \dots, N$. The ascending update allows repeated copies of c . An independent recurrence is

$$np(n) = \sum_{j=1}^n b(j)p(n-j), \quad b(j) = \sum_{c_k | j} c_k, \quad (9.1)$$

obtained from $zF'(z)/F(z)$. Agreement of these two algorithms is checked through $n = 300$. The first 20 terms agree with the inspected OEIS entry. The stored small coefficient file is independently generated; it is not a redistributed download of another author's table.

The complete run used Python 3.13.5 and mpmath 1.3.0 at 60 decimal digits. Exact coefficients were generated through $N = 10^6$. The Barnes product identity was checked at $M = 0, \dots, 20$. Fourier and real-lattice formulas for both Ψ_α and its partial α -derivative were compared at 10 phases; their maximum recorded discrepancy was less than $4.8 \cdot 10^{-59}$. Rational Gaussian algebra gives the first three constants in (3.17) exactly.

These statements describe what was actually executed. Decimal output and finite tail cutoffs are not rigorous interval enclosures, and the tests do not prove the asymptotic estimates. A separate symbolic audit verifies the normalization and first-correction cancellation using exact algebra.

9.2 Coefficient accuracy

Let P_0 denote the saddle prefactor in (3.7), $P_1 = P_0(1 + E_1)$, and $P_2 = P_0(1 + E_1 + E_2)$ in this subsection only. The table reports signed relative error, $P_j/p(n) - 1$.

Table 1: Exact-cumulant saddle approximations. Errors are not assumed to be monotone.

n	m_n	P_0 error	P_1 error	P_2 error
10^2	4.5337	$2.126 \cdot 10^{-2}$	$-4.169 \cdot 10^{-3}$	$-3.016 \cdot 10^{-3}$
10^3	6.2400	$1.832 \cdot 10^{-2}$	$7.139 \cdot 10^{-4}$	$4.696 \cdot 10^{-4}$
10^4	7.9492	$1.380 \cdot 10^{-2}$	$2.681 \cdot 10^{-4}$	$3.448 \cdot 10^{-5}$
10^5	9.6549	$9.577 \cdot 10^{-3}$	$-3.205 \cdot 10^{-4}$	$-1.160 \cdot 10^{-4}$
10^6	11.3564	$8.557 \cdot 10^{-3}$	$6.471 \cdot 10^{-5}$	$4.661 \cdot 10^{-5}$

For reference, some exact values from the computation are

$$\begin{aligned} p(100) &= 2031, \\ p(1000) &= 18299196, \\ p(10000) &= 8125868781802, \\ p(100000) &= 180507423003606259924, \\ p(1000000) &= 198409222879647727339514603123. \end{aligned}$$

The explicit μ formula is more transparent but converges more slowly at these arguments. Exponentiating the right side of (2.6) without, and with, D_α/μ gives the following signed relative errors.

Table 2: Explicit periodic formulas at moderate arguments.

n	Through the periodic constant term	Including D_α/μ
10^2	-0.122792	-0.069993
10^3	-0.072947	-0.036697
10^4	-0.048618	-0.018121
10^5	-0.040289	-0.012929
10^6	-0.032146	-0.010559

As a check independent of coefficient extraction, substitute $m = 10.2, 30.2, 100.2, 300.2$ directly into the exact product logarithm and subtract (4.10) through P_1/m . The residual multiplied by m^2 is respectively about $-0.1474, -0.2747, -0.3078, -0.3166$. This is consistent with the next periodic coefficient at the fixed phase 0.2. It is not used to infer the remainder theorem.

9.3 Inverse and largest-index checks

At $y = p(n)$ and $n \geq 4$, the true threshold is exactly $N(y) = n$. This permits an unambiguous inverse test.

Table 3: Signed relative error of the inverse approximations at $y = p(n)$.

n	Leading equivalent (2.13)	Corrected formula (2.12)
10^3	2.042368	0.133978
10^4	1.548609	0.071730
10^5	1.256439	0.044236
10^6	1.061185	0.029094

The large leading errors are displayed deliberately: an asymptotic equivalent can be mathematically correct while still being a poor numerical approximation at a million. The corrected formula is much better in this test, but is not a certified integer inversion rule.

At $n = 10^6$ we have $M = 11$ and $\theta_n \approx 0.35639211$. Prefix dynamic programs give exact numerators for the event in theorem 7.1. The comparison is The exact probabilities in the table mean ratios of exactly

Table 4: Largest-index probabilities at $n = 10^6$.

h	Cutoff $M + h$	Exact probability	$G_h(\theta_n)$
-2	9	0.05550439	0.05898171
-1	10	0.39681499	0.41691590
0	11	0.89851234	0.91283590
1	12	0.99999711	0.99994242

computed integers, displayed as decimals. Their limits retain the phase because only a bounded number of indices occupy the transition region.

9.4 Numerical safeguards and build instructions

The cumulant implementation works with the scaled quantities $B_j = t^j K_j$ to avoid unnecessary overflow. It evaluates $1 - e^{-y}$ using `-expm1(-y)` near zero. The geometric sums in (3.1) are computed as rational functions using Eulerian polynomials, rather than by summing a slowly convergent inner series.

For the Fourier coefficients, the supplied code explicitly requests the Euler–Maclaurin method for $\zeta(1 + 2\pi i \ell / \log 4)$ and its derivative. At even ℓ , the denominator $1 - 2^{1-s}$ of an eta-based formula for $\zeta(s)$ vanishes at the evaluation point, so a naive numerical quotient is ill-conditioned. In the tested mpmath environment, forcing the Euler–Maclaurin route removed a discrepancy exposed by the independent real-lattice check. This is a safeguard for the tested implementation, not a claim about every current zeta library. The relevant evaluation methods are documented by mpmath [8].

From the package directory, run

```
python code/verify.py --max-n 10000000 --dps 60
python code/symbolic_audit.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The CSV files are convenient for inspecting the tables, and `data/verification.json` records the environment, exact checks, constants, and additional Gaussian coefficients. The code performs no network access. The bibliography and source-audit notes record the external provenance separately.

[Write note, batch 7301, 1 October 2026.] In the collection these commands must not be run inside the report directory. `code/verify.py` writes by default to the `data/` directory beside `code/`, that is to the shipped data, unless `-output` names another directory; `code/symbolic_audit.py` always writes `../data/symbolic_audit.json` relative to its own location and has no option; and `code/build.sh` changes into `code/`, where there is no `article.tex`, so it fails as shipped. Rerun on a copy of the report directory (the collection README gives the commands). At intake both scripts were rerun on such a copy: the three CSV files were byte-identical to the delivery, `a033552_0_300.txt` and `symbolic_audit.json` equal apart from Windows line ends, and `verification.json` equal apart from line ends and the recorded Python and platform strings.

10 Further research questions and concrete next steps

The following directions are deliberately separated from the proved statements. Each would improve either the mathematical scope or the reliability of computation.

10.1 Certified errors and exact inverse decisions

Replace the unspecified constants in the contour and moving-lattice remainders by explicit, computable bounds. The proof already divides the problem into positive sums, elementary arc inequalities, and Taylor remainders. An interval implementation could enclose the saddle, truncate both lattice tails with proven bounds, and return a rigorous interval for $\log p(n)$. The decisive inverse question is then whether that interval separates two neighboring integer thresholds. This is substantially stronger than a fixed-order asymptotic equivalent.

10.2 A symbolic compiler for periodic asymptotic coefficients

Implement (4.14), (5.6), (3.8), and the m -to- μ reversion in one coefficient algebra. A useful output would include L_2, L_3, L_4 in a normalized basis of phase derivatives, together with exact rational audits. The central design issue is to avoid treating $\Psi_\alpha(m)$ as a constant under differentiation and to track the partial α -derivative separately from the phase derivative.

10.3 Beyond-all-orders contributions

The proof discards the compact outer arcs at a relative scale $O(t\sqrt{m})$, much smaller than every inverse power of m . Their actual contributions may have an arithmetic organization at roots of unity. Identify the dominant secondary arcs, their exponential actions, and how they interact with the analytic tail of the Barnes expansion. A complete transseries including such sectors would go beyond the all-orders Poincaré expansion established here.

10.4 Joint statistics near the largest part

For finitely many indices $M + j$, the unconditioned geometric variables suggest a phase-dependent local product limit. Prove a joint conditional limit with an explicit first correction, rather than only the tail-absence probability in [theorem 7.1](#). Extend the estimates uniformly in h far enough to justify moment convergence and asymptotics for the expected largest index. Neither extension follows automatically from the fixed- h statement.

[Write note, batch 98, 5 October 2026.] Answered in Part II: the joint conditional limit of the multiplicities near the largest part with an explicit first correction, in weighted L^1 for the whole upper process ([theorem 16.1](#)), with its first covariances ([corollary 16.3](#)) and, for finite index sets, in transform form jointly with the number of parts ([theorem 16.4](#)); estimates uniform in h , moment convergence, and the expected largest index to $O(m^{-2})$ (see the note after [Theorem 7.1](#)). Higher orders and the growing range beyond $m^{1/8}$ remain open ([section 23](#)).

10.5 The number of parts and mixed conditioning

Introduce a marking variable for the total number of parts. Small allowed parts and the bounded-width cutoff have very different fluctuation scales. Determine the correct normalization and a joint limit with J_n . A two-variable saddle may need a different treatment from a direct Gaussian approximation because the unit-part multiplicity can remain macroscopic on the scale $1/t$.

[Write note, batch 98, 5 October 2026.] Answered in Part II: the normalization is $t_n N_n$, whose limit is the non-Gaussian $W = \sum_k \mathbf{E}_k / c_k$, with every moment and a first correction ([theorem 18.1](#)); jointly with J_n the limit is independent ([theorem 18.5](#)), and the first dependence is explicit ([theorems 16.6](#) and [18.6](#)). As anticipated here, the small parts are macroscopic on the scale $1/t$; the proofs mark the generating function by the number of parts ([theorems 14.1](#) and [18.1](#)) rather than use a two-variable Gaussian. A lattice local limit for N_n remains open ([section 23](#)).

10.6 Fuss–Catalan and other gamma-quotient part sequences

Apply [proposition 8.1](#) to integer part sequences with a gamma-quotient asymptotic, after checking the initial-part gcd condition and removing any duplicated values. Calculate the normalization constant by Barnes or multiple-gamma products when possible. The proposition predicts a universal periodic shape for each limiting logarithmic gap, but the power and the normalization require a new calculation in every family.

10.7 Exact counting complexity

The supplied dynamic program uses $O(n \log n)$ integer additions to compute all coefficients through n , because there are $O(\log n)$ allowed parts up to n . That is not a polynomial-in- $\log n$ algorithm for a single exact value. Determine whether Catalan-part partitions admit a substantially faster exact counting method of the kind motivated by Pak’s discussion [2]. The asymptotic formulas may provide useful bounds or numerical guidance, but they do not settle this question.

10.8 Formalization and a periodic coefficient algebra

The finite coefficient identities, the rational Gaussian multi-index formula, the gamma-quotient recurrence for Catalan numbers, and the monotonicity argument for $N(y)$ are natural first formalization targets. The analytic frontier is a reusable theorem for positive Euler products with a controlled central arc and fixed-factor minor-arc suppression. For ProveIt’s transseries development, an algebra of smooth or analytic periodic coefficient germs would encode the phase correctly. One should not label the present contour proof formalized merely because some underlying algebraic operations have Lean counterparts.

11 Conclusion

A033552 admits a concrete, computable all-orders asymptotic description whose most visible unusual feature is a persistent periodic amplitude. The dominant size coordinate is not simply $\log_4 n$: the Catalan polynomial factor shifts the saddle and changes both the logarithmic power and the inverse growth. Keeping that shift, the exact Barnes constant, the local lattice correction, and the Gaussian coefficient corrections in one calculation yields (2.6) and its inverse.

The main analytic issues have explicit proofs: noncentral coefficient arcs are negligible to every fixed inverse-logarithmic order; local lattice Taylor expansions have uniform remainders; saddle reversion has a derivative bounded away from zero; and integer threshold errors are separated from analytic inverse errors. The largest-index theorem gives a probabilistic consequence of the same local phase. What remains is not a missing leading argument, but sharper certification, higher explicit coefficients, secondary exponential sectors, wider statistics, and a systematic formalization.

[Write note, batch 98, 5 October 2026.] The wider statistics near both endpoints of a partition are the subject of Part II.

Part II

Conditioning, joint limit laws and two-endpoint statistics

12 Provenance, scope and notation of Part II

[Write note, batch 98, 5 October 2026.] Part II was added in the write of batch 98. It is built from two manuscripts, merged: manuscript 06 of batch 98 (the base) and manuscript 10 of the same batch. Both continue Part I at the same place: its Theorem 7.1, which asserts no growing-offset or moment theorem, and its open questions 10.4 and 10.5. Each result of either manuscript is printed; a theorem proved in both is printed once, says so, and keeps the other proof as a marked second route (table 6). Text written in the write, rather than taken from a manuscript, is marked [write]. Part I is unchanged except for dated notes.

12.1 Provenance

Manuscript 06 (base). The archive `catalan_partition_laws.zip` (17 files) arrived in commit 2172df76a and was placed in this report by commit 0fba5167f. Its article *Sharp Conditioning and Joint Limit Laws for Catalan Partitions: Upper multiplicities, a corrected edge distribution, and the total number of parts* (26 pages, dated October 5, 2026) gives the author line “A research article on OEIS A033552”; its PDF metadata names “ChatGPT”. Its pinned repository commit is ec4b972dba46c5f0557cdc55964f286c2bcd4899 (4 October 2026, 20:05 Pacific). It contributes the

exact conditioning toolkit at the precision needed (section 13), the complete total-variation criterion for the upper multiplicity process and its sharp rate (section 15), the corrected joint law of the upper process in weighted L^1 (section 16), the maximum expansion summed over all thresholds with the periodic mean and variance coefficients (section 17), the joint limit of the number of parts with the upper process (section 18), the explicit distribution function of W (section 19), the mass profile and Brownian bridge (section 20) and a lacunary universality class (section 21). Files prefixed 02-laws-.

Manuscript 10. The archive `Catalan_Partition_Statistics_Research.zip` (23 files, one of them the checksum ledger `SHA256SUMS.txt`, which lists the other 22) arrived in the same commit and was placed by the same commit. Its article *Two-Endpoint Statistics of Catalan Partitions: Non-Gaussian Length, All-Orders Conditioning, Phase-Dependent Extremes, and Inverse Tail Expansions* (24 pages, dated October 4, 2026) names “AI-assisted research report prepared for Vladimir Reshetnikov” as author on its title page and in its PDF metadata. Its pinned repository commit is `0f93381e8932d3aee3d5c2abfb5c98d128d49b9c` (4 October 2026, 19:54 Pacific), and it records the blob it read of this report’s `article.tex`, `39d225cc2`, which is that file’s blob at both pins and immediately before this write. It contributes the conditioned Gaussian coefficient operator to every order (section 14), the sharp theory of the number of parts with the first correction, the cancellation in the mean and the small-part energies (section 18), the cutoff vector jointly with the number of parts and the first endpoint dependence (section 16), the growing-offset estimate (section 17), the spectral coefficients with two-sided bounds, the natural boundary and the convergent inverse upper-tail expansion (section 19). Files prefixed 03-endpoint-.

How the two relate. The two manuscripts were written independently, against the same text of Part I, and in the same model: the same exact saddle t_n , the same continuous cutoff m , the same $M = \lfloor m \rfloor$ and phase $\theta = m - M$ as Theorem 7.1. Neither has a hypothesis the other lacks. Three theorems (the first correction of the maximum law, the two-sided tails and the joint limit law) and four smaller results are proved in both and are printed once (table 6); everything else is complementary. Each answers questions the other leaves open (table 5). Two independent codes agree to all printed digits where they compute the same number: at $n = 10^6$ manuscript 06’s maximal uncorrected and corrected CDF errors, 0.020100913 and 0.003262362, are the $h = -1$ differences of manuscript 10’s table 14. The batch-98 dossier’s hand checks and spot computations are summarized in section 22.5.

Where the merge had to choose. (1) Base: manuscript 06, the more general text (a criterion for every cutoff sequence, the weighted- L^1 density expansion of the infinite upper process, the all-threshold maximum expansion, and a universality class). (2) Notation: one convention for both, fixed in section 12.3 before any use; the manuscripts use the same letters for different objects. (3) A theorem proved in both keeps manuscript 06’s label and statement; where manuscript 10’s statement is stronger in some respect (the remainder in the survival expansion, the full-sequence form of the moment limits), its statement is printed. (4) Questions of one manuscript answered by the other are not printed as open (section 23).

12.2 What Part II answers

Re-proofs of Part I. Both manuscripts re-prove, at the precision they need, material of Part I, and neither claims it as new: the conditioning identity in the proof of Theorem 7.1; the cumulant sizes of lemma 3.1 (lemma 13.2); the noncentral-arc estimates of Section 3, which manuscript 10 adapts explicitly and manuscript 06 extends uniformly to prefixes (lemma 13.3); and the all-orders saddle estimate theorem 3.2, of which manuscript 10’s operator (theorem 14.1) is the extension to marked amplitudes. The leading maximum law G_h is Part I’s (7.2). Manuscript 06’s “longer expansion for $p(n)$ is not assumed as an unverified black box”; manuscript 10 “do[es] not rebrand its counting theorem or its fixed-offset leading maximum law as new results”.

Question	Answer in Part II
Part I, Theorem 7.1: “a growing- h or moment theorem is not being asserted”	uniform two-sided tails (lemma 17.3), all polynomial and exponential moments, the maximum expansion summed over every threshold with an $O(m^{-2})$ weighted remainder (theorem 17.5), the mean and variance of J_n to $O(m^{-2})$ (17.11), a growing-offset estimate for $ h \leq m^{1/8}$ (theorem 17.1)
Part I, question 10.4 (joint statistics near the largest part, an explicit first correction, moments)	the corrected joint law of the whole upper process (theorem 16.1), its first covariances (corollary 16.3), the corrected maximum law (corollary 16.5), the cutoff vector as a transform (theorem 16.4); and the exact boundary of independence (theorems 15.1 and 15.2)
Part I, question 10.5 (the number of parts and mixed conditioning)	$t_n N_n \Rightarrow W = \sum_k \mathbf{E}_k / c_k$ with every moment and a first correction (theorem 18.1); the joint limit with the upper process and the maximum, with independence (theorem 18.5); the first endpoint dependence (theorem 16.6); and [write] the limit of $m \operatorname{Cov}(t_n N_n, J_n)$ (theorem 18.6)
manuscript 06, question “first finite-size interactions”	the $1/m$ term of $\mathbb{E}(t_n N_n)$ vanishes (corollary 18.2 , manuscript 10); $m \operatorname{Cov}(t_n N_n, J_n)$ converges ([write], theorem 18.6)
manuscript 10, questions on all-orders largest-index moments, the expanding cutoff field and universality	first order answered by (17.11)–(17.12) (manuscript 06); the expanding upper window in total variation by theorems 15.1 and 15.2 ; universality in part by proposition 21.1

Table 5: Questions answered in Part II. Partly answered questions stay open, re-scoped, in [section 23](#).

Result	Printed as	Second route kept
First correction of the maximum law (06 Corollary 5.4, 10 Theorem 6.1, first line; $C_h = D_h$ identically)	corollary 16.5 , proof from the L^1 edge expansion (06)	the tail-deletion amplitude and the coefficient operator (10), after it
Two-sided uniform tails (06 Lemma 6.1, 10 Theorem 7.1)	lemma 17.3	the same argument in both; 10’s exponential-moment consequence (17.4) added
Joint limit of $(t_n N_n, J_n - M)$ (10 Corollary 7.3 is a marginal of 06 Theorem 7.1)	theorem 18.5 (06)	joint Laplace transforms at every cutoff (10), after it
Survival function of W (06 (19.3), 10 Theorem 8.2)	theorem 19.2 , 10’s stronger remainder	06’s partial-fraction bound, in the proof
$S_1 = 1 + 4\pi/(9\sqrt{3})$ (both)	(18.11)	two evaluations of the same beta integral
Mean and variance of J_n (06 (17.11), 10 Corollary 7.2)	06’s $O(m^{-2})$ expansion	10’s full-sequence $o(1)$ form, corollary 17.4
Ratio bound $p(n-v)/p(n) \leq Ce^{-tv}$ (06 Corollary 3.2, 10 (3.12))	corollary 13.4	10’s one-line derivation from the operator

Table 6: Results proved in both manuscripts.

Non-claims kept from manuscript 06. The probabilistic framework is classical: the independent-process representation and the reduction of total variation to the law of the weighted sum are the general theory of Arratia and Tavaré [11, Theorem 3], whose concrete form is proved here only to keep the arguments self-contained; Mutařchiev [12, Theorem 1(i)] obtains non-Gaussian part-count products under hypotheses including a positive Meinardus pole, which do not cover this case because $\sum_k c_k^{-s}$ converges for every $\Re s > 0$; the reciprocal-Catalan constant is OEIS A121839 [13] and is not introduced as new. “A targeted search does not establish priority over all possible general partition theorems.” No exhaustive bibliographic coverage, external peer review or proof-assistant verification is claimed, and the results are not formalized. [Theorem 16.1](#) is an L^1 statement: the corrected density need not be positive, and no L^2 density ratio is asserted. The independence in [theorem 18.5](#) is a limit statement; the finite- n covariance is not zero. [Proposition 21.1](#)’s gcd condition is sufficient, not necessary, and pure powers are not covered. The numerical work has no interval certificates; finite-range maxima are not certified global suprema; the Fourier audit targets a continuous canonical mean, not an integer coefficient; the JSON fields marked exploratory are not claims, and its phase covariance is a limit of truncated-energy covariances (no L^2 total energy exists). Manuscript 06’s ledger divides its claims into three levels: *proved* (every numbered statement under its printed hypotheses), *numerically checked* (the exact integer computations and floating evaluations of [section 22](#), which do not certify asymptotic constants) and *proposed* (its further questions and the exploratory JSON fields).

Non-claims kept from manuscript 10. AI-assisted, unrefereed, not Lean-formalized; numerical values are not interval certificates. The conditioning approach is classical, with Fristedt [14] a central reference; Gaussian coefficient expansions, finite exponential partial fractions and Lagrange inversion are classical too. No publication priority is claimed: the search “was not an exhaustive MathSciNet, zbMATH, or journal-archive priority investigation”. No new exact counting algorithm is given, so Pak’s question [2, §2.5] is untouched, and no conjecture is attributed to the OEIS entry. The operator expansion is an all-orders Poincaré expansion, not a complete transseries: the $O(tm)$ marking corrections and the noncentral arcs are discarded. [Theorem 14.1](#) is not a general transfer theorem for locally analytic amplitudes ([remark 14.2](#)). The exponent $1/8$ of [theorem 17.1](#) is deliberately conservative, with no moderate-deviation optimality. The inverse tail is a sequential limit ($n \rightarrow \infty$, then $\epsilon \downarrow 0$), with no uniform finite- n quantile and no integer-quantile reconstruction. Hadamard’s gap theorem is the one external input of the natural-boundary theorem, and no differential-algebraicity conclusion is drawn. Its transform comparison uses double precision; its independent cross-checks run only through the stated small ranges, not through 10^6 ; a truncated corrected CDF may exceed one. Part I “is itself AI-assisted and unrefereed; its role here is to identify prior work and questions, not to provide independent validation of our answers.”

12.3 Notation

The two manuscripts use the same letters for different objects, and both reuse letters of Part I. [Table 7](#) fixes one convention for Part II, before any use; every renamed symbol is listed, and no normalization is changed. Symbols with the same meaning in Part I and in both manuscripts keep Part I’s names: c_k , F , $p(n)$, $H(t)$, K_r , $t = t_n$, $r(x)$, $\alpha = \log 4$, $m = m_n$, $M = \lfloor m \rfloor$, $\theta = \theta_n = m - M$, $B_r = t^r K_r(t)$, E_1 , J_n and G_h . All logarithms are natural; in particular manuscript 06’s $\log 4 n / \log n$ in (18.15) means $(\log 4) n / \log n$, not $\log_4 n$ divided by anything.

The ideal edge family is indexed by the integer offset j from M , the actual sizes by $k = M + j$. K always denotes a cutoff index, $K_r(t)$ a cumulant; Part I’s Barnes constant K does not occur in Part II. $\mathbb{Q}$ is the measure, never Part I’s function $Q(m)$, which equals the scaled mean B_1 of $\mathbb{Q}_t$.

Notion	Part II	ms. 06	ms. 10	not to be confused with
independent geometric (Boltzmann) law	Q_t	Q_t	$\mathbb{P}_t$	Part I's $Q(m) = tK_1(t)$
uniform law on partitions of n	$\mathbb{P}_n$	$\mathbb{P}_n$	$\mathbb{P}_n$	
number of parts	N_n	N_n	L_n	Part I's threshold $N(y)$; Part I's L_j
largest part size	c_{J_n}	L_n	c_{J_n}	
exponential variables of mean one	$\mathbf{E}_k$	E_k	E_k	Part I's Gaussian corrections E_j
$\mathbb{E}W^q$	ω_q	$r! h_r$	μ_q	Part I's size coordinate μ
Laplace transform of W	$\mathcal{L}(s)$	$\mathbb{E}e^{sW} = \mathcal{L}(-s)$	$\mathcal{L}(s)$	Part I's L_j
ideal edge rate $4^{j-\theta}$	u_j	u_j	v_j	
actual edge rate $t c_{M+j}$	v_j	v_j	$t c_{M+j}$	manuscript 10's v_j
marking variable of Z_{M+j}	x_j	—	u_j	
mean, variance of $Y_j^{(\theta)}$	$\bar{y}_j, w_j$	μ_j, w_j	—	
limit of $J_n - M$	$\mathcal{J}_\theta$	H_θ	Y_θ	Part I's $H(t)$, $\mathcal{H}(m)$
$\mathbb{E}\mathcal{J}_\theta$	$\tilde{\mathcal{J}}(\theta)$	$b(\theta)$	$\gamma(\theta)$	Euler's γ ; Part I's b_j
periodic mean coefficients of J_n	F_0, F_1	F_0, F_1	$\psi (= F_0)$	Part I's Ψ_α , $F(z)$
periodic variance coefficients	$\mathcal{V}_0, \mathcal{V}_1$	V_0, V_1	$\sigma^2 (= \mathcal{V}_0)$	the tail sum V_h at $h = 0$
tail sums	A_h, V_h, U_h	A_h, V_h, B_h	$S_{1,h}, S_{2,h}, U_h$	B_r, S_r ; Part I's $A, U(m)$
first correction of the maximum law	C_h	C_h	D_h	Part I's C, D_α, D_j
exact tail cumulants after K	$\hat{T}_{r,K}$	A_v, V_v	T_r	06's tail mass T
tail-deletion amplitude	$A_K(\tau)$	$Q_v(0)$ at $\tau = t$	$A_K(\tau)$	
general and length amplitudes; jets	$A(\tau; t), A_s; a_\ell$	—	$A, A_s; a_\ell$	Part I's $\mathcal{A}$; 06's fraction a
Gaussian coefficient operator	Ω_J	—	$\mathcal{D}_J$	
cutoff transform; its Euler operator	$\mathcal{G}_I(\mathbf{u}, \mathbf{x}); \mathcal{D}_I$	—	$\mathcal{Q}(\mathbf{v}, \mathbf{u}); \mathcal{D}$	
limiting cylinder probabilities	$\pi_{\mathbf{r}}$	—	$Q_{\mathbf{r}}$	
small-part energies; their transform	$X_{k,n}; \Lambda_d$	—	$Y_{k,n}; Q(\mathbf{s})$	
spectral coefficients of W	$\mathfrak{d}_j$	b_k	d_j	06's TV curve $d(a)$; Part I's b_j
$\prod_{\ell \geq 1} (1 - 2^{-\ell})$	ϖ	—	q_0	Part I's $q_0(\theta)$
survival series; $\mathcal{T}/(\mathfrak{d}_1 z)$	$\mathcal{T}(z); \mathcal{T}_1(z)$	—	$T(z); V(z)$	
upper ϵ -quantile of W	ξ_ϵ	—	q_ϵ	
periodic constant terms of B_r	$\mathfrak{b}_r(\theta)$	—	$b_{r,0}(\theta)$	Part I's b_j
ideal upper product law	$\Pi_{\theta,h}$	$\mu_{\theta,h}$	—	
Brownian bridge	$\mathbb{B}$	B	—	B_r
$v/(e^v - 1), v^2 e^v / (e^v - 1)^2$	ϱ_1, ϱ_2	a, b	—	
likelihood remainder; normalized tail	$\varepsilon_v; \widetilde{W}$	$\mathcal{R}_v; W$	—	the limit W
point and corrected point probabilities of $\mathcal{J}_\theta$	$\mathfrak{p}_h, \mathfrak{r}_h$	q_h, r_h	—	Part I's $q_0, q_1, r(x)$
compact parameter set; angle	$\mathcal{K}; y$	—; y	$K; \varphi$	the cutoff index K
$\sum_h G_h(\theta) A_h(\theta)$	$\Xi(\theta)$	JSON “gamma”	—	

Table 7: Notation of Part II against the two manuscripts and Part I. Part II uses the second column throughout.

13 Exact conditioning, saddle normalization and local estimates

13.1 The model and the two statistics

With c_k , F and $p(n)$ as in (1.1), the unit part is included once: $c_0 = c_1 = 1$ does not give two kinds of part. For example $p(5) = 4$, from 5 , $2 + 2 + 1$, $2 + 1 + 1 + 1$ and $1 + 1 + 1 + 1 + 1$. Let Z_k be the multiplicity of c_k under the uniform measure $\mathbb{P}_n$ on the $p(n)$ partitions of n , and

$$N_n = \sum_{k \geq 1} Z_k, \quad J_n = \max\{k : Z_k > 0\} \quad (n > 0). \quad (13.1)$$

N_n counts parts with multiplicity; J_n is the largest *index*, and the largest part size is c_{J_n} . “Large” refers to the size c_k , or its index k , not to the number of copies.

There are about $\log_4 n$ active sizes, and a typical active size carries mass of order $n/\log n$, although the number of copies varies enormously between small and large sizes. Three limits therefore coexist (manuscript 06): at a fixed small size the multiplicity is of order $1/t_n$ with an exponential limit; a positive fraction of the active sizes has Gaussian mass fluctuations, which fixing the total mass turns into a bridge; and within a bounded number of indices of the largest used size the multiplicities stay discrete with a phase-dependent geometric law. Manuscript 10 puts the same point as two fluctuation scales: the saddle is Gaussian in the *total size*, but both statistics keep non-Gaussian limits after conditioning on it, so a Gaussian approximation to either is structurally wrong.

13.2 Exact conditioning

For $t > 0$ let $\mathbb{Q}_t$ make the Z_k independent with

$$\mathbb{Q}_t(Z_k = z) = (1 - e^{-tc_k})e^{-tc_k z}, \quad z \in \{0, 1, 2, \dots\}, \quad (13.2)$$

counting failures before the first success, and put $S = \sum_k c_k Z_k$. All moments of S are finite, and almost surely only finitely many Z_k are nonzero, since $\sum_k \mathbb{Q}_t(Z_k > 0) = \sum_k e^{-tc_k} < \infty$.

Lemma 13.1 (Conditioning and shifts). *For every $t > 0$,*

$$\mathbb{P}_n = \mathbb{Q}_t(\cdot \mid S = n), \quad \mathbb{Q}_t(S = n) = \frac{p(n)e^{-tn}}{F(e^{-t})}. \quad (13.3)$$

For distinct indices $i_1, \dots, i_d$ and nonnegative integers $b_1, \dots, b_d$,

$$\mathbb{P}_n(Z_{i_\ell} \geq b_\ell, 1 \leq \ell \leq d) = \frac{p(n-v)}{p(n)}, \quad v = \sum_{\ell=1}^d c_{i_\ell} b_\ell, \quad (13.4)$$

with $p(q) = 0$ for $q < 0$.

Proof. Every multiplicity vector with weighted sum n has $\mathbb{Q}_t$ -probability $e^{-tn}/F(e^{-t})$. For the second identity remove b_ℓ copies of each specified part: this is a bijection with the partitions of $n - v$. $\square$

The identity is exact, not an equivalence-of-ensembles assumption; it is the one used in the proof of Theorem 7.1. With $H(t) = \log F(e^{-t})$ and $K_r(t) = (-1)^r H^{(r)}(t)$ as in (3.1), the K_r are the cumulants of S under $\mathbb{Q}_t$, the saddle $t = t_n$ is the unique solution of $K_1(t_n) = \mathbb{E}_{\mathbb{Q}_{t_n}} S = n$ (3.2), and, with $r(x)$ as in (3.3),

$$r(m) = \log(1/t), \quad M = \lfloor m \rfloor, \quad \theta = m - M \in [0, 1), \quad B_r = t^r K_r(t). \quad (13.5)$$

This is Part I’s m_n of (3.4) and θ_n of Theorem 7.1; the phase is *not* obtained by rounding $\log_4 n$. In interpolated notation $tc_m = 1$, and exactly M allowed sizes are at most $1/t$. Since $r'(1) = 1/2$ and

$r''(x) = \psi_1(x + \frac{1}{2}) - \psi_1(x + 2) > 0$ (the trigamma function ψ_1 decreases), $0 < r'(x) < \alpha$ for $x \geq 1$, $r'(x) = \alpha - 3/(2x) + O(x^{-2})$ and $r''(x) = O(x^{-2})$. With the ratio (3.6), $c_{k+1}/c_k = 2(2k+1)/(k+2) \in [2, 4]$,

$$r(m+d) - r(m) = \alpha d - \frac{3d}{2m} + O\left(\frac{1+|d|+d^2}{m^2}\right) \quad (|d| \leq m/2), \quad (13.6)$$

by the differentiable Stirling expansion; its full expansion to each fixed order is Part I's (4.12).

Lemma 13.2 (Active cumulants). *For each fixed integer $r \geq 1$,*

$$B_r = (r-1)! m + O_r(1). \quad (13.7)$$

For a prefix $S_K = \sum_{k \leq K} c_k Z_k$ write $B_{r,K}$ for its scaled cumulants. If $K \leq M$, then, uniformly in K ,

$$B_{r,K} = (r-1)! K + O_r(1). \quad (13.8)$$

Consequently

$$nt = m + O(1), \quad \text{Var}_{\mathbb{Q}_t} S = \frac{m + O(1)}{t^2}, \quad (13.9)$$

$$m = \frac{\log n + \frac{1}{2} \log \log n + \frac{1}{2} \log(\pi/\alpha)}{\alpha} + O\left(\frac{\log \log n}{\log n}\right), \quad (13.10)$$

so that $m \sim \log n / \log 4$ and $t \sim \log n / (n \log 4)$.

Proof. For $f_r(v) = v^r \sum_{\ell \geq 1} \ell^{r-1} e^{-\ell v} = v^r \text{Li}_{1-r}(e^{-v})$, $f_r(v) = (r-1)! + O_r(v)$ on $0 < v \leq 1$, and f_r decays exponentially times a polynomial on $v \geq 1$. There are exactly M active terms $tc_k \leq 1$, and by the ratio bound the sums of tc_k over them and of the inactive $f_r(tc_k)$ are uniformly bounded. This proves both cumulant estimates. Stirling's expansion gives $r(m) = \alpha m - \frac{3}{2} \log m - \frac{1}{2} \log \pi + O(m^{-1})$; combine it with $\log n = r(m) + \log(m + O(1))$ and invert once. $\square$

Lemma 13.2 restates Part I's lemma 3.1 (proved independently in both manuscripts, manuscript 10 with $f_r(v) = v^r \text{Li}_{1-r}(e^{-v})$); the prefix form and (13.10) are manuscript 06's additions. At the edge, for every fixed integer j ,

$$v_j := tc_{M+j} = u_j \left(1 - \frac{3(j-\theta)}{2m} + O_j(m^{-2})\right), \quad u_j = 4^{j-\theta}. \quad (13.11)$$

The coefficient $3/2$ reappears in every corrected edge law.

13.3 Fourier bounds with uniform control

Weak convergence of S alone would not suffice: conditioning on one lattice value needs a local estimate and control of all competing arcs. For the full sum ($K = \infty$) or a prefix, the characteristic function has modulus

$$|\phi_K(y)| = \prod_{k \leq K} \left(1 + \frac{4e^{-tc_k} \sin^2(c_k y/2)}{(1 - e^{-tc_k})^2}\right)^{-1/2}, \quad (13.12)$$

which for $K = \infty$ is Part I's (3.12).

Lemma 13.3 (Integrated Fourier control). *Fix $\varepsilon > 0$. Uniformly for $K \geq \varepsilon m$,*

$$\sup_{\ell \in \mathbb{Z}} \mathbb{Q}_t(S_K = \ell) \leq C_\varepsilon t / \sqrt{m}. \quad (13.13)$$

If $K \leq M$, put $\sigma_K^2 = \text{Var } S_K$. Then, uniformly for $\varepsilon m \leq K \leq M$,

$$\sup_{\ell \in \mathbb{Z}} \left| \sigma_K \mathbb{Q}_t(S_K = \ell) - \phi\left(\frac{\ell - \mathbb{E} S_K}{\sigma_K}\right) \right| = o(1), \quad \phi(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}}. \quad (13.14)$$

The full sum has the same local limit theorem.

Proof. Choose a fixed large B . For $|y| \leq tm^B$ use all factors with $k \leq K$ and $c_k \leq 1/(tm^B)$; at least $\varepsilon m - O(\log m)$ are available when $0 < \varepsilon \leq 1$. The elementary inequalities for $\sin$ and $1 - e^{-v}$ give $|\phi_K(y)| \leq (1 + c(y/t)^2)^{-c\varepsilon m}$, and the integral outside $|y| \leq t \log m / \sqrt{m}$ within this range is smaller than tm^{-L} for every fixed L . For $tm^B \leq |y| \leq \delta$, choose a fixed number q of small part factors and then $\delta > 0$ with $\delta c_q < 1$; their product is at most $C_q(t/|y|)^q$, with integral $O_q(tm^{-B(q-1)})$. For $\delta \leq |y| \leq \pi$ the unit factor is $O_\delta(t)$, and the part-2 and part-5 factors cannot both resonate, since $\sin y$ and $\sin(5y/2)$ have no common zero on this compact set; one more $O_\delta(t)$ factor makes this integral $O_\delta(t^2)$. All the fixed factors used are eventually in every admissible prefix.

On the central interval $\log \mathbb{E} e^{iy(S_K - \mathbb{E} S_K)} = -\frac{1}{2} \sigma_K^2 y^2 + \sum_{r \geq 3} (iy)^r \kappa_{r,K} / r!$, with normalized cumulants $O_r(m^{1-r/2})$. After $z = \sigma_K y$ the integrand converges in L^1 to $e^{-z^2/2}$ by the bounds just proved, and Fourier inversion gives (13.14): the factor $e^{-iz(\ell - \mathbb{E} S_K)/\sigma_K}$ has modulus one, so the error is uniform in ℓ . Integrating the absolute value proves (13.13); for $\varepsilon > 1$ retain the prefix $K = \lfloor m/2 \rfloor$. $\square$

The three arc ranges are those of Part I, Section 3, which manuscript 10 adapts in the same way for the full sum (theorem 14.1); manuscript 06's addition is the uniformity over prefixes that keep a fixed fraction of the active sizes, which manuscript 10 also states for its tail bounds.

Corollary 13.4 (Uniform coefficient ratios). *There is a constant C such that for all large n and every integer $v \geq 0$,*

$$\frac{p(n-v)}{p(n)} \leq C e^{-tv}. \quad (13.15)$$

For each fixed $A < \infty$, uniformly for $0 \leq tv \leq A$,

$$\frac{p(n-v)}{p(n)} = e^{-tv} (1 + o(1)). \quad (13.16)$$

Proof. By lemma 13.1 the ratio is $e^{-tv} \mathbb{Q}_t(S = n-v) / \mathbb{Q}_t(S = n)$. The denominator is $t(2\pi m)^{-1/2} (1 + o(1))$. Use (13.13) for the uniform bound and (13.14) for bounded tv , since $v/\sigma = O(m^{-1/2})$. $\square$

Second route (manuscript 10). The coefficient operator gives $\mathbb{Q}_t(S = n) = t(2\pi B_2)^{-1/2} (1 + O(m^{-1}))$ (theorem 14.1); combined with (13.13) for the full sum this is (13.15), with $p(b) = 0$ for $b < 0$.

13.4 The displacement expansion needed at the edge

The following refinement records the term an uncentred normal approximation would miss.

Lemma 13.5 (A finite-tail likelihood expansion). *Fix $h \in \mathbb{Z}$, put $K = M + h$, and let*

$$T = \sum_{k > K} c_k Z_k, \quad \widehat{T}_{1,K} = t \mathbb{E}_{\mathbb{Q}_t} T, \quad \widehat{T}_{2,K} = t^2 \text{Var}_{\mathbb{Q}_t} T, \quad \delta_v = tT - \widehat{T}_{1,K}.$$

For any upper multiplicity configuration z , its conditional likelihood relative to its $\mathbb{Q}_t$ -law is

$$R_v(z) = 1 + \frac{1}{m} \left\{ \delta_v + \frac{\widehat{T}_{2,K} - \delta_v^2}{2} \right\} + \varepsilon_v(z), \quad |\varepsilon_v(z)| \leq \frac{C_h}{m^2} (1 + |\delta_v|^4). \quad (13.17)$$

In particular $\mathbb{E}_{\mathbb{Q}_t} |\varepsilon_v| = O_h(m^{-2})$. All constants can be chosen uniformly in $\theta \in [0, 1]$.

Proof. The exact likelihood is $R_v(z) = \mathbb{Q}_t(S_K = n - T(z)) / \mathbb{Q}_t(S = n)$. Every fixed cumulant of the scaled tail tT is bounded uniformly: its first rate $t c_{K+1}$ is bounded below and later rates grow at least geometrically. So the scaled prefix cumulants satisfy $B_{2,K} = B_2 - \widehat{T}_{2,K} = m + O_h(1)$ and $B_{3,K} = 2m + O_h(1)$. With $y = t(\ell - \mathbb{E} S_K)$, central Fourier expansion gives

$$\mathbb{Q}_t(S_K = \ell) = \frac{t}{\sqrt{2\pi B_{2,K}}} \left[1 + E_{1,K} - \frac{y^2}{2B_{2,K}} - \frac{B_{3,K} y}{2B_{2,K}^2} + O_h\left(\frac{1 + |y|^4}{m^2}\right) \right], \quad (13.18)$$

where $E_{1,K} = B_{4,K}/(8B_{2,K}^2) - 5B_{3,K}^2/(24B_{2,K}^3) = -1/(12m) + O_h(m^{-2})$ is the prefix analogue of Part I's (3.9). This is a polynomially bounded *absolute* error expansion, not a relative approximation at a distant Gaussian tail. To see the remainder, set $x = y_{\text{Fourier}}/t$, expand the displacement factor through degree three and use $|e^{ia} - \sum_{q=0}^3 (ia)^q/q!| \leq |a|^4/4!$. The constant integral has its usual fourth/sixth cumulant correction; the linear displacement integral has leading term $-B_{3,K}y/(2B_{2,K}^2)$; the quadratic one gives $-y^2/(2B_{2,K})$ with next error $O(y^2m^{-2})$; the cubic one is $O(|y|^3m^{-2})$ by parity and the first odd cumulant; the fourth-order remainder is $O(y^4m^{-2})$, and further centred expansion errors are $O(m^{-2})$. The noncentral bounds of lemma 13.3, at any fixed precision, are uniform in the displacement, whose Fourier factor has modulus one.

For the numerator $y = \widehat{T}_{1,K} - tT = -\delta_v$. The denominator is $t(2\pi B_2)^{-1/2}(1 - 1/(12m) + O(m^{-2}))$. The common Edgeworth constant cancels, $\sqrt{B_2/B_{2,K}} = 1 + \widehat{T}_{2,K}/(2m) + O_h(m^{-2})$, and the linear term becomes $+\delta_v/m$. Bounded fourth moments of tT give the L^1 version. $\square$

Remark 13.6 (Why the linear term matters). The term $+\delta_v/m$ comes from the positive skewness of the geometric variables. A purely Gaussian likelihood gives only $(\widehat{T}_{2,K} - \delta_v^2)/(2m)$ and the wrong first correction. The exact saddle normalization is also essential to the coefficient.

14 A conditioned Gaussian coefficient operator

This section is manuscript 10's. It extends Part I's all-orders saddle estimate theorem 3.2, whose coefficients E_j it contains as the unmarked case, to marked amplitudes. Its noncentral-arc analysis is that of Part I, Section 3, and of lemma 13.3; in manuscript 10 it yields the uniform bound $\sup_a \mathbb{Q}_r(S = a) \leq Ct/\sqrt{m}$, which is (13.13) for the full sum, and the same bound for every prefix keeping a fixed positive fraction of the active sizes and the parts 1, 2, 5.

14.1 The finite operator

For a marked coefficient write

$$\mathcal{R}_n(\mathbf{A}) = \frac{[z^n]\{F(z)\mathbf{A}(-\log z; t)\}}{p(n)}. \quad (14.1)$$

The notation is local on the coefficient contour: in each application the full marked product is a single-valued power series in z . Set

$$\mathbf{a}_\ell = t^\ell \partial_\tau^\ell \mathbf{A}(\tau; t)|_{\tau=t}. \quad (14.2)$$

For $J \geq 0$ and a jet $\mathbf{a} = (\mathbf{a}_0, \mathbf{a}_1, \dots)$ define

$$\Omega_J[\mathbf{a}] = \sum_{\substack{\ell, k_3, \dots, k_{2J+2} \geq 0 \\ \ell + \sum_{r=3}^{2J+2} (r-2)k_r \leq 2J \\ \ell + R \text{ even}}} \frac{(-1)^{(\ell-R)/2} (\ell + R - 1)!! \mathbf{a}_\ell}{\ell! B_2^{(\ell+R)/2}} \prod_{r=3}^{2J+2} \frac{B_r^{k_r}}{k_r! (r!)^{k_r}}, \quad R = \sum_{r=3}^{2J+2} r k_r. \quad (14.3)$$

Here $(-1)!! = 1$ and $\mathbf{1}$ is the constant jet $(1, 0, 0, \dots)$. The index set is finite. A term of weight $w = \ell + \sum (r-2)k_r$ has order $m^{-w/2}$ when the jet is bounded.

Theorem 14.1 (Conditioned Gaussian coefficient expansion). *For each fixed J , the marked products used in Part II satisfy*

$$\mathcal{R}_n(\mathbf{A}) = \frac{\Omega_J[\mathbf{a}]}{\Omega_J[\mathbf{1}]} + O_J(m^{-J-1}). \quad (14.4)$$

The error is uniform on compact admissible sets of marking parameters and on $0 \leq \theta \leq 1$ for fixed cutoff offsets. This applies to the number-of-parts marks of section 18, to finitely many fixed-small-part marks, to finitely many cutoff marks (theorem 16.4) and to fixed-offset tail-deletion amplitudes, including their products. For a tail deletion with $|h| \leq m^{1/8}$ the relative first-order remainder is instead bounded as in theorem 17.1.

Proof. Use $z = e^{-(t+iy)}$ and $v = y/t$. At the saddle,

$$H(t + iy) - H(t) + iny = -\frac{B_2 v^2}{2} + \sum_{r \geq 3} \frac{(-i)^r B_r v^r}{r!}, \quad A(t + iy; t) = \sum_{\ell \geq 0} \frac{a_\ell (iv)^\ell}{\ell!}. \quad (14.5)$$

Expand the second exponential and integrate against the centred Gaussian. Odd powers vanish. A term with indices (ℓ, k_r) has Gaussian moment $(\ell + R - 1)!! B_2^{-(\ell+R)/2}$ and phase $i^\ell (-i)^R = (-1)^{(\ell+R)/2}$; these are the terms of (14.3).

The expansion also has an analytic remainder. In $|y| \leq t/2$, derivatives of H have Cauchy bounds from the absolutely convergent geometric sums (3.1); each fixed-order scaled derivative is $O(m)$. The amplitude jets of the stated classes are bounded on a fixed complex neighbourhood of $\tau/t = 1$. On $|v| \leq \log m/\sqrt{m}$, Taylor-expand to an order above $2J + 2$, integrate the remainder against a Gaussian majorant and use parity: the first omitted even weight integrates to $O(m^{-J-1})$, and higher remainders are bounded by taking a few more Taylor terms. The Gaussian may be extended to the real line at a superpolynomially small error.

Localization for the marks is proved, not inferred from local analyticity. A number-of-parts mark replaces the geometric rate tc_k by $t(c_k + s)$; on a compact subset of $\operatorname{Re} s > -1$ the total normalizing ratio is bounded above and below. For a complex mark the angular phase has an additional bounded multiple of t . In the central-tail estimate keep only the $m - O(\log m)$ sizes with $c_k \geq m$ and $c_k \leq 1/(tm^B)$; for them, outside the central window, $c_k y$ dominates the extra $O(t)$ phase, so the central bound (3.13) holds with changed constants. The intermediate fixed-factor and outer 1, 2, 5 estimates also survive the $O(t)$ phase perturbation. Finitely many small-part marks behave the same after their rates are shifted; finitely many cutoff marks can be omitted from the modulus bound, the $m - O(1)$ remaining factors still localizing the integral; a fixed-offset deletion retains $m - O(1)$ active sizes and all fixed small factors. So the noncentral integrals of every listed class are negligible to every inverse power. The unmarked denominator is the same Gaussian integral with jet 1, with leading term one after normalization; dividing the two expansions proves (14.4). $\square$

Remark 14.2 (The scope of the amplitude condition). Being analytic near t alone does not imply (14.4): an arbitrary amplitude could be large on another arc. The theorem asserts the estimate for the specified product amplitudes, with the noncentral bounds proved above; it does not assume a general coefficient-transfer theorem for locally analytic functions.

Remark 14.3 ([write] Re-reading the localization for marks). The batch-98 dossier flagged this proof as compressed. The write checked its two steps. (1) For a number-of-parts mark the marked factor is $|1 - e^{-x-i\psi}|^{-1}$ with $x = t(c_k + \operatorname{Re} s) > 0$ and $\psi = c_k y + t\Im s$; normalized by its value at $y = 0$ it is $(1 + 4e^{-x} \sin^2(\psi/2)/(1 - e^{-x})^2)^{-1/2}$, of the same form as the unmarked factor. For the retained sizes $c_k \geq m$ one has $|\psi| \geq c_k |y|/2$ as soon as $|y| \geq 2t|s|/m$, far inside the central window, so the central bound survives; the small fixed factors keep the bound $O(t/|y|)$ with a constant depending on $1 + \operatorname{Re} s \geq c_K > 0$. (2) “A fixed complex neighbourhood of $\tau/t = 1$ ” must be allowed to depend on the compact set $\mathcal{K}$: the jets involve $\mathcal{L}(s/(1+v))$ (theorem 18.1), and $s/(1+v)$ stays in a compact subset of $\operatorname{Re} s > -1$ only for $|v| \leq \rho_K$. Since the central window is $|v| \leq \log m/\sqrt{m} \rightarrow 0$, any $\rho_K > 0$ suffices. With these two readings the proof is complete as stated.

At first order,

$$E_1 = \frac{B_4}{8B_2^2} - \frac{5B_3^2}{24B_2^3}, \quad (14.6)$$

$$\Omega_1[a] = a_0(1 + E_1) - \frac{a_2}{2B_2} - \frac{B_3 a_1}{2B_2^2}, \quad (14.7)$$

where E_1 is Part I’s (3.9). In particular $\mathbb{Q}_t(S = n) = t(2\pi B_2)^{-1/2}(1 + O(m^{-1}))$, which gives the second route to corollary 13.4.

Remark 14.4 ([write] The periodic constants of the scaled cumulants). Manuscript 10 writes (section 18) $B_r = (r-1)!m + \mathfrak{b}_r(\theta) + O_r(m^{-1})$ with

$$\mathfrak{b}_r(\theta) = -(r-1)! \theta + \sum_{j \in \mathbb{Z}} \{f_r(4^{j-\theta}) - (r-1)! \mathbf{1}_{\{j \leq 0\}}\}. \quad (14.8)$$

For $r = 1, 2$ these are Part I's coefficients: $\mathfrak{b}_1 = q_0$ of (5.3) and $\mathfrak{b}_2 = q_0 + (1 + q'_0)/\alpha$ of (5.5). Indeed, differentiating the lattice form (4.9), with $g'(v) = -f_1(e^v) + \mathbf{1}_{\{v < 0\}}$, gives $\Psi'_\alpha(\theta) = \frac{\alpha}{2}(1 - 2\theta) + \alpha \sum_j \{f_1(u_j) - \mathbf{1}_{\{j \leq 0\}}\}$, so $q_0 = -\frac{1}{2} + \Psi'_\alpha/\alpha = \mathfrak{b}_1$; and since $f_2(v) = f_1(v) - v f'_1(v)$, $(1 + q'_0)/\alpha = \sum_j (f_2 - f_1)(u_j)$, which adds up to $\mathfrak{b}_2$. The two manuscripts and Part I thus use one and the same periodic coefficient algebra; the recursion (5.6) produces every $\mathfrak{b}_r$ from q_0 .

15 The sharp threshold for an independent upper process

This section is manuscript 06's. For an integer $K \geq 0$ let $\nu_{n,K}$ be the law of $(Z_k)_{k > K}$ under $\mathbb{P}_n$ and $\mathbb{Q}_{t,K}$ its independent geometric law under $\mathbb{Q}_t$, on the countable space of finitely supported upper multiplicity vectors, and

$$\Delta_n(K) = d_{\text{TV}}(\nu_{n,K}, \mathbb{Q}_{t,K}), \quad d_{\text{TV}}(P, Q) = \frac{1}{2} \sum_z |P(z) - Q(z)|.$$

Theorem 15.1 (Complete cutoff criterion). *Let Φ be the standard normal distribution function, and*

$$d(a) = 2 \left[\Phi \left(\sqrt{\frac{-\log a}{1-a}} \right) - \Phi \left(\sqrt{\frac{-a \log a}{1-a}} \right) \right], \quad 0 < a < 1, \quad d(0) = 1, \quad d(1) = 0. \quad (15.1)$$

For every sequence of nonnegative integer cutoffs $K = K_n$,

$$\Delta_n(K) - d(\min\{K/M, 1\}) \longrightarrow 0. \quad (15.2)$$

In particular

$$\Delta_n(K) \rightarrow 0 \iff (M - K)_+ = o(M) : \quad (15.3)$$

an upper window is asymptotically independent exactly when it contains $o(\log n)$ active sizes. If $K/M \rightarrow a \in (0, 1)$ and $T_K = \sum_{k > K} c_k Z_k$, then

$$\frac{t(T_K - \mathbb{E}_{\mathbb{Q}_t} T_K)}{\sqrt{m}} \xrightarrow[\mathbb{P}_n]{d} \mathcal{N}(0, a(1-a)), \quad (15.4)$$

with convergence of every fixed moment; under $\mathbb{Q}_t$ the limiting variance is $1 - a$.

Proof. Put $X = S_K$ and $T = S - X$. Independence under $\mathbb{Q}_t$ gives the exact likelihood and total-variation identities

$$R_n(T) = \frac{\mathbb{Q}_t(X = n - T)}{\mathbb{Q}_t(S = n)}, \quad \Delta_n(K) = \frac{1}{2} \mathbb{E}_{\mathbb{Q}_t} |R_n(T) - 1|, \quad (15.5)$$

the concrete Arratia–Tavaré reduction.

Suppose first $K/M \rightarrow a \in (0, 1)$. Write $\sigma^2 = \text{Var } S$, $\sigma_X^2 = \text{Var } X$, $\sigma_T^2 = \text{Var } T$ and $\tilde{W} = (T - \mathbb{E}T)/\sigma_T$. Then $\sigma_X^2/\sigma^2 \rightarrow a$ and $\tilde{W} \Rightarrow \mathcal{N}(0, 1)$, for instance by Lyapunov's condition: the third absolute centred moments sum to $O((m - K)/t^3)$, while $\sigma_T^3 \asymp (m - K)^{3/2}/t^3$; the bounded set of inactive rates does not change this. The uniform local limit theorem and $n = \mathbb{E}X + \mathbb{E}T$ give, uniformly in T ,

$$R_n(T) = \frac{\sigma}{\sigma_X} \exp\left(-\frac{\sigma_T^2 \tilde{W}^2}{2\sigma_X^2}\right) + o(1).$$

The leading term is bounded, so the limit of (15.5) is $\frac{1}{2} \mathbb{E}|a^{-1/2} \exp(-\frac{1-a}{2a} Z^2) - 1|$ with $Z \sim \mathcal{N}(0, 1)$. Multiplying the standard normal density by the likelihood inside the absolute value gives the density of

$\mathcal{N}(0, a)$; the two densities cross at $|z| = \sqrt{-a \log a / (1 - a)}$, and integrating their difference between the crossings gives (15.1).

If $K/M \rightarrow 1$ with $K \leq M$, then $\sigma_T^2/\sigma^2 \rightarrow 0$, the likelihood tends to one in probability and stays bounded, so $\Delta_n(K) \rightarrow 0$. For $K \geq M$ project the upper process after M onto that after K : total variation cannot increase. If $K/M \rightarrow 0$, fix $0 < \varepsilon < 1$; eventually $K \leq \lfloor \varepsilon M \rfloor$, and projection gives $\Delta_n(K) \geq \Delta_n(\lfloor \varepsilon M \rfloor)$, whose lower limit is at least $d(\varepsilon)$; let $\varepsilon \downarrow 0$. Since $d(\varepsilon) \rightarrow 1$, $\Delta_n(K) \rightarrow 1$. This needs no local theorem for a suffix of Catalan sizes. For an arbitrary cutoff sequence extract a convergent subsequence of $\min(K/M, 1)$; the three cases prove (15.2) along it. Continuity of d and the fact that its only zero is at one give (15.3).

Finally the same bounded likelihood weights the canonical tail normal limit, changing its variance from $1 - a$ to $a(1 - a)$. Uniform moment bounds for the canonical normalized tail follow from its cumulants; the likelihood is bounded by lemma 13.3; applying these bounds at an order above the moment considered proves moment convergence. $\square$

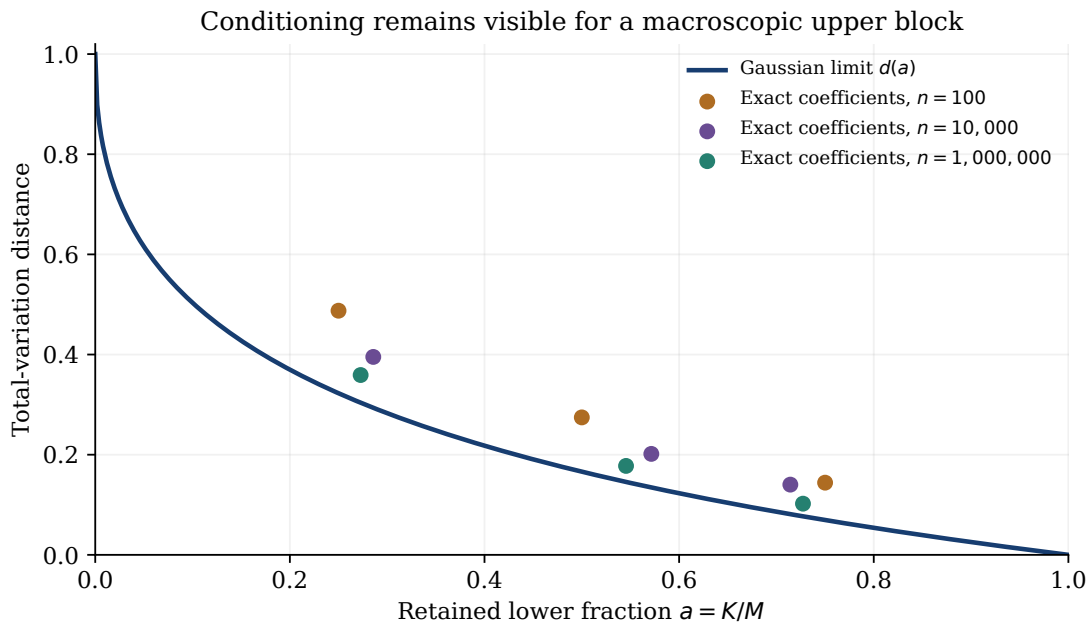


Figure 1: The limiting total-variation distance $d(a)$ as a function of the unobserved prefix fraction a (manuscript 06). The comparison is between complete upper multiplicity processes, not one statistic. The curve is computed from (15.1); the markers are finite coefficient calculations at the indicated n (table 10).

15.1 A sharp rate for growing windows

Theorem 15.2 (Rate inside the independent regime). *Suppose $L = M - K \rightarrow \infty$ and $L = o(M)$. Then*

$$\Delta_n(K) \sim \phi(1) \frac{L}{m}, \quad \phi(1) = \frac{e^{-1/2}}{\sqrt{2\pi}}. \quad (15.6)$$

No assumption such as $L \gg \sqrt{m}$ is needed.

Proof. Let $b = \sigma_T^2/\sigma^2 \sim L/m$ and $\tilde{W} = (T - \mathbb{E}T)/\sigma_T$. Keeping the first odd term in the Fourier proof gives the uniform local Edgeworth formula

$$\sigma_X \mathbb{Q}_t(X = \ell) = \phi(x) \left(1 + \frac{\kappa_3(X)}{6\sigma_X^3} (x^3 - 3x) \right) + O(m^{-1}), \quad x = \frac{\ell - \mathbb{E}X}{\sigma_X}. \quad (15.7)$$

On the central interval the logarithmic characteristic remainder after the cubic is $O(m^{-1}z^4)$; the cubic is imaginary and $|e^{iu} - 1 - iu| \leq u^2/2$, so the integrated error is at most $Cm^{-1}(z^4 + z^6)e^{-cz^2}$, and the three noncentral arc estimates are smaller. This proves (15.7) uniformly in the lattice target, for the prefix $X = S_K$ and for the full sum $X = S$. At $n = \mathbb{E}S$ its odd term vanishes, so $\mathbb{Q}_t(S = n) = \phi(0)\sigma^{-1}(1 + O(m^{-1}))$.

At $\ell = n - T$, $x = -\sqrt{b/(1-b)}\tilde{W}$. Its Edgeworth term has expected absolute size $O(m^{-1/2}\sqrt{b}) = O(\sqrt{L}/m)$, by bounded first and third moments of $\tilde{W}$, and the uniform remainder contributes $O(m^{-1})$; both are $o(L/m)$. So

$$\Delta_n(K) = \frac{1}{2}\mathbb{E}|g_b(\tilde{W}) - 1| + o(L/m), \quad g_b(w) = (1-b)^{-1/2}e^{-bw^2/(2(1-b))}.$$

For $0 \leq b \leq 1/2$, $|g_b(w) - 1 - \frac{b}{2}(1-w^2)| \leq Cb^2(1+w^4)$. Since $\tilde{W} \Rightarrow \mathcal{N}(0, 1)$ with uniformly bounded higher moments, $\mathbb{E}|1 - \tilde{W}^2| \rightarrow 4\phi(1)$, and $\Delta_n(K) \sim b\phi(1)$. $\square$

For example, observing the upper $M^{1/3}$ active indices gives distance asymptotic to $\phi(1)M^{-2/3}$; observing half of them gives distance tending to $d(1/2) = 0.1660640749 \dots$ (figure 1). These regimes cannot be told apart by convergence of a single fixed edge coordinate.

16 Corrected laws at the upper edge

16.1 The corrected joint law of the upper process

For $\theta \in [0, 1]$ let $(Y_j^{(\theta)})_{j \in \mathbb{Z}}$ be independent with

$$\mathbb{P}(Y_j^{(\theta)} = z) = (1 - e^{-u_j})e^{-u_j z}, \quad u_j = 4^{j-\theta}. \quad (16.1)$$

For each fixed h the restriction to $j > h$ has almost surely finitely many nonzero coordinates; denote that upper law by $\Pi_{\theta,h}$. Define

$$\bar{y}_j = \frac{1}{e^{u_j} - 1}, \quad w_j = \frac{e^{u_j}}{(e^{u_j} - 1)^2}, \quad (16.2)$$

$$A_h = \sum_{j>h} u_j \bar{y}_j, \quad V_h = \sum_{j>h} u_j^2 w_j, \quad U_h = \sum_{j>h} (j - \theta) u_j \bar{y}_j, \quad (16.3)$$

$$T_h(z) = \sum_{j>h} u_j z_j, \quad \delta_h(z) = T_h(z) - A_h.$$

Every sum converges, and all fixed moments of T_h are bounded uniformly in $\theta \in [0, 1]$. (These are manuscript 10's $S_{1,h}$, $S_{2,h}$ and U_h ; $\bar{y}_j$ and w_j are the mean and variance of $Y_j^{(\theta)}$.)

Theorem 16.1 (Corrected joint edge law). *Fix $h \in \mathbb{Z}$. Let $\nu_{n,h}$ be the law of $(Z_{M+j})_{j>h}$ under $\mathbb{P}_n$, and*

$$\mathcal{P}_{\theta,h}(z) = \delta_h(z) + \frac{V_h - \delta_h(z)^2}{2} + \frac{3}{2} \sum_{j>h} (j - \theta) u_j (z_j - \bar{y}_j). \quad (16.4)$$

Then, with $\theta = \theta_n$,

$$\int \left| \frac{d\nu_{n,h}}{d\Pi_{\theta,h}}(z) - 1 - \frac{\mathcal{P}_{\theta,h}(z)}{m} \right| d\Pi_{\theta,h}(z) = O_h(m^{-2}). \quad (16.5)$$

For each fixed $q \geq 0$ the same holds after multiplying the integrand by $(1 + T_h(z))^q$, with an $O_{h,q}(m^{-2})$ bound. In particular

$$d_{TV}(\nu_{n,h}, \Pi_{\theta,h}) = \frac{1}{2m} \mathbb{E}_{\Pi_{\theta,h}} |\mathcal{P}_{\theta,h}| + O_h(m^{-2}), \quad (16.6)$$

and the leading coefficient is positive at every phase: the approximation rate $1/\log n$ is sharp.

Proof. First keep the actual geometric rates $v_j = tc_{M+j}$; the conditional law relative to that product is given by lemma 13.5. We transport the product from v_j to u_j in L^1 . Put $\eta_j = \log(v_j/u_j)$. The gamma-ratio expansion and its derivative bounds give

$$|\eta_j| \leq C_h(1 + |j|)/m, \quad \left| \eta_j + \frac{3(j-\theta)}{2m} \right| \leq C_h(1 + j^2)/m^2, \quad (16.7)$$

for all $j > h$ once m is large: for positive $j - \theta$ use $0 \leq x - \log(1+x) \leq x^2/2$; the finitely many negative offsets use ordinary Taylor expansion.

Interpolate logarithmically, $\lambda_j(s) = u_j e^{s\eta_j}$, $0 \leq s \leq 1$; all these rates are at least $c_h 2^j$ for $j \geq 0$. For a finite product with density p_s the score is

$$L_s = \frac{\partial}{\partial s} \log p_s = \sum_{j>h} \eta_j \lambda_j(s) \left(\frac{1}{e^{\lambda_j(s)} - 1} - Z_j \right),$$

a sum of independent centred terms. Exponential decay of geometric moments at large rates and (16.7) give $\mathbb{E}_s L_s^2 = O_h(m^{-2})$ and $\mathbb{E}_s |L'_s| = O_h(m^{-2})$, uniformly in the finite truncation: all controlling series are polynomial factors in j and the rates times an exponentially decaying function of the rate. Since $p''_s = p_s(L_s^2 + L'_s)$, Taylor's integral remainder in L^1 yields

$$\mathbb{Q}_t((Z_{M+j})_{j>h} \in \cdot) = \Pi_{\theta,h} + \frac{3}{2m} \left[\sum_{j>h} (j-\theta) u_j (Z_j - \bar{y}_j) \right] \Pi_{\theta,h} + O_{L^1,h}(m^{-2}),$$

and finite products converge to the infinite upper products in total variation, so the uniform estimate passes to the limit.

The same argument works with polynomial energy weights: $u_j \leq C_h 4^j$ while $\lambda_j(s) \geq c_h 2^j$, so the weighted geometric moments still have summable bounds. Replacing the actual centred energy and its variance in the quadratic conditioning score by δ_h, V_h changes that score times its product measure by $O_h(m^{-1})$ in weighted L^1 : differences of the energy are sums of $(v_j - u_j)Z_j$, whose fixed polynomial moments are $O_h(m^{-1})$, and the differences of mean and variance obey the same bound; the factor $1/m$ makes this $O_h(m^{-2})$. Combining the two expansions proves (16.5) and its weighted version.

The score has mean zero, since $\mathbb{E}\delta_h = 0$ and $\mathbb{E}\delta_h^2 = V_h$; the reverse triangle inequality gives (16.6). It is not the zero variable: fixing all other coordinates at zero leaves a quadratic polynomial in any one coordinate with nonzero coefficient $-u_j^2/2$, and each such configuration has positive product probability. $\square$

Remark 16.2 (The corrected approximation is signed). $(1 + \mathcal{P}_{\theta,h}/m)\Pi_{\theta,h}$ is a signed first-order approximation of total mass one; it need not be nonnegative on every remote configuration. The theorem is the L^1 bound (16.5); it asserts no global positivity of the truncated density, and the proof needs no L^2 density ratio between the two infinite product measures.

Corollary 16.3 (First correlations of distinct large sizes). *For distinct fixed integers i, j ,*

$$\text{Cov}_{\mathbb{P}_n}(Z_{M+i}, Z_{M+j}) = -\frac{u_i u_j w_i w_j}{m} + O_{i,j}(m^{-2}). \quad (16.8)$$

Proof. Choose $h < \min(i, j)$ in theorem 16.1. Under the product law the centred coordinates are independent; in the expectation of $(Z_{M+i} - \bar{y}_i)(Z_{M+j} - \bar{y}_j)$ all linear terms of the score vanish and only the cross term of $-\delta_h^2/2$ survives, giving $-u_i u_j w_i w_j/m$. The product of the two mean displacements is $O(m^{-2})$, and the weighted remainder justifies the quadratic observable. $\square$

So the asymptotically independent upper variables keep a specific negative interaction at first order. *Second route.* The cross term of $-\frac{1}{2}\mathcal{D}_I^2 \mathcal{G}_I$ in manuscript 10's operator (16.10) gives the same $-u_i u_j w_i w_j/m$; the dossier checked this agreement by hand.

16.2 The cutoff vector jointly with the number of parts

Manuscript 10 states the edge law as a transform, for a finite set of offsets, jointly with the number of parts. Its edge part is the transform form of [theorem 16.1](#) (which is stronger: L^1 , the infinite upper process, polynomial weights); the joint statement with N_n is new. Let $I \subset \mathbb{Z}$ be finite and put, with $\mathcal{L}$ the Laplace transform of W ([18.2](#)),

$$\mathcal{G}_I(\mathbf{u}, \mathbf{x}) = \prod_{j \in I} \frac{1 - e^{-u_j}}{1 - x_j e^{-u_j}}, \quad \mathcal{D}_I = \sum_{j \in I} u_j \partial_{u_j}. \quad (16.9)$$

In a derivative the u_j are independent variables, set to $4^{j-\theta}$ afterwards. For $\mathbf{x} \in [0, 1]^I$, and on a small complex neighbourhood of that cube, the denominators stay away from zero. Define

$$C_I \mathcal{G} = -\frac{3}{2} \sum_{j \in I} (j - \theta) u_j \partial_{u_j} \mathcal{G} - \frac{1}{2} (\mathcal{D}_I^2 + \mathcal{D}_I) \mathcal{G}. \quad (16.10)$$

Theorem 16.4 (Cutoff vector and its joint law with the number of parts). *For fixed I and compact admissible parameter sets,*

$$\mathbb{E} \prod_{j \in I} x_j^{Z_{M+j}} = \mathcal{G}_I + \frac{C_I \mathcal{G}_I}{m} + O(m^{-2}), \quad (16.11)$$

$$\mathbb{E} \left[e^{-st N_n} \prod_{j \in I} x_j^{Z_{M+j}} \right] = \mathcal{L} \mathcal{G}_I + \frac{1}{m} \left\{ \mathcal{L} C_I \mathcal{G}_I - \frac{s^2 \mathcal{L}'' \mathcal{G}_I}{2} + s \mathcal{L}' \mathcal{D}_I \mathcal{G}_I \right\} + O(m^{-2}), \quad (16.12)$$

uniformly in $0 \leq \theta \leq 1$. Along any subsequence $\theta_n \rightarrow \theta$ the cutoff multiplicities tend to independent geometric variables with parameters $e^{-4^{j-\theta}}$, and this finite vector is independent of W in the joint limit.

Proof. The exact cutoff amplitude is a finite product of the factors of (16.9) with u_j replaced by τc_{M+j} . At $\tau = t$, (13.11) gives $t c_{M+j} = u_j (1 - 3(j - \theta)/(2m) + O_I(m^{-2}))$, which produces the first term of (16.10). At the limiting arguments, the jets under $\tau = t(1 + v)$ are $\mathbf{a}_0 = \mathcal{G}_I$, $\mathbf{a}_1 = \mathcal{D}_I \mathcal{G}_I$ and $\mathbf{a}_2 = (\mathcal{D}_I^2 - \mathcal{D}_I) \mathcal{G}_I$, and (14.7) produces the second term.

For the joint mark, the product of the length and cutoff amplitudes differs from the exact joint amplitude by $O_I(t)$, with the same bound for scaled derivatives: in the finitely many marked cutoff factors the extra mark changes $e^{-\tau c}$ to $e^{-\tau c - st}$, and $t c$ is bounded above and away from zero there. This is below every inverse power of m . The first two jets of $\mathcal{L}(s/(1+v)) \mathcal{G}_I((1+v)\mathbf{u})$ give

$$-\frac{\mathbf{a}_2 + 2\mathbf{a}_1}{2} = -\frac{s^2 \mathcal{L}'' \mathcal{G}_I}{2} + s \mathcal{L}' \mathcal{D}_I \mathcal{G}_I - \frac{\mathcal{L}}{2} (\mathcal{D}_I^2 + \mathcal{D}_I) \mathcal{G}_I;$$

add the geometric-curvature correction to obtain (16.12). The leading product transform proves the joint limit and independence; its continuity at zero and at $\mathbf{x} = \mathbf{1}$ gives tightness. No independence at finite n is assumed. $\square$

Exact cylinder events. For fixed nonnegative integers r_j the limiting probability of $\{Z_{M+j} = r_j, j \in I\}$ is

$$\pi_{\mathbf{r}}(\mathbf{u}) = \prod_{j \in I} (1 - e^{-u_j}) e^{-r_j u_j}, \quad (16.13)$$

with first correction $C_I \pi_{\mathbf{r}}/m$, by coefficient extraction in the marking variables or by the same amplitude argument applied to $\prod_{j \in I} (1 - z^{c_{M+j}}) z^{r_j c_{M+j}}$. With $T = \sum_{j \in I} (u_j/(e^{u_j} - 1) - r_j u_j)$ and $V_2 = \sum_{j \in I} u_j^2 e^{u_j} / (e^{u_j} - 1)^2$,

$$\frac{C_I \pi_{\mathbf{r}}}{\pi_{\mathbf{r}}} = -\frac{3}{2} \sum_{j \in I} (j - \theta) \left(\frac{u_j}{e^{u_j} - 1} - r_j u_j \right) + \frac{V_2 - T^2 - 2T}{2}. \quad (16.14)$$

An independent finite- n identity, used for verification, is

$$\mathbb{P}_n(Z_{M+j} = r_j \ (j \in I)) = \sum_{A \subseteq I} (-1)^{|A|} \frac{p(n - \sum_{j \in I} (r_j + \mathbf{1}_{\{j \in A\}}) c_{M+j})}{p(n)}, \quad (16.15)$$

obtained by removing the specified copies and excluding further copies of the selected sizes; it needs no saddle approximation.

16.3 The corrected largest-index law

Define, as in Part I's (7.2) and with the tail sums (16.3),

$$G_h(\theta) = \prod_{j>h} (1 - e^{-4^{j-\theta}}), \quad C_h(\theta) = -A_h + \frac{V_h - A_h^2}{2} - \frac{3}{2}U_h, \quad h \in \mathbb{Z}. \quad (16.16)$$

G_h tends to one as $h \rightarrow +\infty$ and to zero as $h \rightarrow -\infty$; it is the distribution function of $\mathcal{J}_\theta = \max\{j : Y_j^{(\theta)} > 0\}$. All products and sums converge absolutely; for fixed h they and all their phase derivatives are bounded on $[0, 1]$, and G_h is bounded away from zero there.

Corollary 16.5 (First correction to the maximum). *For every fixed $h \in \mathbb{Z}$, uniformly in the saddle phase,*

$$\mathbb{P}_n(J_n - M \leq h) = G_h(\theta) \left(1 + \frac{C_h(\theta)}{m}\right) + O_h(m^{-2}). \quad (16.17)$$

Proof (manuscript 06). The event is that all coordinates $j > h$ vanish. There $\delta_h = -A_h$, and the last score term equals $-\frac{3}{2}U_h$. Apply [theorem 16.1](#). $\square$

Second proof (manuscript 10). Set $K = M + h$. Removing all factors with index above K gives the exact amplitude

$$A_K(\tau) = \prod_{k>K} (1 - e^{-\tau c_k}), \quad (16.18)$$

with scaled tail cumulants

$$\widehat{T}_{r,K} = t^r \sum_{k>K} c_k^r \text{Li}_{1-r}(e^{-t c_k}). \quad (16.19)$$

At fixed h these are bounded, and the first jets are $a_0 = A_K(t)$, $a_1 = A_K(t) \widehat{T}_{1,K}$, $a_2 = A_K(t) (\widehat{T}_{1,K}^2 - \widehat{T}_{2,K})$. The first conditioning correction is therefore

$$A_K(t) \left\{ \frac{\widehat{T}_{2,K} - \widehat{T}_{1,K}^2}{2B_2} - \frac{B_3 \widehat{T}_{1,K}}{2B_2^2} \right\}. \quad (16.20)$$

The local expansion of the Catalan sizes gives

$$A_K(t) = G_h(\theta) \left(1 - \frac{3U_h(\theta)}{2m} + O_h(m^{-2})\right), \quad \widehat{T}_{1,K} = A_h + O_h(m^{-1}), \quad \widehat{T}_{2,K} = V_h + O_h(m^{-1}). \quad (16.21)$$

To sum the size expansion, split $j - \theta$ at $m/2$: below it the derivatives of $\log(1 - e^{-u_j})$ have a summable superexponential majorant at the positive end and only finitely many negative j occur; above it the whole tail is superexponentially small. So the errors in (16.21) are uniform in the phase, and (16.20)–(16.21) with $B_2 = m + O(1)$, $B_3 = 2m + O(1)$ give (16.17), in manuscript 10's form $C_h = -\frac{3}{2}U_h + \frac{1}{2}(V_h - A_h^2 - 2A_h)$. $\square$

The correction has two sources. The term $-\frac{3}{2}U_h$ records the polynomial perturbation of the geometric scale 4^k in the Catalan numbers; the other three terms are the effect of conditioning on the exact total size. Dropping either changes the first correction. In the alternative description of manuscript 06, let

$u_j = \exp(\alpha(j - \theta))$ with variable $\alpha > 0$; termwise differentiation gives $\partial_\theta G_h = -\alpha A_h G_h$, $\partial_\theta^2 G_h = \alpha^2(A_h^2 + A_h - V_h)G_h$ and $\partial_\alpha G_h = U_h G_h$, so that

$$G_h C_h = \left(\frac{1}{2\alpha} \partial_\theta - \frac{1}{2\alpha^2} \partial_\theta^2 - \frac{3}{2} \partial_\alpha \right) G_h \Big|_{\alpha=\log 4}, \quad (16.22)$$

the α -derivative keeping θ, h fixed. This separates the conditioning derivatives from the polynomial factor $k^{-3/2}$ in Catalan growth, in the same way as $-\frac{3}{2} \partial_\alpha \Psi_\alpha$ enters Part I's (2.4).

Theorem 16.6 (The maximum jointly with the number of parts). *For fixed $h \in \mathbb{Z}$, compact-uniformly in $\operatorname{Re} s > -1$,*

$$\mathbb{E}[e^{-stN_n} \mathbf{1}_{\{J_n \leq M+h\}}] = G_h(\theta) \left[\mathcal{L}(s) + \frac{1}{m} \left\{ C_h(\theta) \mathcal{L}(s) - \frac{s^2 \mathcal{L}''(s)}{2} + A_h(\theta) s \mathcal{L}'(s) \right\} \right] + O_{h,K}(m^{-2}). \quad (16.23)$$

In particular

$$\operatorname{Cov}(tN_n, \mathbf{1}_{\{J_n \leq M+h\}}) = \frac{S_1 G_h(\theta) A_h(\theta)}{m} + O_h(m^{-2}), \quad (16.24)$$

and the leading coefficient is strictly positive.

Proof. For the joint mark use the amplitude $A_s(\tau; t) \prod_{k>K} (1 - e^{-\tau c_k - st})$, with A_s the length amplitude (18.5). Its difference from $\mathcal{L}(st/\tau) A_K(\tau)$, together with fixed scaled derivatives, is $O_h(tm)$ on the local neighbourhood. The product-jet calculation of theorem 16.4 gives (16.23), the cross term being $s \mathcal{L}' G_h A_h / m$. Differentiate at $s = 0$ and subtract the product of the marginal expectations to obtain (16.24); positivity is immediate from (16.16) and (16.3). $\square$

17 Growing offsets, uniform tails and moments of the largest index

A fixed-threshold distributional limit does not imply a moment theorem, and a fixed- h estimate does not justify substituting $h = h(n)$. This section supplies what Part I's Theorem 7.1 withholds.

17.1 A uniform growing-window statement

The next result (manuscript 10) is formulated with the exact tail quantities (16.19) precisely to avoid an uncontrolled replacement of them by their phase limits.

Theorem 17.1 (Growing-offset estimate). *There is a constant C such that for all large n and all integers $|h| \leq m^{1/8}$, with $K = M + h$,*

$$\frac{\mathbb{P}_n(J_n \leq K)}{A_K(t)} = 1 + \frac{\widehat{T}_{2,K} - \widehat{T}_{1,K}^2}{2B_2} - \frac{B_3 \widehat{T}_{1,K}}{2B_2^2} + O\left(\frac{(1 + |h|)^4}{m^2}\right), \quad (17.1)$$

the implicit constant being independent of h and of the phase.

Proof. Put $H_- = 1 + \max(0, -h)$. The ratio bound on consecutive sizes gives $\widehat{T}_{r,K} = O_r(H_-)$ for every fixed r , and the same geometric-sum estimates apply to derivatives on $|\tau/t - 1| \leq 1/4$. Direct differentiation of $\log A_K$ and the exponential Bell recurrence give

$$\frac{|t^\ell A_K^{(\ell)}(t)|}{A_K(t)} \leq C_\ell H_-^\ell. \quad (17.2)$$

On the central window the derivative remainder carries at most a factor $\exp(CH_-|y|/t)$, bounded because $H_- \log m / \sqrt{m} \rightarrow 0$ in the stated range. Expand the relative amplitude and the phase as in (14.5). The

retained terms have even weights zero and two; the first omitted even weight is four, and by (17.2) it integrates to $O(H_-^4/m^2)$. Higher weights are bounded by continuing the Taylor expansion before estimating the remainder; their extra factors are powers of $H_-/\sqrt{m}$ and $m^{-1/2}$, uniformly small. The denominator correction changes the retained nonconstant terms by $O(H_-^2/m^2)$.

Localization is uniform without dividing a crude full-product bound by a tiny $A_K(t)$: the numerator divided by $A_K(t)F(e^{-t})$ is exactly the normalized prefix characteristic function. The prefix has at least $m - O(m^{1/8})$ active sizes and every fixed small factor, so the three arc estimates hold with uniform constants for the *relative* integral. With the central estimate and (14.7) this proves (17.1). $\square$

The exponent $1/8$ is deliberately conservative: the theorem gives a proved window growing to infinity, not an optimal moderate-deviation range or a uniform phase-polynomial formula for arbitrarily large $|h|$.

Remark 17.2 ([write] Re-reading the central-window remainder). The dossier flagged this proof as compressed. The write checked that the relative amplitude is analytic and zero-free on $|\tau/t - 1| \leq 1/4$ with $t(\log A_K)^{(\ell)}$ -bounds $O_\ell(H_-)$: for $k > K$ the points τc_k lie in the sector $|\arg z| \leq \arcsin \frac{1}{4}$, where $1 - e^{-z}$ vanishes only at $z = 0$; the factors with $tc_k \leq 1$ (at most $H_- + 1$ of them) contribute $O(1)$ each to $t \, d \log A_K / d\tau$, and the factors with $tc_k > 1$ a geometrically convergent $O(1)$ in total. Bell's formula then gives (17.2); on the central window $|\log A_K(t + iy) - \log A_K(t)| \leq CH_-|y|/t \leq CH_- \log m / \sqrt{m} \rightarrow 0$. A term of weight w carries at most w amplitude derivatives, hence is $O(H_-^w m^{-w/2})$; weight four is the stated error. The proof is complete as stated. The same argument applies verbatim when the amplitude also carries the number-of-parts mark of [theorem 18.1](#), whose jets are bounded; this is used in [theorem 18.6](#).

17.2 Uniform two-sided tails

Lemma 17.3 (Uniform two-sided tails). *There are positive constants C, c such that for all large n and all integers $r, h \geq 0$,*

$$\mathbb{P}_n(J_n - M > h) \leq Ce^{-c2^h}, \quad \mathbb{P}_n(J_n - M \leq -r) \leq Ce^{-cr^2}. \quad (17.3)$$

The same bounds hold for $\mathcal{J}_\theta$, uniformly in $\theta \in [0, 1]$. Consequently, for every fixed $a > 0$,

$$\sup_{n \geq n_0} \mathbb{E} \exp(a|J_n - M|) < \infty. \quad (17.4)$$

Proof. Right tail: the event that a size c_k occurs has exact probability $p(n - c_k)/p(n)$ (remove one copy), so the union bound, [lemma 13.1](#) and [corollary 13.4](#) give $\mathbb{P}_n(J_n > M + h) \leq C \sum_{k > M+h} e^{-tc_k}$. Since $tc_M > 1/4$ and successive sizes have ratio at least two, $tc_{M+h} \geq 2^h/4$, and the sum is at most Ce^{-c2^h} .

Left tail: first take $0 \leq r \leq M/2$ and $K = M - r$. Independence before conditioning gives $\mathbb{P}_n(J_n \leq K) = A_K(t) \mathbb{Q}_t(S_K = n) / \mathbb{Q}_t(S = n)$; the event fixes the upper process at zero. The uniform prefix atom bound, divided by the full atom at its mean, is bounded by a constant, so

$$\mathbb{P}_n(J_n \leq K) \leq C \prod_{k > K} (1 - e^{-tc_k}) \leq C \prod_{\ell=0}^{r-1} tc_{M-\ell} \leq C 2^{-r(r-1)/2},$$

since $tc_{M-\ell} \leq 2^{-\ell}$. For $M/2 < r < M$ bound the event by the one at $r = \lfloor M/2 \rfloor$ and reduce the constant in the Gaussian exponent; for $r \geq M$ the event is empty because $J_n \geq 1$. The product-law bounds follow from the same elementary products and sums without conditioning. Summing the two tails against e^{ah} proves (17.4). $\square$

Both manuscripts prove this lemma, by the same argument; (17.4) is manuscript 10's statement of its consequence.

Corollary 17.4 (All maximum moments and a periodic mean). *If $\theta_n \rightarrow \theta$, then $J_n - M \Rightarrow \mathcal{J}_\theta$, and all polynomial and real exponential moments converge. Define*

$$\begin{aligned}\bar{\mathcal{J}}(\theta) &= \mathbb{E}\mathcal{J}_\theta = \sum_{h \geq 0} (1 - G_h(\theta)) - \sum_{h \leq -1} G_h(\theta), \\ F_0(\theta) &= \bar{\mathcal{J}}(\theta) - \theta, \quad \mathcal{V}_0(\theta) = \text{Var } \mathcal{J}_\theta = \sum_{h \in \mathbb{Z}} (h - \bar{\mathcal{J}}(\theta))^2 \mathfrak{p}_h(\theta),\end{aligned}\tag{17.5}$$

where $\mathfrak{p}_h = G_h - G_{h-1}$. Then F_0 and $\mathcal{V}_0$ are smooth one-periodic functions and, along the full sequence,

$$\mathbb{E}J_n = m + F_0(\theta_n) + o(1), \quad \text{Var } J_n = \mathcal{V}_0(\theta_n) + o(1).\tag{17.6}$$

Proof. For each phase $G_h \rightarrow 0$ as $h \rightarrow -\infty$ and $G_h \rightarrow 1$ as $h \rightarrow +\infty$, so it defines a proper integer law. [Corollary 16.5](#) gives convergence at every integer cutoff, and [lemma 17.3](#) uniform integrability for every polynomial and exponential function. Tail summation proves (17.5), and the variance formula follows from the point probabilities. Uniform phase convergence on bounded h -ranges, followed by the uniform tails, proves (17.6) on the full sequence. Note that G_h itself is *not* one-periodic:

$$G_h(\theta + 1) = G_{h-1}(\theta), \quad \mathcal{J}_{\theta+1} \stackrel{d}{=} \mathcal{J}_\theta + 1,\tag{17.7}$$

so $\bar{\mathcal{J}}(\theta + 1) = \bar{\mathcal{J}}(\theta) + 1$, while F_0 and $\mathcal{V}_0$ are periodic; it would be wrong to call $\bar{\mathcal{J}}$ periodic. Differentiated lattice products have the same summable tails up to polynomial factors in $|h|$, so termwise differentiation proves smoothness. $\square$

This (manuscript 10's) corollary supplies the moment conclusion missing from a fixed- h statement and distinguishes a phase-dependent family of integer laws from a continuous extreme-value approximation. [Theorem 17.5](#) below (manuscript 06) sharpens (17.6) to an expansion with $O(m^{-2})$ remainder.

Every phase is realized by a subsequence: implicit differentiation of the saddle equations gives $dm/dn = t/(r'(m)B_2) \sim 1/(n \log 4)$, so $m(n)$ is increasing, unbounded and has increments tending to zero, and rounding the real inverse of $m = j + \theta$ realizes any limiting phase (compare the proof of Part I's [corollary 2.2](#)). The phase family is a feature of the integer sequence, not an artificial continuous parameter.

17.3 The first correction summed over all thresholds

Theorem 17.5 (A weighted expansion for the maximum). *Put $\mathfrak{p}_h(\theta) = G_h(\theta) - G_{h-1}(\theta)$ and $\mathfrak{r}_h(\theta) = G_h(\theta)C_h(\theta) - G_{h-1}(\theta)C_{h-1}(\theta)$. For every fixed $\lambda > 0$,*

$$\sum_{h \in \mathbb{Z}} e^{\lambda|h|} \left| \mathbb{P}_n(J_n - M = h) - \mathfrak{p}_h(\theta) - \frac{\mathfrak{r}_h(\theta)}{m} \right| = O_\lambda(m^{-2}).\tag{17.8}$$

So the first correction is valid for every polynomially or exponentially bounded function of the centred maximum.

Proof. Let $E_h^{\max}$ be the error in (16.17). We prove $\sum_h e^{\lambda|h|} |E_h^{\max}| = O_\lambda(m^{-2})$; successive differences then prove the theorem.

Negative thresholds $h = -r$, $0 \leq r \leq M/2$. The prefix has at least $M/2$ active sizes, so the Fourier constants are uniform, and its removed scaled cumulants are $O_q(r + 1)$. Repeating the displacement computation of [lemma 13.5](#), now expanding the variance ratio for $\widehat{T}_{2,K}/B_2 \leq 1/2 + o(1)$, gives at the all-zero configuration

$$R_v(0) = 1 + \frac{-\widehat{T}_{1,K} + (\widehat{T}_{2,K} - \widehat{T}_{1,K}^2)/2}{m} + O\left(\frac{1 + r^4}{m^2}\right).$$

The inverse-variance replacements contribute $O((r+1)/m^2)$ before multiplication by the displacement, and all resulting terms fit the bound. Write $\varrho_1(v) = v/(e^v - 1)$ and $\varrho_2(v) = v^2 e^v/(e^v - 1)^2$. For $r \leq M/2$ and $j > -r$ one has $m + j - \theta \geq m/2 - O(1)$, so the proof of (16.7), with the logarithmic expansion on $(j - \theta)/m \geq -1/2 - O(m^{-1})$, gives those bounds with a constant independent of r . The logarithmic derivatives of ϱ_1 and ϱ_2 vanish at zero to orders one and two; with (16.7) and the geometric rate bounds this gives, uniformly in this range,

$$|\widehat{T}_{1,K} - A_{-r}| + |\widehat{T}_{2,K} - V_{-r}| \leq C/m.$$

For the all-zero product itself, Taylor expansion of $\log(1 - e^{-v})$ gives

$$\log \frac{A_K(t)}{G_{-r}} = -\frac{3U_{-r}}{2m} + O\left(\frac{1+r^3}{m^2}\right), \quad \left| \log \frac{A_K(t)}{G_{-r}} \right| \leq C(1+r^2)/m,$$

the cubic bound coming from summing the second-order rate errors over the r small rates. Since $G_{-r} \leq Ce^{-cr^2}$, exponentiating and multiplying the two expansions gives

$$|E_{-r}^{\max}| \leq Cm^{-2}(1+r^4)G_{-r} \exp(C(1+r^2)/m) \leq Cm^{-2}e^{-c'r^2}.$$

For $r > M/2$ the midpoint argument of lemma 17.3 and the same Gaussian bound for $G_{-r}C_{-r}$ give this with smaller exponent constants; for $r \geq M$ the event is empty. All negative errors are summable with every exponential weight.

Positive thresholds. Use the weighted version of theorem 16.1 for the upper process $j > 0$. For every $h \geq 0$,

$$\int_{\{z_j=0 \text{ for all } j>h\}} \mathcal{P}_{\theta,0}(z) \, d\Pi_{\theta,0}(z) = G_h(\theta)C_h(\theta) :$$

the free coordinates $1, \dots, h$ have centred means, and their contribution to the expected quadratic energy cancels the corresponding variance term. So $E_h^{\max}$ is exactly the error of this marginal of the upper-process expansion. Its signed remainder measure has total mass zero, so the error on the event that all $j > h$ vanish is minus the error on the complement. On the complement the energy T_0 is at least $4^{h-\theta}$, so for fixed λ the sum of the weights $e^{\lambda h}$ over thresholds below the largest occupied upper index is at most $C_\lambda(1+T_0)^q$ for a fixed large q . The weighted L^1 bound of theorem 16.1 proves $\sum_{h \geq 0} e^{\lambda h} |E_h^{\max}| = O_\lambda(m^{-2})$. $\square$

Remark 17.6 ([write] Re-reading the negative thresholds). The dossier flagged the negative-threshold step as compressed (“repeating the displacement computation” uniformly for $r \leq M/2$). The write checked it term by term. With $K = M - r \geq M/2$, the prefix scaled cumulants are $B_{2,K} = B_2 - \widehat{T}_{2,K}$ with $\widehat{T}_{2,K} = O(r+1)$, so $\sqrt{B_2/B_{2,K}} = (1 - \widehat{T}_{2,K}/B_2)^{-1/2} = 1 + \widehat{T}_{2,K}/(2m) + O(r^2/m^2)$, $B_{3,K}/(2B_{2,K}^2) = 1/m + O(r/m^2)$, $1/B_{2,K} = 1/m + O(r/m^2)$ and $E_{1,K} - E_1 = O(r/m^2)$; at the all-zero configuration $y = \widehat{T}_{1,K} = O(r+1)$. The displacement expansion (13.18) is an absolute bound uniform over prefixes with at least $M/2$ active sizes (the prefix normalized cumulants are $O(m^{1-r/2})$ uniformly), and for $|y| \geq \sqrt{m}$ both sides are $O(1 + y^4/m^2)$, so it holds for every $r \leq M/2$. The comparison $|\eta_j| \leq C(1+|j|)/m$ for $j \geq -m/2$ follows from $\eta_j = -\frac{3}{2} \log(1 + (j - \theta)/m) + O(|j - \theta|/m^2)$. Collecting the terms gives the displayed $R_v(0)$, and the step is correct as stated. The two comparison bounds displayed in the proof are used again in theorem 18.6.

17.4 Explicit mean and variance functions

With $\tilde{\mathcal{J}}$ and $F_0, \mathcal{V}_0$ as in corollary 17.4, let

$$F_1(\theta) = - \sum_{h \in \mathbb{Z}} G_h(\theta) C_h(\theta), \tag{17.9}$$

$$\mathcal{V}_1(\theta) = - \sum_{h \in \mathbb{Z}} (2h+1) G_h(\theta) C_h(\theta) - 2\tilde{\mathcal{J}}(\theta) F_1(\theta). \tag{17.10}$$

The sums and their fixed derivatives converge uniformly. The weighted theorem and summation by parts give

$$\mathbb{E}J_n = m + F_0(\theta) + \frac{F_1(\theta)}{m} + O(m^{-2}), \quad (17.11)$$

$$\text{Var } J_n = \mathcal{V}_0(\theta) + \frac{\mathcal{V}_1(\theta)}{m} + O(m^{-2}). \quad (17.12)$$

By (17.7), $F_0, F_1, \mathcal{V}_0, \mathcal{V}_1$ are smooth one-periodic functions; figure 2 plots F_0 and $\mathcal{V}_0$.

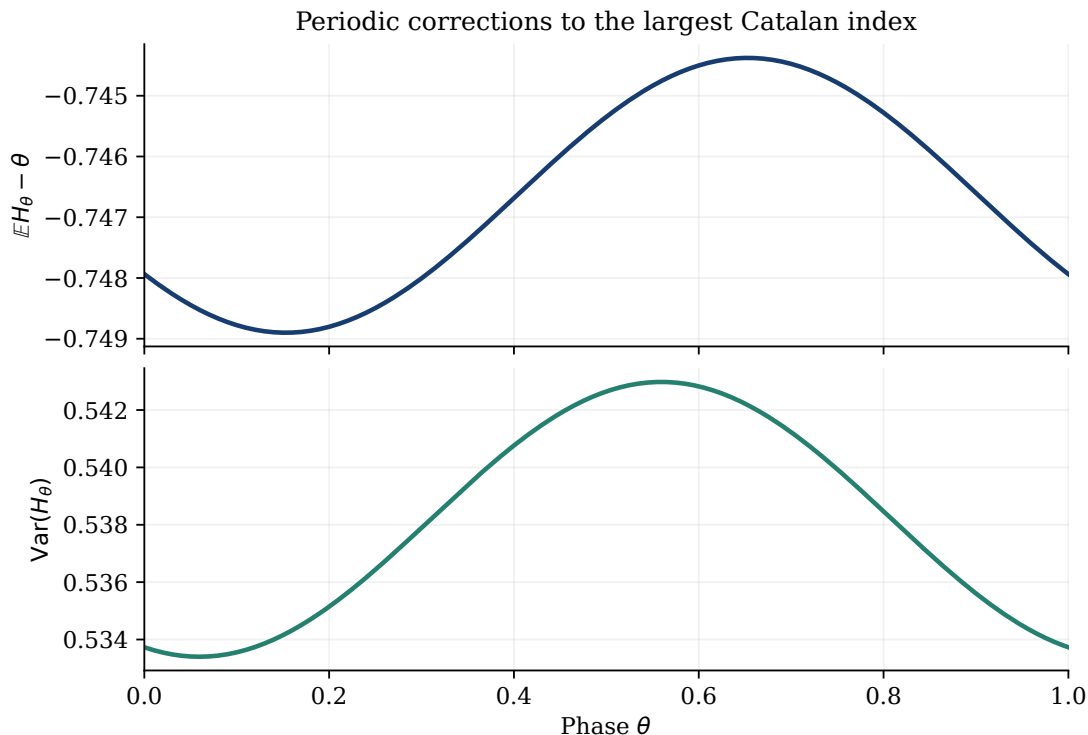


Figure 2: The leading periodic correction $F_0(\theta) = \mathbb{E}J_\theta - \theta$ and the limiting variance $\mathcal{V}_0(\theta)$ (manuscript 06's figure, whose axis labels write H_θ for J_θ). The phase dependence is small but persists as n grows. These are numerical evaluations of convergent products and sums.

For orientation, numerical evaluations (illustrative, not interval certificates) give

θ	0	1/2
$F_0(\theta)$	-0.747936333	-0.745346630
$F_1(\theta)$	0.321333399	0.372111961
$\mathcal{V}_0(\theta)$	0.533729833	0.542646703
$\mathcal{V}_1(\theta)$	0.126361559	0.187492246

17.5 The largest part size itself

The largest size is c_{J_n} . Along any subsequence $\theta_n \rightarrow \theta$,

$$t_n c_{J_n} \Rightarrow 4^{\mathcal{J}_\theta - \theta}, \quad (17.13)$$

and every fixed real power has convergent expectation. For positive and negative powers alike, $0 < r'(x) < \log 4$ for $x \geq 1$ gives $|\log(t_n c_{M+h})| = |r(M+h) - r(m)| \leq \log 4 |h - \theta|$ for every possible $J_n = M + h$, and the uniform exponential moments (17.4) give uniform integrability. So the largest part size has the same order $1/t_n \asymp n/\log n$ as the number of parts, although their limit laws differ (section 18).

18 The number of parts

18.1 The limiting variable

Let $\mathbf{E}_1, \mathbf{E}_2, \dots$ be independent exponential variables of mean one, and

$$W = \sum_{k \geq 1} \frac{\mathbf{E}_k}{c_k}, \quad S_r = \sum_{k \geq 1} c_k^{-r}, \quad \omega_q = \mathbb{E}W^q. \quad (18.1)$$

The series converges almost surely and in every L^q , since $\sum_k 1/c_k < \infty$ and Minkowski's inequality applies (equivalently, the geometric lower bound on c_k gives all moment bounds). Its Laplace transform is

$$\mathcal{L}(s) = \mathbb{E}e^{-sW} = \prod_{k \geq 1} \frac{c_k}{c_k + s}, \quad \operatorname{Re} s > -1, \quad (18.2)$$

converging locally uniformly on this half-plane, whose boundary is forced by the first exponential.

Theorem 18.1 (Number-of-parts law, first correction and all moments). *As $n \rightarrow \infty$,*

$$\mathbb{E}e^{-stN_n} = \mathcal{L}(s) - \frac{s^2 \mathcal{L}''(s)}{2m} + O_{\mathcal{K}}(m^{-2}) \quad (18.3)$$

uniformly for s in every compact $\mathcal{K} \subset \{\operatorname{Re} s > -1\}$. In particular $tN_n \Rightarrow W$, and for every fixed integer $q \geq 0$

$$t^q \mathbb{E}N_n^q = \omega_q \left(1 - \frac{q(q-1)}{2m}\right) + O_q(m^{-2}). \quad (18.4)$$

For every real $a < 1$, $\mathbb{E}e^{atN_n} \rightarrow \mathbb{E}e^{aW}$, so all fixed absolute moments converge as well.

Proof. With the marking variable e^{-st} , the generating function marked by the number of parts is $\prod_k (1 - e^{-st} z^{c_k})^{-1}$; its amplitude relative to F is

$$A_s(\tau; t) = \prod_{k \geq 1} \frac{1 - e^{-\tau c_k}}{1 - e^{-\tau c_k - st}}. \quad (18.5)$$

Uniformly on a small fixed complex neighbourhood of $\tau/t = 1$ (depending on $\mathcal{K}$, [remark 14.3](#)) and on $\mathcal{K}$,

$$A_s(\tau; t) = \mathcal{L}(st/\tau) \{1 + O_{\mathcal{K}}(tm)\}; \quad (18.6)$$

an exponentially small error such as tm is below every fixed inverse power of m . To prove it, put $g_0(x) = \log((1 - e^{-x})/x)$, with its analytic branch in the relevant right half-plane. After factoring the linear terms out of each factor, the remaining logarithm is $\sum_k [g_0(\tau c_k) - g_0(\tau c_k + st)]$. For $tc_k \leq 1$ each difference is $O_{\mathcal{K}}(t)$, and there are $O(m)$ terms; for $tc_k > 1$ the bound $g'_0(x) = O(1/|x|)$ in the chosen sector gives a summable $O_{\mathcal{K}}(1/c_k)$, whose tail is $O_{\mathcal{K}}(t)$. Cauchy estimates give the same accuracy for any fixed number of scaled τ -derivatives. (The kernel g_0 is the Laplace-product kernel of the Fabius-tree compilation cited after Part I's [lemma 4.1](#); it is not Part I's g .)

So the jet may be replaced by

$$\mathbf{a}_{\ell}(s) = \frac{d^{\ell}}{dv^{\ell}} \mathcal{L}\left(\frac{s}{1+v}\right) \Big|_{v=0}, \quad \mathbf{a}_0 = \mathcal{L}, \quad \mathbf{a}_1 = -s\mathcal{L}', \quad \mathbf{a}_2 = 2s\mathcal{L}' + s^2\mathcal{L}''. \quad (18.7)$$

Apply (14.7) and divide by $1 + E_1$. With $B_2 = m + O(1)$ and $B_3 = 2m + O(1)$ the two terms in $s\mathcal{L}'$ cancel, and the remaining term is (18.3). Laplace convergence gives the distributional assertion; compact complex uniformity permits differentiation at $s = 0$ by Cauchy's formula, and since the q th derivative of $s^2\mathcal{L}''(s)$ at zero is $q(q-1)\mathcal{L}^{(q)}(0)$ we obtain (18.4). Evaluation at $s = -a$, $a < 1$, proves exponential-moment convergence, and a slightly larger admissible exponential moment gives uniform integrability for every fixed power. $\square$

An explicit all-orders construction. By the proof and [theorem 14.1](#), for every fixed J ,

$$\mathbb{E}e^{-stN_n} = \frac{\Omega_J[\mathbf{a}(s)]}{\Omega_J[\mathbf{1}]} + O_{J,\kappa}(m^{-J-1}), \quad (18.8)$$

where (18.7) and (14.3) are finite constructive rules at each order. For the q th raw moment the jet is

$$\mathbf{a}_\ell = \omega_q(-1)^\ell (q)_\ell, \quad (q)_\ell = q(q+1) \cdots (q+\ell-1), \quad (18.9)$$

since differentiating the leading amplitude in the mark gives $\omega_q(t/\tau)^q$. The exact B_r keep the local phase without further expansion; to display an ordinary inverse- m expansion write $B_r = (r-1)!m + \mathbf{b}_r(\theta) + O_r(m^{-1})$ with $\mathbf{b}_r$ as in (14.8). That series converges absolutely with all phase derivatives and defines a smooth one-periodic function; each higher coefficient comes from Taylor-expanding (13.6) inside f_r and summing polynomially weighted derivatives over the same lattice. Remainder justification: terms with $k < m/2$ differ from the small- ν constant by an exponentially small summable error; terms with $k > 3m/2$ have a superexponentially small sum; in between, the differentiable Stirling expansion of $r(m+d) - r(m)$ has remainder bounded by a fixed polynomial in d times the requested inverse power of m , and the derivatives of $f_r(4^d)$ minus its small- ν constant decay exponentially at the negative end and superexponentially at the positive end. These bounds permit summing each remainder and termwise phase differentiation. So every fixed inverse-power coefficient of (18.8) is determined by smooth one-periodic functions and derivatives of $\mathcal{L}$: an all-orders Poincaré expansion in m^{-1} with a phase coefficient algebra, the same algebra as Part I's ([remark 14.4](#)). It does not include the exponentially small noncentral-arc sectors of a complete transseries.

18.2 The first-order cancellation in the mean

Corollary 18.2 (Mean through its first nonzero conditioning term). *With $\omega_1 = S_1$,*

$$t\mathbb{E}N_n = S_1 \left(1 + \frac{B_3 - 2B_2}{2B_2^2} \right) + O(m^{-3}). \quad (18.10)$$

In particular $t\mathbb{E}N_n = S_1 + O(m^{-2})$: there is no m^{-1} term. Moreover

$$S_1 = 1 + \frac{4\pi}{9\sqrt{3}} = 1.806133050770763489 \dots, \quad S_2 = 1.295732295098655075 \dots \quad (18.11)$$

Proof. For $q = 1$, (18.9) is $\mathbf{a}_\ell = S_1(-1)^\ell \ell!$. The weight-two terms of the operator give $S_1(B_3 - 2B_2)/(2B_2^2)$, already $O(m^{-2})$ by [lemma 13.2](#). The weight-four terms have zero leading m^{-2} coefficient: by the gamma benchmark ([proposition 18.3](#)), on setting $B_r = (r-1)!m$ the full formal moment quotient for $q = 1$ is identically one, and replacing any leading cumulant by its $O(1)$ correction in a weight-four term adds a factor $O(m^{-1})$. The denominator correction changes the displayed $O(m^{-2})$ term only by $O(m^{-3})$. Equation (18.8) with $J = 2$ proves (18.10).

For the constant, the beta integral gives $1/c_k = (k+1)(2k+1) \int_0^1 [x(1-x)]^k dx$. Summing from $k = 0$ (temporarily including $c_0 = 1$) with $\sum_{k \geq 0} (k+1)(2k+1)y^k = (1+3y)/(1-y)^3$,

$$\sum_{k \geq 0} \frac{1}{c_k} = \int_0^1 \frac{1+3x(1-x)}{(1-x(1-x))^3} dx = 16 \int_0^1 \frac{7-3y^2}{(3+y^2)^3} dy = 2 + \frac{4\pi}{9\sqrt{3}}.$$

Route of manuscript 06: an antiderivative of the last integrand is $\frac{64y}{3(3+y^2)^2} + \frac{8y}{3(3+y^2)} + \frac{8}{3\sqrt{3}} \arctan(y/\sqrt{3})$.

Route of manuscript 10: the middle integral is elementary after $x = (1+u)/2$. Subtract the $k = 0$ term. $\square$

S_1 is the reciprocal-Catalan constant, OEIS A121839 [13], whose entry sums from $k \geq 1$ as here; it is not a new constant. [Corollary 18.2](#) proves the cancellation that manuscript 06 asked about ([section 23](#)) and that its recorded run shows numerically: $m^2(t\mathbb{E}N_n - S_1) = 0.578$ at $n = 10^6$.

Proposition 18.3 (Gamma benchmark for the universal coefficients). *If B_r is formally replaced by $(r-1)!m$, the moment quotient for (18.9) has the expansion of*

$$\omega_q \frac{m^q \Gamma(m)}{\Gamma(m+q)} = \omega_q \prod_{j=0}^{q-1} (1 + j/m)^{-1} \quad (18.12)$$

for every integer $q \geq 1$.

Proof. These are the cumulants of the local phase $-m \log \tau$ at t , with $nt = m$. The inverse-Laplace Gaussian expansion for the unmarked model τ^{-m} has exact coefficient-density factor $n^{m-1}/\Gamma(m)$; multiplication by $\omega_q (t/\tau)^q$ changes the ratio to $\omega_q t^q n^q \Gamma(m)/\Gamma(m+q)$, which is (18.12). This exact model determines the Gaussian formal coefficients; equivalently, expanding the finite product on the right and the finite sums of (14.3) verifies the identity to every prescribed order by formal Gaussian integration. $\square$

18.3 Cumulants, non-normality and lack of concentration

Expanding $\log \mathbb{E}e^{sW} = -\sum_k \log(1 - s/c_k)$ at zero,

$$\text{cum}_r(W) = (r-1)! S_r \quad (r \geq 1). \quad (18.13)$$

For a finite recurrence for all moments, put $h_0 = 1$ and

$$r h_r = \sum_{j=1}^r S_j h_{r-j}; \quad \text{then} \quad \omega_r = \mathbb{E}W^r = r! h_r. \quad (18.14)$$

$\text{Var } W = S_2 > 0$ and $\text{cum}_3(W) = 2S_3 > 0$, so W is not Gaussian; numerically $2S_3/S_2^{3/2} \approx 1.53685181130881994$. The number of parts therefore satisfies

$$\mathbb{E}N_n \sim S_1 \frac{(\log 4) n}{\log n}, \quad \text{Var } N_n \sim S_2 \left(\frac{(\log 4) n}{\log n} \right)^2, \quad (18.15)$$

that is $\text{Var } N_n \sim S_2/t_n^2$. The limiting relative variance S_2/S_1^2 is positive: the number of parts does not concentrate around its mean on its natural scale, a Gaussian approximation would give the wrong limit law, and replacing the count by its mean is not an accurate description of a typical partition.

Corollary 18.4 ([write] The first variance correction). $t^2 \text{Var } N_n = S_2 - \omega_2/m + O(m^{-2})$, where $\omega_2 = S_2 + S_1^2$.

Proof. By (18.4) with $q = 2$, $t^2 \mathbb{E}N_n^2 = \omega_2(1 - 1/m) + O(m^{-2})$, and by corollary 18.2 $(t\mathbb{E}N_n)^2 = S_1^2 + O(m^{-2})$; subtract. $\square$

This is the field `variance_first_heuristic` of manuscript 06's recorded run (0.8943857364718 at $n = 10^6$, where the exact value is 0.938223, table 8), which that manuscript marked as an exploratory, unproved refinement; its field `m_times_variance_error`, $m(t^2 \text{Var } N_n - S_2) = -4.06$ at $n = 10^6$, tends to $-\omega_2 = -4.5578 \dots$ by the corollary.

18.4 Fixed small parts

For fixed d let $X_{k,n} = t c_k Z_k$, $1 \leq k \leq d$. Their joint Laplace transform has limit $\Lambda_d(\mathbf{s}) = \prod_{k=1}^d (1 + s_k)^{-1}$ and first correction

$$\mathbb{E} \exp\left(-\sum_{k=1}^d s_k X_{k,n}\right) = \Lambda_d(\mathbf{s}) - \frac{1}{2m} \sum_{i,j=1}^d s_i s_j \partial_i \partial_j \Lambda_d(\mathbf{s}) + O(m^{-2}). \quad (18.16)$$

To prove it, mark only these d factors in (18.5); their limiting arguments are $s_k/(1+v)$. With $\mathcal{D}_s = \sum_k s_k \partial_k$, $\mathbf{a}_1 = -\mathcal{D}_s \Lambda_d$ and $\mathbf{a}_2 = (\mathcal{D}_s^2 + \mathcal{D}_s) \Lambda_d$, and $-(\mathbf{a}_2 + 2\mathbf{a}_1)/(2m)$ is the correction in (18.16). So the $X_{k,n}$ tend to independent mean-one exponentials, while for $i \neq j$

$$\text{Cov}(X_{i,n}, X_{j,n}) = -m^{-1} + O(m^{-2}). \quad (18.17)$$

The independence is a limit statement; conditioning leaves a universal negative first dependence among fixed small-part energies.

18.5 Joint limit with the upper edge

Theorem 18.5 (Joint count and edge limit). *Along a subsequence with $\theta_n \rightarrow \theta \in [0, 1]$, for every fixed integer h ,*

$$(t_n N_n, (Z_{M+j})_{j>h}, J_n - M) \Rightarrow (W, (Y_j^{(\theta)})_{j>h}, \mathcal{J}_\theta), \quad (18.18)$$

the upper process on its countable space of finitely supported vectors and the maximum defined by the full two-sided family. W is independent of that whole family. Every fixed mixed moment of $t_n N_n$ and $J_n - M$ converges: $\mathbb{E}[(t_n N_n)^p (J_n - M)^q] \rightarrow \omega_p \mathbb{E} \mathcal{J}_\theta^q$. Moreover, without a phase subsequence,

$$\lim_{n \rightarrow \infty} \mathbb{E}_{\mathbb{P}_n} e^{s t_n N_n} = \mathbb{E} e^{s W} = \prod_{k \geq 1} (1 - s/c_k)^{-1} \quad \text{for every real } s < 1. \quad (18.19)$$

Proof (manuscript 06). Fix finitely many small indices $i_1, \dots, i_a$ and edge offsets $j_1, \dots, j_b$. For nonnegative real thresholds x_ℓ prescribe $Z_{i_\ell} \geq \lceil x_\ell/t \rceil$, and for edge coordinates arbitrary fixed integer thresholds $d_\ell \geq 0$. The shift v of (13.4) satisfies $tv \rightarrow \sum_\ell c_{i_\ell} x_\ell + \sum_\ell 4^{j_\ell - \theta} d_\ell$, so by (13.16) the joint survival probability tends to the exponential of minus this sum, which factorizes into the survival functions of independent $\mathbf{E}_{i_\ell}/c_{i_\ell}$ and $Y_{j_\ell}^{(\theta)}$. Finite inclusion–exclusion gives the joint finite-coordinate limit.

For uniform control before summing all small multiplicities, (13.4) and (13.15) give $\mathbb{P}_n(Z_k \geq z) \leq C e^{-t c_k z}$, hence $\mathbb{E}_{\mathbb{P}_n} Z_k^r \leq C \Gamma(r+1)/(t c_k)^r$ for every real $r \geq 1$, and, uniformly in large n ,

$$\left\| t \sum_{k>K} Z_k \right\|_{L^r(\mathbb{P}_n)} \leq (C \Gamma(r+1))^{1/r} \sum_{k>K} \frac{1}{c_k} \rightarrow 0 \quad (K \rightarrow \infty). \quad (18.20)$$

With the finite-coordinate limit this proves $t N_n \Rightarrow W$ with all positive moments. For the upper process, truncate at a fixed large positive offset: a nonzero coordinate beyond it is uniformly unlikely by lemma 17.3, and geometric tails give tightness on the remaining finite set. For the maximum also truncate at a fixed large negative offset; the left tail makes losing the maximum arbitrarily unlikely. These truncations, the finite joint survival limit and (18.20) prove (18.18) and the independence; uniform higher moments from (18.20) and lemma 17.3, with Hölder's inequality, give every mixed moment.

For the moment generating function choose $0 < s < s' < 1$ and $y = e^{s't}$. The power series $\prod_k (1 - z^{c_k})/(1 - y z^{c_k})$ has nonnegative coefficients, and convolution with (13.15) gives

$$\mathbb{E}_{\mathbb{P}_n} e^{s' t N_n} \leq C \prod_{k \geq 1} \frac{1 - e^{-t c_k}}{1 - e^{-t(c_k - s')}} \leq C \prod_{k \geq 1} \frac{c_k}{c_k - s'} < \infty,$$

since $(1 - e^{-x})/x$ decreases on $x > 0$. This exponential uniform integrability and the distributional convergence give (18.19) for $0 < s < 1$; for $s \leq 0$ bounded convergence suffices. The product converges because $\sum_k 1/c_k < \infty$. $\square$

Second proof of the $(t_n N_n, J_n - M)$ marginal (manuscript 10). Theorem 16.6 gives the joint Laplace transforms with every integer cutoff, and marginal tightness follows from theorem 18.1 and lemma 17.3; these determine the joint law and its product form. For moments choose $0 < a < 1/2$ and $b > 0$; by Cauchy–Schwarz and the exponential bounds, $\sup_n \mathbb{E} e^{a t N_n + b |J_n - M|} \leq (\sup_n \mathbb{E} e^{2 a t N_n})^{1/2} (\sup_n \mathbb{E} e^{2 b |J_n - M|})^{1/2} < \infty$, which is the needed mixed uniform integrability. $\square$

The independence is a limit assertion. At finite n many small parts compete for mass with large parts, and the exact covariance need not vanish (table 8): its first order is the next theorem.

Theorem 18.6 ([write] The first mixed covariance). *Let $\Xi(\theta) = \sum_{h \in \mathbb{Z}} G_h(\theta) A_h(\theta)$, a smooth one-periodic positive function. Along the full sequence,*

$$m \operatorname{Cov}(t_n N_n, J_n) = -S_1 \Xi(\theta_n) + O\left(\frac{(\log m)^{5/2}}{m}\right). \quad (18.21)$$

Proof. Since $J_n - M = \sum_{h \geq 0} \mathbf{1}_{\{J_n > M+h\}} - \sum_{h \leq -1} \mathbf{1}_{\{J_n \leq M+h\}}$ and M is deterministic, $\operatorname{Cov}(tN_n, J_n) = -\sum_{h \in \mathbb{Z}} \operatorname{Cov}(tN_n, \mathbf{1}_{\{J_n \leq M+h\}})$; the series converges absolutely by the tails below.

Step 1: a per-threshold covariance uniform in $|h| \leq m^{1/8}$. Mark the tail-deleted product with the number-of-parts mark. Its amplitude is $A_s(\tau; t) \prod_{k > K} (1 - e^{-\tau c_k - st})$, which equals $\mathcal{L}(st/\tau) A_K(\tau)$ up to a relative error $O(tm) + O(\sum_{k > K} c_k^{-1})$, exponentially small in m for $|h| \leq m^{1/8}$ (since $c_{M-r} \geq 4^{-r} c_M$), with the same bound for scaled derivatives. Relative to $A_K(t)$ its jets are products of the bounded jets (18.7) with the relative jets (17.2), so they are $O(H_-^\ell)$, and the proof of theorem 17.1 (remark 17.2) gives, uniformly for $|h| \leq m^{1/8}$, s on the circle $|s| = \frac{1}{2}$ and the phase,

$$\frac{\mathbb{E}[e^{-stN_n} \mathbf{1}_{\{J_n \leq K\}}]}{A_K(t)} = \mathcal{L} - \frac{X(s)}{2B_2} - \frac{B_3 Y(s)}{2B_2^2} + O\left(\frac{(1+|h|)^4}{m^2}\right),$$

with the product jets $X = 2s\mathcal{L}' + s^2\mathcal{L}'' - 2s\mathcal{L}'\widehat{T}_{1,K} + \mathcal{L}(\widehat{T}_{1,K}^2 - \widehat{T}_{2,K})$ and $Y = -s\mathcal{L}' + \mathcal{L}\widehat{T}_{1,K}$. The left side and the main terms are analytic in $|s| < 1$, so Cauchy's formula transfers the bound to the derivative at $s = 0$. With $\mathcal{L}'(0) = -S_1$ this gives

$$\frac{\mathbb{E}[tN_n \mathbf{1}_{\{J_n \leq K\}}]}{A_K(t)} = S_1 \left[1 + \frac{\widehat{T}_{2,K} - \widehat{T}_{1,K}^2}{2B_2} - \frac{B_3 \widehat{T}_{1,K}}{2B_2^2} \right] + \frac{S_1 \widehat{T}_{1,K}}{B_2} + S_1 \frac{B_3 - 2B_2}{2B_2^2} + O\left(\frac{(1+|h|)^4}{m^2}\right).$$

Subtract $\mathbb{E}[tN_n] \mathbb{P}_n(J_n \leq K)/A_K(t)$, using (18.10) and (17.1): the bracket and the last displayed main term cancel up to $O(m^{-2}) \cdot O((1+|h|)^2/m)$, so

$$\operatorname{Cov}(tN_n, \mathbf{1}_{\{J_n \leq K\}}) = A_K(t) \left[\frac{S_1 \widehat{T}_{1,K}}{B_2} + O\left(\frac{(1+|h|)^4}{m^2}\right) \right].$$

At fixed h this is (16.24).

Step 2: replacing exact tail quantities by phase functions. For $h = -r$ with $0 \leq r \leq M/2$, the comparison bounds in the proof of theorem 17.5 (remark 17.6) give $|\log(A_K(t)/G_h)| \leq C(1+h^2)/m$ and $|\widehat{T}_{1,K} - A_h| \leq C/m$; for $h \geq 0$ the same bounds hold with C/m , by (16.7) and the geometric growth of the rates. With $A_h = O(1+|h|)$ and $1/B_2 = 1/m + O(m^{-2})$, for $h^2 \leq m$,

$$\left| \operatorname{Cov}(tN_n, \mathbf{1}_{\{J_n \leq M+h\}}) - \frac{S_1 G_h A_h}{m} \right| \leq C G_h \frac{(1+|h|)^4}{m^2}.$$

Step 3: summation. Let $H = \lceil C_0 \sqrt{\log m} \rceil$, so that $H \leq m^{1/8}$ and $H^2 \leq m$ for large m . For $|h| > H$ use $|\operatorname{Cov}(tN_n, \mathbf{1}_E)| \leq \|tN_n - \mathbb{E}tN_n\|_2 \mathbb{P}_n(E)^{1/2}$ with E or its complement: by theorem 18.1 and lemma 17.3 this is at most $Ce^{-c_2 h^2}$ for $h > H$ and Ce^{-ch^2} for $h < -H$, and the sum over $|h| > H$ is $O(m^{-3})$ once C_0 is large. The same holds for $\sum_{|h| > H} G_h A_h$, since $A_h \leq Ce^{-c_2 h^2}$ for $h \geq 0$ and $G_{-r} \leq 4^{-r(r-1)/2}$. Over $|h| \leq H$ the errors of Step 2 sum to $Cm^{-2}(\sum_{0 \leq h \leq H} (1+h)^4 + \sum_{r \geq 1} e^{-cr^2} (1+r)^4) = O(H^5/m^2)$. Altogether $m \operatorname{Cov}(tN_n, J_n) = -S_1 \sum_h G_h A_h + O((\log m)^{5/2}/m)$.

Finally $\Xi > 0$ termwise; it converges with all phase derivatives by the bounds just used, and it is one-periodic because (17.7) and $A_h(\theta + 1) = A_{h-1}(\theta)$ shift the summation index. $\square$

Remark 18.7 (Status and numbers). [Theorem 18.6](#) proves the value manuscript 06 recorded in its run as `mixed_covariance_candidate_at_phase`, “explicitly not claims”: $-S_1\Xi(\theta)$ is -1.3205626659 at the phase $\theta = 0.35639\dots$ of $n = 10^6$, against the exact $m \operatorname{Cov}(t_n N_n, J_n) = -1.4368$ (its field `m_times_cov_tN_J`). The batch-98 dossier observed that the candidate is the sum over thresholds of manuscript 10’s per-threshold covariance ([16.24](#)); the write supplies the summable uniform remainder that this needs. The proof reuses the joint-mark form of [theorem 17.1](#) ([remark 17.2](#)), which neither manuscript states. The next coefficient, and whether the remainder is $O(m^{-1})$ without the logarithm, remain open ([section 23](#)). Manuscript 06’s phase-function file records Ξ as “gamma” with the caution that it is a limit of covariances of truncated energies, not the covariance of an L^2 total energy, which does not exist.

19 The law of W : spectral expansion, natural boundary and inverse tail

The law of W can be evaluated without simulation, and much more than a leading tail equivalent holds. Manuscript 06 gives the exponential expansion with a one-sided coefficient bound; manuscript 10 gives two-sided bounds, an explicit remainder and the analytic consequences. Define

$$\mathfrak{d}_j = \prod_{\substack{k \geq 1 \\ k \neq j}} \frac{c_k}{c_k - c_j} = \prod_{k \neq j} \left(1 - \frac{c_j}{c_k}\right)^{-1}, \quad \varpi = \prod_{\ell \geq 1} (1 - 2^{-\ell}) > 0. \quad (19.1)$$

For each j the product converges, and $\operatorname{sign}(\mathfrak{d}_j) = (-1)^{j-1}$.

Lemma 19.1 (Two-sided bounds on the spectral coefficients). *For every $j \geq 1$,*

$$4^{-j(j-1)/2} \leq |\mathfrak{d}_j| \leq \varpi^{-1} 2^{-j(j-1)/2}. \quad (19.2)$$

In particular $\sum_j |\mathfrak{d}_j| c_j^r < \infty$ for every fixed $r \geq 0$.

Proof. For $k < j$ write the absolute factor as $(c_k/c_j)/(1 - c_k/c_j)$. The bounds $4^{-(j-k)} \leq c_k/c_j \leq 2^{-(j-k)}$ make its product at least $4^{-j(j-1)/2}$ and at most $\varpi^{-1} 2^{-j(j-1)/2}$. For $k > j$ each factor exceeds one, and their product is at most $\prod_{\ell \geq 1} (1 - 2^{-\ell})^{-1} = \varpi^{-1}$. The last assertion follows from $c_j \leq 4^j$. $\square$

Theorem 19.2 (Exact survival expansion). *For every real $x \geq 0$,*

$$\mathbb{P}(W > x) = \sum_{j \geq 1} \mathfrak{d}_j e^{-c_j x}, \quad f_W(x) = \sum_{j \geq 1} c_j \mathfrak{d}_j e^{-c_j x}. \quad (19.3)$$

The series and each fixed derivative converge absolutely and uniformly on $x \geq 0$. For $J \geq 1$,

$$\left| \mathbb{P}(W > x) - \sum_{j=1}^J \mathfrak{d}_j e^{-c_j x} \right| \leq \frac{\varpi^{-2} 2^{-J(J+1)/2}}{1 - 2^{-(J+1)}} e^{-c_{J+1} x}. \quad (19.4)$$

The law has a strictly positive smooth density on $x > 0$, and

$$\sum_{j \geq 1} \mathfrak{d}_j = 1, \quad \sum_{j \geq 1} \mathfrak{d}_j c_j^r = 0 \quad (r = 1, 2, \dots). \quad (19.5)$$

Proof. For $W_N = \sum_{k \leq N} \mathbf{E}_k / c_k$, partial fractions of the Laplace transform $\prod_{k \leq N} c_k / (s + c_k)$ give $\mathbb{P}(W_N > x) = \sum_{j \leq N} \mathfrak{d}_j^{(N)} e^{-c_j x}$ with $\mathfrak{d}_j^{(N)} = \prod_{k \leq N, k \neq j} c_k / (c_k - c_j)$. The proof of [lemma 19.1](#) bounds these uniformly in N (manuscript 06’s route: $|\mathfrak{d}_j^{(N)}| \leq C 2^{-j(j-1)/2}$ from $c_{\ell+1}/c_\ell \geq 2$). For fixed j , $\mathfrak{d}_j^{(N)} \rightarrow \mathfrak{d}_j$, and $W_N \rightarrow W$ almost surely; $W = \mathbf{E}_1 + \sum_{k \geq 2} \mathbf{E}_k / c_k$ has a continuous law, so dominated convergence proves (19.3), and the last assertion of [lemma 19.1](#) gives the derivative bounds (since $c_k \leq 4^{k-1}$). For (19.4) factor out $e^{-c_{J+1} x}$ and bound the coefficient tail by a geometric series: consecutive majorants have ratios at most $2^{-(J+1)}$.

Flatness at zero: for every positive integer R , $W \geq W_R$, and integration over the simplex gives

$$\mathbb{P}(W \leq x) \leq \mathbb{P}(W_R \leq x) \leq \frac{x^R}{R!} \prod_{k=1}^R c_k \quad (x \geq 0). \quad (19.6)$$

With the smoothness already proved and R arbitrary, all right derivatives of the distribution function vanish at zero; evaluating the uniform derivative series at zero gives (19.5).

Positivity: with $V = \sum_{k \geq 2} \mathbf{E}_k / c_k$, convolution with $\mathbf{E}_1$ gives $f_W(x) = e^{-x} \mathbb{E}[e^V \mathbf{1}_{\{V < x\}}]$ for $x > 0$. For each $x > 0$, $\mathbb{P}(V < x) > 0$: choose a far tail with expectation below $x/4$, use Markov's inequality for it, and require each of the finitely many preceding variables to be small. $\square$

In particular $\mathbb{P}(W > x) = \mathfrak{d}_1 e^{-x} + O(e^{-2x})$ with $\mathfrak{d}_1 = \prod_{k \geq 2} (1 - 1/c_k)^{-1}$, and the first coefficients are

$$\begin{aligned} \mathfrak{d}_1 &= 2.78827476752297890, & \mathfrak{d}_2 &= -2.08690959322975775, \\ \mathfrak{d}_3 &= 0.31102280892305314, & \mathfrak{d}_4 &= -0.01252727293482829, \\ \mathfrak{d}_5 &= 0.00013971276281143. \end{aligned}$$

The alternating signs are essential: the expansion is not a mixture of positive exponential laws. Near zero it cancels severely, exactly as (19.5) quantifies, and the distribution function is flat there: every derivative vanishes at $x = 0$ (19.6). At large x the expansion is very effective, the rates growing geometrically. These are properties of the familiar infinite exponential convolution; their role here is to make the Catalan limit law explicit.

19.1 Smoothness without analytic continuation

The change of variable $z = e^{-x}$ turns (19.3) into

$$\mathcal{T}(z) = \sum_{j \geq 1} \mathfrak{d}_j z^{c_j}, \quad \mathbb{P}(W > x) = \mathcal{T}(e^{-x}). \quad (19.7)$$

$\mathcal{T}$ is not the partition generating function $F(z)$ and not the Laplace transform $\mathcal{L}(s)$: three different analytic objects.

Theorem 19.3 (Natural boundary and non-D-finiteness). *$\mathcal{T}$ has radius of convergence exactly one and is smooth on the closed unit disk: all its fixed holomorphic derivatives extend continuously to it. Nevertheless every point of $|z| = 1$ is singular for analytic continuation, so $\mathcal{T}$ is not D-finite, and the analytic function $\mathbb{P}(W > x) = \mathcal{T}(e^{-x})$ on $\operatorname{Re} x > 0$ has the imaginary axis as a natural boundary. The Laplace transform $\mathcal{L}(s)$ is meromorphic with simple poles at $-c_j$, $j \geq 1$, and is also not D-finite.*

Proof. The upper bound of (19.2) gives uniform convergence on $|z| \leq 1$ of every differentiated series. The two-sided bound gives $\lim_j |\mathfrak{d}_j|^{1/c_j} = 1$, because $j^2/c_j \rightarrow 0$; so the radius is exactly one. We invoke one classical result: a power series of finite positive radius supported on exponents λ_j with $\lambda_{j+1}/\lambda_j \geq 1 + \delta$ for some $\delta > 0$ has its circle of convergence as a natural boundary, Hadamard's gap theorem [15, §13, pp. 116–117]. Its hypotheses hold with $\lambda_j = c_j$ and $\delta = 1$, by (3.6). A germ satisfying a linear differential equation with polynomial coefficients continues analytically along paths avoiding the finitely many zeros of the leading coefficient, so it cannot have a whole circle as natural boundary; $\mathcal{T}$ is not D-finite. Since $x \mapsto e^{-x}$ is locally biholomorphic, a continuation across an imaginary-axis point would continue $\mathcal{T}$ across the corresponding unit-circle point. Finally $\prod_j (1 + s/c_j)$ converges locally uniformly on the whole s -plane ($\sum 1/c_j < \infty$), with exactly the distinct simple zeros $-c_j$; its reciprocal is $\mathcal{L}$, and infinitely many finite poles again contradict a polynomial-coefficient linear differential equation. $\square$

Even the singularity at $z = 1$ has a direct reason: $\mathcal{T}(1) = 1$ and all its positive-order derivatives there vanish by (19.5), so a continuation near 1 would be locally the constant one, contradicting $\mathcal{T}(0) = 0$. The gap theorem strengthens this to *every* boundary point. This is a sequence-specific consequence of a classical theorem, not a new gap theorem, and no conclusion about differential *algebraicity* is drawn from non-D-finiteness.

19.2 A convergent inverse upper-tail expansion

For $0 < \epsilon < 1$ let ξ_ϵ be the unique solution of

$$\mathbb{P}(W > \xi_\epsilon) = \epsilon; \quad (19.8)$$

existence and uniqueness follow from theorem 19.2, and $\xi_\epsilon \rightarrow \infty$ as $\epsilon \downarrow 0$. Put

$$\mathcal{T}_1(z) = \frac{\mathcal{T}(z)}{\mathfrak{d}_1 z} = 1 + \frac{\mathfrak{d}_2}{\mathfrak{d}_1} z + \frac{\mathfrak{d}_3}{\mathfrak{d}_1} z^4 + \frac{\mathfrak{d}_4}{\mathfrak{d}_1} z^{13} + \cdots; \quad (19.9)$$

the sparsity of the exponents is inherited from the Catalan rates.

Theorem 19.4 (Inverse tail, all orders and first four coefficients). *For small positive ϵ the following expansion converges:*

$$\xi_\epsilon = \log \frac{\mathfrak{d}_1}{\epsilon} - \sum_{r \geq 1} \frac{1}{r} \left(\frac{\epsilon}{\mathfrak{d}_1} \right)^r [z^r] \mathcal{T}_1(z)^{-r}. \quad (19.10)$$

In particular

$$\xi_\epsilon = \log \frac{\mathfrak{d}_1}{\epsilon} + \frac{\mathfrak{d}_2}{\mathfrak{d}_1^2} \epsilon - \frac{3\mathfrak{d}_2^2}{2\mathfrak{d}_1^4} \epsilon^2 + \frac{10\mathfrak{d}_2^3}{3\mathfrak{d}_1^6} \epsilon^3 + \left(\frac{\mathfrak{d}_3}{\mathfrak{d}_1^5} - \frac{35\mathfrak{d}_2^4}{4\mathfrak{d}_1^8} \right) \epsilon^4 + O(\epsilon^5). \quad (19.11)$$

At any fixed order the coefficients use only finitely many $\mathfrak{d}_j$.

Proof. Put $y = \epsilon/\mathfrak{d}_1$ and $z = e^{-\xi_\epsilon}$; the equation is $y = z\mathcal{T}_1(z)$ with $\mathcal{T}_1(0) = 1$. The analytic inverse theorem gives a convergent inverse germ $z = z(y)$, so $\xi_\epsilon = \log(1/y) + \log \mathcal{T}_1(z(y))$ with the logarithm vanishing at $y = 0$. The residue substitution $y = z\mathcal{T}_1(z)$, or one integration by parts in it, gives for $r \geq 1$

$$[y^r] \log \mathcal{T}_1(z(y)) = \frac{1}{r} [z^{r-1}] \frac{\mathcal{T}_1'(z)}{\mathcal{T}_1(z)^{r+1}} = -\frac{1}{r} [z^r] \mathcal{T}_1(z)^{-r},$$

which proves (19.10) with convergence, not merely formal equality. Through order four replace $\mathcal{T}_1$ by $1 + az + bz^4 + O(z^{13})$ with $a = \mathfrak{d}_2/\mathfrak{d}_1$, $b = \mathfrak{d}_3/\mathfrak{d}_1$; the coefficients of y, y^2, y^3, y^4 are $a, -3a^2/2, 10a^3/3$ and $b - 35a^4/4$. $\square$

The first new rate after 1 and 2 is 5, so it first changes the inverse expansion at order ϵ^4 ; before that every correction comes from reverting the first two exponentials, an example of sparse forward scales creating denser inverse scales. The natural boundary is no obstacle to the inverse germ at zero, since $\mathcal{T}'(0) = \mathfrak{d}_1 \neq 0$. The maximal convergence radius of (19.10) is not identified; complex critical values of $\mathcal{T}$ may limit it before any boundary value does (section 23).

What this says about finite partitions. For fixed $0 < \epsilon < 1$, convergence in distribution and strict positivity of f_W give convergence of the $(1 - \epsilon)$ -quantile of tN_n (left-continuous inverse) to ξ_ϵ ; the lattice spacing t tends to zero. This is a fixed- ϵ statement. Applying (19.11) afterwards is a sequential regime, first $n \rightarrow \infty$ and then $\epsilon \downarrow 0$; it proves no uniform formula for $\epsilon = \epsilon(n) \rightarrow 0$ and reconstructs no exact integer quantile by rounding a truncation. That needs further tail and local-limit estimates (section 23).

20 A logarithmic mass profile and a Gaussian bridge

This section is manuscript 06's. The total mass is spread over many active sizes, although the number of parts is dominated by the smallest sizes. Define the cumulative mass profile

$$\mathcal{F}_n(x) = \frac{1}{n} \sum_{k \leq \lfloor xM \rfloor} c_k Z_k, \quad 0 \leq x \leq 1. \quad (20.1)$$

Theorem 20.1 (Mass profile and finite-dimensional fluctuations). *Under $\mathbb{P}_n$,*

$$\sup_{0 \leq x \leq 1} |\mathcal{F}_n(x) - x| \rightarrow 0 \quad \text{in probability.} \quad (20.2)$$

For every $0 < x_1 < \dots < x_d < 1$,

$$(\sqrt{m}(\mathcal{F}_n(x_i) - x_i))_{i=1}^d \Rightarrow (\mathbb{B}(x_i))_{i=1}^d, \quad (20.3)$$

the right side centred Gaussian with $\text{Cov}(\mathbb{B}(x), \mathbb{B}(y)) = \min(x, y) - xy$. Every fixed joint polynomial moment converges as well. This is a finite-dimensional Brownian-bridge theorem; a functional central limit theorem is not needed for (20.2).

Proof. Put $K_i = \lfloor x_i M \rfloor$ and, under $\mathbb{Q}_t$, $R_i = (t/\sqrt{m}) \sum_{k > K_i} (c_k Z_k - \mathbb{E}_{\mathbb{Q}_t} c_k Z_k)$. The disjoint index bands between the cutoffs, and the final tail, are independent; their scaled second cumulants tend to their index-length fractions and their higher normalized cumulants vanish. So $(R_1, \dots, R_d)$ converges under $\mathbb{Q}_t$ to a centred Gaussian vector $\mathbf{G}$ with covariance $C_{ij} = 1 - \max(x_i, x_j)$.

The whole vector is measurable with respect to the tail after the earliest cutoff K_1 . By the uniform prefix local theorem its conditioning likelihood is $x_1^{-1/2} \exp(-R_1^2/(2x_1)) + o(1)$ uniformly over that tail, and the likelihoods are uniformly bounded. On the conditioning event $S = n = \mathbb{E}_{\mathbb{Q}_t} S$, $(t/\sqrt{m}) \sum_{k \leq K_i} (c_k Z_k - \mathbb{E}_{\mathbb{Q}_t} c_k Z_k) = -R_i$ exactly. The limit law is therefore that of $-\mathbf{G}$ weighted by $x_1^{-1/2} \exp(-\mathbf{G}_1^2/(2x_1))$, which integrates to one since $C_{11} = 1 - x_1$. Completing the square changes the covariance to

$$C_{ij} - \frac{C_{i1}C_{1j}}{x_1 + C_{11}} = 1 - \max(x_i, x_j) - (1 - x_i)(1 - x_j) = \min(x_i, x_j) - x_i x_j.$$

No local theorem for an arithmetically restricted suffix is used. By lemma 13.2, $t\mathbb{E}S_{K_i} = K_i + O(1)$ and $nt = m + O(1)$, so replacing canonical centring by x_i , and the normalization by that of (20.3), changes the vector by $o(1)$; uniform higher canonical moments and the bounded likelihood give convergence of joint moments.

Pointwise convergence of the profile follows. At zero it vanishes identically. At one, the scaled upper mass $t(S - S_M)$ is tight by the fixed-upper-process theorem and its moment bounds, so $1 - \mathcal{F}_n(1) = O_{\mathbb{P}_n}(m^{-1})$. A finite grid and monotonicity of $\mathcal{F}_n$ prove (20.2). $\square$

For example, the first half of the active indices carries asymptotically half the total mass, with variance $1/4$ after the $\sqrt{m}$ normalization. This differs from the total part count, which has a nondegenerate limit after scaling by t : the distinction is whether multiplicities are weighted by c_k .

21 What is universal and what is Catalan-specific

The mechanism extends beyond one OEIS sequence, but its arithmetic hypotheses must be stated. The following proposition (manuscript 06) complements Part I's proposition 8.1, which treats the counting asymptotics under the hypotheses of Section 3; it does not re-prove it.

Proposition 21.1 (A sufficient lacunary class). *Let $a_1 < a_2 < \dots$ be positive integers such that*

1. $a_1 = 1$, and some finite subset of $\{a_2, a_3, \dots\}$ has greatest common divisor one;
2. for all large k , $1 < q_- \leq a_{k+1}/a_k \leq q_+ < \infty$.

For partitions with these allowed sizes let t solve the weighted-mean equation and $M = \#\{k : ta_k \leq 1\}$. Then [theorems 15.1](#) and [15.2](#) hold with the same function $d(a)$ and constant $\phi(1)$, with M in place of m . The total-count limit is $tN_n \Rightarrow \sum_{k \geq 1} \mathbf{E}_k/a_k$ with all moments, and the normalized index mass profile and its finite-dimensional bridge have the same form.

Suppose in addition an eventual smooth interpolation $\log a_k = r(k)$, $r(x) = \alpha x - \beta \log x + c + O(x^{-1})$ with $\alpha > 0$, the remainder differentiable three times with bounds $O(x^{-1-\ell})$. Set $m = r^{-1}(\log(1/t))$ and $M = \lfloor m \rfloor$. Then the fixed-edge laws and their first correction have exactly the forms above, with $u_j = e^{\alpha(j-\theta)}$ and $\frac{3}{2}$ replaced by β , and the joint small-part/edge independence holds.

Proof. The ratio bounds give scaled mean and variance $M + O(1)$ and other scaled cumulants $(r-1)!M + O_r(1)$. The central and intermediate Fourier estimates use a positive fraction of the active sizes and finitely many fixed factors. On the compact outer arcs the unit factor gives one $O(t)$ factor; by the finite gcd hypothesis the sines of the other selected sizes have no common zero there, so one gives a second. The proofs of the local theorem, the cutoff transition and the bridge therefore apply without change. The ratio bounds give $\sum_k 1/a_k < \infty$, so the shift identity and moment domination prove the count limit. Under the interpolation hypothesis $ta_{M+j} = e^{\alpha(j-\theta)}(1 - \beta(j-\theta)/m + O_j(m^{-2}))$, and the L^1 product interpolation gives exactly the coefficient β in the edge score; the same survival identities prove the joint independence. The scalar cumulant constants depending on the initial sizes cancel at first order. $\square$

Remark 21.2 ([write] Re-reading “apply without change”). The dossier flagged this proof as compressed. The write checked each transfer. (1) Outer arcs: a common zero of $\sin(a_k y/2)$ over the finite gcd-one set F needs $ya_k \in 2\pi\mathbb{Z}$ for all $k \in F$, hence $y \in 2\pi\mathbb{Z}$, so on $\delta \leq |y| \leq \pi$ one factor of F is $O_\delta(t)$, and with the unit factor the arc integral is $O(t^2) = o(t/\sqrt{M})$ exponentially, as $t \leq Cq_-^M$. (2) The likelihood expansion [\(13.17\)](#) uses only $B_2 = m + O(1)$, $B_3 = 2m + O(1)$ and $E_1 = -1/(12m) + O(m^{-2})$, which hold for every such sequence because $B_r = (r-1)!M + O_r(1)$ and $m - M \in [0, 1]$; for large n every initial a_k is at most $1/t$ and r is eventually increasing, so $M = \lfloor m \rfloor$ is the number of active sizes. (3) The transport bound [\(16.7\)](#) needs, for $d = j - \theta \geq -m/2$, $\eta_j = -\beta \log(1 + d/m) + \varepsilon(m + d) - \varepsilon(m)$ with $\varepsilon(x) = O(1/x)$, $\varepsilon'(x) = O(x^{-2})$; both bounds of [\(16.7\)](#) follow, with $\frac{3}{2}$ replaced by β , using one derivative of the remainder. (4) The tails of [lemma 17.3](#) hold with 2^h and $2^{-r(r-1)/2}$ replaced by q_-^h and $q_-^{-r(r-1)/2}$. With these readings the proof is complete. Its scope is the first part’s statistics and the first-order edge laws; the higher-order statements of [sections 17](#) and [19](#) are not asserted for the class.

The finite gcd condition is sufficient, not necessary. Pure powers of a fixed integer fail this condition; their resonant arcs need a separate argument, and the proposition does not justify applying the present outer-arc proof to them. For Catalan numbers the witnesses 1, 2, 5 make the arithmetic check immediate (Part I, Section 3). The division between universal and specific data is now clear: the Gaussian transition d depends on the fraction of active variances retained; the growth rate 4 sets the edge lattice, and the Catalan exponent $\frac{3}{2}$ its first drift term; the entire small-size sequence determines the exponential-sum law W .

22 Computation and numerical diagnostics

Each manuscript ships its own programs: manuscript 06 code/[02-laws-verify.py](#) and code/[02-laws-audit-correction.py](#), manuscript 10 code/[03-endpoint-catalan-statistics.py](#), code/[03-endpoint-verify.py](#) and code/[03-endpoint-verify_cutoff.py](#). Neither is used as a substitute for an asymptotic proof. Manuscript 06’s recorded run used Python 3.12.14, mpmath 1.3.0 and matplotlib 3.10.8 at 60 decimal digits; manuscript 10’s used Python 3.13.5, mpmath 1.3.0 and SymPy 1.14.0 at 60 digits.

22.1 Exact coefficients and integer moment numerators

Both programs use arbitrary-precision integers for the finite Euler products needed up to n (manuscript 10 for every $0 \leq n \leq 10^6$). When a part of size c is added, the ascending update of the arrays (p_s, A_s, B_s) , which sum $1, N, N^2$ over the partitions of s , is

$$p_s \leftarrow p_s + p_{s-c}, \quad A_s \leftarrow A_s + A_{s-c} + p_{s-c}, \quad B_s \leftarrow B_s + B_{s-c} + 2A_{s-c} + p_{s-c};$$

the entries at $s - c$ already include the new part type, which allows repeated copies. Then $A_n/p_n = \mathbb{E}N_n$ and $B_n/p_n = \mathbb{E}N_n^2$. Prefix snapshots give the exact law of J_n and the numerators for $N_n J_n$ and the mixed first moments; manuscript 06 keeps all integer numerators in `data/02-laws-verification.json`. The exact counts $p(10^k)$, $k = 2, \dots, 6$, both programs print equal those of Part I, [section 9](#).

For the total variation after K , let $p_K(q)$ count prefix partitions, b_s tail partitions, and $F_{\text{tail}} = \prod_{k>K} (1 - e^{-tc_k})^{-1}$. The exact reduction to tail mass gives

$$\Delta_n(K) = \frac{1}{2} \left\{ \sum_{s=0}^n \left| \frac{b_s p_K(n-s)}{p(n)} - \frac{b_s e^{-ts}}{F_{\text{tail}}} \right| + 1 - \sum_{s=0}^n \frac{b_s e^{-ts}}{F_{\text{tail}}} \right\}. \quad (22.1)$$

The last term accounts for independent tail configurations whose mass exceeds n ; omitting it would understate the distance. The integer arrays are exact; saddles, infinite products and final ratios are numerical, at 60 digits for the high-precision part and in floating summation for the long total-variation sums, with product and phase cutoffs recorded in the JSON.

n	m	$t\mathbb{E}N$	$t^2 \text{Var } N$	$\text{Cov}(tN, J)$
100	4.5337	1.728648	0.526247	-0.304718
1,000	6.2400	1.802456	0.699413	-0.239709
10,000	7.9492	1.815525	0.807284	-0.178759
100,000	9.6549	1.812683	0.881071	-0.146483
1,000,000	11.3564	1.810617	0.938223	-0.126515

Table 8: Exact integer count and moment numerators at the numerical saddle (manuscript 06; N is N_n , J is J_n). The limits of the last three columns are $S_1 = 1.80613305\dots$, $S_2 = 1.29573230\dots$ and zero ([theorem 18.6](#) gives the rate of the last). The variance converges slowly, as [corollary 18.4](#) predicts.

n	θ	Leading error	Corrected error	m^2 scaled
100	0.5337	0.056693	0.022622	0.465
1,000	0.2400	0.043840	0.010381	0.404
10,000	0.9492	0.033055	0.006074	0.384
100,000	0.6549	0.024930	0.004571	0.426
1,000,000	0.3564	0.020101	0.003262	0.421

Table 9: Maximum CDF errors over the recorded thresholds $-28 \leq h \leq 9$ (manuscript 06). The leading approximation is G_h , the corrected one $G_h + G_h C_h/m$; the last column multiplies the corrected maximum error by m^2 . Finite-range numerical maxima, not certified global error bounds.

At $n = 10^6$ there are only $M = 11$ active sizes and $m = 11.356392\dots$. The observed CDF error falls from $0.020100913\dots$ to $0.003262362\dots$ after the correction ([table 9](#)), and the scaled count variance is still about 0.9382 against its limit 1.2957 . A large n is not by itself evidence of a large effective asymptotic parameter.

22.2 An independent central Fourier diagnostic

`code/02-laws-audit_correction.py` works at fixed $\theta = 0.37$ and $m = M + \theta$ for $M = 20, 40, 80, 160$, evaluating finite central Fourier integrals for the full and prefix products. Its target is the continuous

n	K	$a = K/M$	coefficient TV	$d(a)$
100	1	0.2500	0.487538	0.322675
100	2	0.5000	0.274489	0.166064
100	3	0.7500	0.144153	0.069491
1,000	2	0.3333	0.345672	0.259359
1,000	3	0.5000	0.226518	0.166064
1,000	5	0.8333	0.077421	0.044086
10,000	2	0.2857	0.395257	0.293624
10,000	4	0.5714	0.201629	0.134535
10,000	5	0.7143	0.140240	0.081225
100,000	2	0.2222	0.433290	0.347777
100,000	5	0.5556	0.186837	0.141213
100,000	7	0.7778	0.097701	0.060731
1,000,000	3	0.2727	0.359085	0.303814
1,000,000	6	0.5455	0.177617	0.145556
1,000,000	8	0.7273	0.102210	0.076894

Table 10: Total variation from exact coefficients and numerical products, against the limiting curve (manuscript 06). For each n the cutoffs are near one quarter, one half and three quarters of M ; the fractions vary because M is an integer.

canonical mean, not a prescribed integer n , so it is a separate check of the expansion and its signs, not an exact coefficient computation; the JSON records its part cutoff, integration endpoint, quadrature settings and the bound $t/2$ on the scaled change from rounding that mean to an integer. For $h = 0$ the values of $m^2(\text{diagnostic}/G_0(0.37) - 1 - C_0(0.37)/m)$ are approximately -0.054852 , -0.049493 , -0.046916 , -0.045653 : bounded, as the second-order remainder requires. The script also tests $h = -2$ and $h = 1$, including a case where the first coefficient is positive. Its NumPy/SciPy quadrature is not interval-certified.

22.3 Number-of-parts and edge comparisons (manuscript 10)

Table 11 compares $t\mathbb{E}N_n$ with its limit S_1 and with (18.10). Every error in this subsection means *approximation minus the comparison value*, not relative error.

Table 11: The scaled mean and its first nonzero conditioning correction (18.10).

n	m_n	$t_n\mathbb{E}N_n$	Limit error	Refined error
10^2	4.5337	1.728648492	$7.75 \cdot 10^{-2}$	$1.11 \cdot 10^{-1}$
10^3	6.2400	1.802456102	$3.68 \cdot 10^{-3}$	$2.00 \cdot 10^{-2}$
10^4	7.9492	1.815524785	$-9.39 \cdot 10^{-3}$	$2.05 \cdot 10^{-3}$
10^5	9.6549	1.812682904	$-6.55 \cdot 10^{-3}$	$8.50 \cdot 10^{-4}$
10^6	11.3564	1.810616979	$-4.48 \cdot 10^{-3}$	$2.91 \cdot 10^{-4}$

The cancellation of the m^{-1} term does not make the limiting mean accurate at every moderate argument: at $n = 100$ and 1000 the refinement is worse than the limit itself, and it improves the error at the larger arguments, as an expansion in $1/\log n$ rather than $1/n$ allows.

For a transform comparison let $R_J(s)$ be the operator quotient of (18.8); table 12 uses $s = 1$. The comparison transform comes from an independent weighted dynamic program, $f_n \leftarrow f_n + e^{-t} f_{n-c}$, normalized by $p(n)$; unlike the integer arrays it uses ordinary double precision, and the displayed precision reflects this.

The small- n discrepancies include the discrete marking correction $O(tm)$ of (18.6), beyond every fixed inverse- m order asymptotically but not negligible in a finite table; more inverse-power terms do not guarantee monotone improvement.

Table 12: Number-of-parts Laplace transform at $s = 1$: signed errors of exact-cumulant operator quotients.

n	Weighted DP	R_1 error	R_2 error	R_3 error
10^2	0.223714912	$-2.05 \cdot 10^{-2}$	$-2.43 \cdot 10^{-2}$	$-2.50 \cdot 10^{-2}$
10^3	0.219620551	$-2.73 \cdot 10^{-3}$	$-4.39 \cdot 10^{-3}$	$-4.37 \cdot 10^{-3}$
10^4	0.223404483	$6.83 \cdot 10^{-4}$	$-5.45 \cdot 10^{-4}$	$-6.71 \cdot 10^{-4}$
10^5	0.228207266	$7.17 \cdot 10^{-4}$	$-8.79 \cdot 10^{-5}$	$-1.28 \cdot 10^{-4}$
10^6	0.231858951	$4.90 \cdot 10^{-4}$	$-2.00 \cdot 10^{-5}$	$-1.98 \cdot 10^{-5}$

At $n = 10^6$ the saddle is $m \approx 11.3563921110$, so $M = 11$. Table 13 compares four exact two-coordinate cylinder probabilities with the leading product law and the correction (16.14), and table 14 the maximum CDF. The leading column of table 14 is Part I's fixed-offset law (compare Part I's table 4); the corrected column and the uniform and moment results are Part II's.

Table 13: Exact multiplicities at indices M and $M + 1$ for $n = 10^6$.

(Z_M, Z_{M+1})	Exact probability	Product limit	First correction
(0, 0)	0.396817585	0.416939910	0.400090956
(1, 0)	0.231522740	0.226512608	0.230976878
(0, 1)	0.049984689	0.036320274	0.048151337
(1, 1)	0.025229001	0.019731860	0.024758330

Table 14: Maximum CDF at $n = 10^6$: exact value, leading phase law, corrected law.

h	Exact CDF	$G_h(\theta)$	$G_h(1 + C_h/m)$
-1	0.396814988	0.416915901	0.400077350
0	0.898512343	0.912835901	0.898940702
1	0.999997109	0.999942416	1.000012495

For $h = 1$ the truncated correction slightly exceeds one; it must not be treated as a probability. Higher exact-cumulant orders need not improve monotonically: at this h the errors of orders one, two and three are about $3.47 \cdot 10^{-4}$, $-4.73 \cdot 10^{-6}$ and $-4.41 \cdot 10^{-5}$. The data keep these unfavourable cases. (The exact $h = -1$ value of table 14, 0.396814988, and Part I's table 4, 0.39681499, agree; the (0, 0) cylinder of table 13, 0.396817585, is a different event, $Z_M = Z_{M+1} = 0$ without the condition on higher indices.)

The covariance prediction (16.24) can be checked from integer prefix length sums. At $n = 10^6$, for $h = -1$ the exact covariance is about 0.068524922 against the first prediction 0.063602248; for $h = 0$ the values are 0.035783824 and 0.033892036. Both show the predicted positive dependence. In the much rarer $h = 1$ tail the first prediction is not relatively accurate at this argument; only its fixed-offset asymptotic status is asserted.

Table 15 concerns the limit variable W , not a finite- n statistic. The comparison quantile comes from high-precision evaluation of (19.3) and a bracketed scalar root finder; the truncation order is the largest power of ϵ kept after the logarithm. The small errors are consistent with convergence but are not interval enclosures; the convergence proof is theorem 19.4.

22.4 Finite checks recorded by the two runs

Manuscript 06's run passes: (1) all 62 displayed terms of the inspected OEIS entry; (2) the logarithmic-derivative recurrence $np(n) = \sum_{s=1}^n (\sum_{c_k|s} c_k) p(n-s)$ (Part I's (9.1)) through $n = 300$; (3) exhaustive multiplicity enumeration of the numerators for $1, N, N^2, J, J^2, NJ$ through $n = 40$; (4) direct enumeration of full upper multiplicity vectors at $n = 100$, $K = 2$ and a separate test parameter $t = 0.08$, agreeing with

Table 15: Limiting upper quantiles of W and signed inverse-series errors.

ϵ	ξ_ϵ	Order 0 error	Order 2 error	Order 4 error
$1 \cdot 10^{-1}$	3.300015038	$2.80 \cdot 10^{-2}$	$6.92 \cdot 10^{-5}$	$3.49 \cdot 10^{-7}$
$1 \cdot 10^{-2}$	5.627898046	$2.70 \cdot 10^{-3}$	$6.49 \cdot 10^{-8}$	$3.24 \cdot 10^{-12}$
$1 \cdot 10^{-3}$	7.932909782	$2.69 \cdot 10^{-4}$	$6.45 \cdot 10^{-11}$	$3.22 \cdot 10^{-17}$
$1 \cdot 10^{-6}$	14.840933331	$2.68 \cdot 10^{-7}$	$6.45 \cdot 10^{-20}$	$3.21 \cdot 10^{-32}$

(22.1) (distance 0.5597111388 . . . , which differs from the saddle value of [table 10](#) because the parameter differs); (5) the tail-mass conditional probabilities sum to one within the recorded tolerance; and (6) its counts at powers of ten equal Part I's.

Manuscript 10's run passes: the first 20 OEIS terms; all 301 counts through $n = 300$ against the same recurrence; exhaustive recursion for the count and both length sums for $0 \leq n \leq 60$ (61 checks); 549 exact cutoff-cylinder inclusion–exclusion identities ([16.15](#)) for small arguments and a symbolic check of the first-order joint product-operator identity (`03-endpoint-verify_cutoff.py`); 15 exact gamma-benchmark comparisons (orders one to three, moments one to five); the finite partial-fraction identity for the first eight rates; the four inverse-tail coefficients; and, in floating point, 13 spectral-transform and boundary-flatness identities and four upper-quantile equations. The million-term computation is exact integer arithmetic, but the independent recurrence cross-check reaches only 300, as the JSON records. No test count is a proof of the analytic remainder estimates; the code checks finite combinatorics and algebra and evaluates the formulas, but does not verify the contour arguments, certify numerical tails, or establish priority, and the million-term program is not a machine proof of the limit law. Conversely the proofs do not rely on extrapolating the tables.

22.5 Checks by the intake

[Write note, batch 98, 5 October 2026.] The batch-98 intake checked by hand the likelihood linear term (from $B_{3,K} = 2m$ and $E_1 = -1/(12m)$), the crossing point and $d(a)$ of [theorem 15.1](#), $\mathbb{E}|1 - Z^2| = 4\phi(1)$ of [theorem 15.2](#), the transport score $\eta_j = -3(j - \theta)/(2m)$, [corollary 16.5](#) and ([16.22](#)), the agreement of [corollary 16.3](#) with manuscript 10's operator, the mgf bound of [theorem 18.5](#), the bridge covariance, the jet algebra of [theorem 18.1](#), $\text{Cov}(X_{i,n}, X_{j,n}) = -1/m$, the covariance sign of [theorem 16.6](#), [lemma 19.1](#), the remainder ([19.4](#)), [theorem 19.3](#) and the Lagrange coefficients of ([19.11](#)); none failed. At 40 digits it recomputed $S_1, S_2, 2S_3/S_2^{3/2}, d(1/2), \mathfrak{d}_1, \dots, \mathfrak{d}_5, \xi_{0.1}$ and $F_0, F_1, \mathcal{V}_0, \mathcal{V}_1$ at $\theta = 0, \frac{1}{2}$, all agreeing to the printed 9–10 digits, and $-S_1\Xi(\theta) = -1.3205626659$ at the phase of $n = 10^6$. The write reran all four delivered programs on copies with the delivered names restored (Windows, Python 3.13.5): every CSV and .tex table and the phase-function JSON equal the shipped files up to line ends; the verification JSON files differ only in Python, platform, time and run-time strings and, for manuscript 06, in last-digit floats of the double-precision limits; the run times were 61, 19, 17 and 8 seconds. The write also verified [remark 14.4](#) numerically: at the phase of $n = 10^6$, $\mathfrak{b}_1 = -0.496803$, $\mathfrak{b}_2 = 0.239194$ and $\mathfrak{b}_3 = 1.149217$.

22.6 Reproduction as delivered

The delivered instructions, from the extracted archive directories, were

```
python code/verify.py                # manuscript 06
python code/audit_correction.py      # manuscript 06
python -m pip install -r requirements.txt # 10
python code/verify.py --max-n 1000000 --dps 60 --output rerun # 10
python code/verify_cutoff.py --max-n 1000000 --dps 60 --output rerun
```

Manuscript 06's first command needs `mpmath`, and `matplotlib` for the figures; the second needs `NumPy` and `SciPy`. Manuscript 10's `-output` option keeps its shipped data unchanged; for a smaller run replace the million by 10000.

[Write note, batch 98, 5 October 2026.] In this report the programs carry the prefixes `02-laws-` and `03-endpoint-`, import each other and read and write data under their delivered, unprefixed names, and by default write into the shipped data/ and figures/; manuscript 06's `verify.py` writes to the package root unless `-output` is given. Do not run them in the report directory. The report README gives the rerun route on a copy with the delivered names restored.

23 Further questions and research

The following are directions, not theorems claimed by Part II; neither manuscript asserts that all are previously unposed. Each item names its source. Questions one manuscript asked and the other (or the write) answered are recorded as answered and not repeated as open.

Answered within Part II. (a) Manuscript 06 asked for “the possible cancellation of the first $1/m$ correction in $\mathbb{E}(tN_n)$ ”: proved by manuscript 10 ([corollary 18.2](#)); as a consequence of ([18.10](#)) and [remark 14.4](#), [write] $m^2(t\mathbb{E}N_n - S_1) = \frac{S_1}{2}(\mathfrak{b}_3(\theta) - 2\mathfrak{b}_2(\theta)) + O(m^{-1})$, which is 0.6058 at the phase of $n = 10^6$ against the recorded 0.578. (b) Its “ $m \text{Cov}(tN_n, J_n - M)$ ”: the limit is $-S_1\Xi(\theta)$ ([write], [theorem 18.6](#)). (c) All five fields of its `exploratory_unproved_refinements` are now accounted for: the mean error as in (a), the two variance fields by [corollary 18.4](#), the two covariance fields by [theorem 18.6](#). (d) Manuscript 10's “all-orders largest-index moments” at first order: ([17.11](#))–([17.12](#)) (manuscript 06). (e) Its “expanding cutoff field” for the upper window in total variation: [theorems 15.1](#) and [15.2](#) (manuscript 06). (f) Its universality question in part: [proposition 21.1](#).

23.1 Uniform higher orders across the cutoff transition (06)

The regime $M - K \rightarrow \infty$, $M - K = o(M)$ has the universal leading constant $\phi(1)$, while a fixed upper cutoff has the phase-sensitive coefficient of ([16.6](#)). Construct a uniform expansion connecting them and determine the next coefficient throughout $1 \leq M - K \leq (1 - \varepsilon)M$. The absolute value in the total-variation norm introduces moving zeros of the density correction, so extending the local Edgeworth expansion is not the whole problem.

23.2 Higher corrected joint laws and the expanding cutoff field (06, 10)

The proof of [theorem 16.1](#) is constructive: further displacement integrals and gamma-ratio coefficients should give $d\nu_{n,h}/d\Pi_{\theta,h} = 1 + \mathcal{P}_1/m + \mathcal{P}_2/m^2 + \cdots$ with weighted L^1 remainders at every fixed order, ideally by a finite symbolic recurrence in centred tail energies and phase derivatives with automatic checks $\mathbb{E}\mathcal{P}_r = 0$ (manuscript 06). Manuscript 10 asks how fast the finite index set of [theorem 16.4](#) may grow while the joint law stays close in total variation to a product of geometric laws, and for an explicit correction operator for the whole transition field. For the upper window this is answered by [theorems 15.1](#) and [15.2](#); quantitative independence of the transition field from the low-index length variables, in total variation rather than for transforms, remains open.

23.3 Functional bridge and a simultaneous three-scale theorem (06)

[Theorem 20.1](#) is finite-dimensional with a uniform law of large numbers. A functional limit needs tightness near both index endpoints; one route reserves finitely many small arithmetic witness factors for the conditioning local limit and controls the remaining increments by moments. A simultaneous theorem

for the count limit, the whole bridge and the two-sided edge process on one product space would make the three-scale description precise; the topology and normalization at the lower endpoint must be specified first.

23.4 Local limits for the number of parts, and finite-size quantiles (06, 10)

Weak convergence of tN_n with moments does not imply a lattice local limit. A natural target (06) is $\mathbb{P}_n(N_n = \ell) = t f_W(t\ell) + o(t)$ in a stated range, with a version conditioned on a largest-index threshold; this is a two-variable coefficient problem whose Fourier analysis must handle the joint lattice generated by $(c_k, 1)$. Manuscript 10 notes that the first *formal* density correction suggested by (18.3) is $-\frac{1}{2}(x^2 f_W(x))''/m$; compact Laplace control does not prove this density statement. A sufficiently uniform local theorem would transport the correction to finite- n quantiles and give admissible simultaneous ranges $\epsilon = \epsilon(n) \downarrow 0$, turning the sequential inversion of theorem 19.4 into a finite-size inverse theorem.

23.5 The next mixed coefficient (06; [write])

Theorem 18.6 has remainder $O((\log m)^{5/2}/m)$. Determine the next coefficient of $m \operatorname{Cov}(tN_n, J_n)$ (at $n = 10^6$ the exact value -1.4368 differs from the limit -1.3206 by about $1.3/m$), and whether the logarithm can be removed: the write's summation uses only polynomial-in- h errors up to $|h| \asymp \sqrt{\log m}$. A joint version of theorem 17.5 with the number-of-parts mark, as manuscript 06 proposed ("a marked coefficient expansion with a uniform remainder in the marking parameter, followed by differentiated and weighted versions of the maximum-law expansion"), would settle both.

23.6 Distinct sizes and near-edge holes (06)

Let $D_n = \#\{k : Z_k > 0\}$. Manuscript 06 proposes, without proof, the edge description of $D_n - M$ as occupied sites above the cutoff minus missing sites below it, $-\sum_{j \leq 0} \mathbf{1}_{\{Y_j^{(\theta)}=0\}} + \sum_{j \geq 1} \mathbf{1}_{\{Y_j^{(\theta)}>0\}}$; both limiting sums are almost surely finite. A theorem with moments needs a uniform small-rate bound on missing-site probabilities under conditioning, a different estimate from the large-multiplicity survival bound used for N_n .

23.7 Universality with minimal hypotheses (06, 10)

Remove or weaken the finite gcd hypothesis of proposition 21.1: pure integer powers have strong root-of-unity resonances but much of the same geometry, and a general theorem should distinguish a genuinely competing lattice contribution from a resonance negligible after integration (06). For $a_k \sim A\rho^k k^{-\beta}$ the local calculation predicts a length limit $\sum \mathbf{E}_k/a_k$, a cutoff scale $\rho^{j-\theta}$ and a curvature coefficient involving β (10); proposition 21.1 proves this at first order for its class, and a general theorem with minimal growth and lattice-smoothing hypotheses, separating the full-sequence dependence of the length law from the local-ratio dependence of the cutoff law, remains open. Fuss–Catalan and other gamma-quotient sequences are test cases whose noncentral-arc hypotheses must be checked, not inferred from their exponential growth; compare Part I, question 10.6.

23.8 The optimal transition for a receding maximum cutoff (10)

Theorem 17.1 uses $|h| \leq m^{1/8}$; its proof points to the ratio $|h|/\sqrt{m}$, the deleted tail having scaled mean of order $|h|$. Determine the optimal perturbative range, then the nonperturbative regime $h \asymp \sqrt{m}$, where the prefix saddle should move on the width scale of the full saddle. For h proportional to m a different large-deviation rate function is "likely" the right object (manuscript 10's expectation, unproved); the present Gaussian tail bound is not sharp enough to identify it.

23.9 All-orders largest-index moments (10)

First order is answered (item (d) above). Derive and normalize explicit higher inverse- m corrections for the mean, variance and higher cumulants of J_n by summing cutoff corrections with a proved summable remainder; the growing-window estimate is a starting point, but the exact tail amplitudes must be expanded uniformly in that window. A useful computational goal is a phase-function library with certified extrema and Fourier tails.

23.10 Small deviations of the limit law (10)

The survival expansion is excellent for large x , but cancellation makes it inefficient for $\mathbb{P}(W \leq x)$ as $x \downarrow 0$. Determine the complete logarithmic small-deviation expansion, with its phase modulation and its inverse. The product $\mathcal{L}(s)$ for large positive s suggests a logarithmic saddle related to the partition-counting saddle of Part I, but the sign and boundary factors differ (manuscript 10's suggestion, unproved).

23.11 The convergence radius of the inverse tail (10)

[Theorem 19.4](#) proves a positive radius without identifying it. Locate the nearest complex critical values of $\mathcal{T}$ and decide whether they, rather than the natural boundary, control the inverse germ; this would give sharp coefficient growth and a principled truncation rule. Neither non-D-finiteness nor the natural boundary answers it.

23.12 Exponential sectors and arithmetic secondary arcs (10)

The operator expansion keeps every algebraic order in m^{-1} but discards the marking correction $O(tm)$ and the noncentral arcs. Determine their leading contributions and whether the roots of unity organize them into an exponentially improved expansion; such sectors can matter numerically before the inverse- m series is accurate ([table 12](#)). This is the statistics counterpart of Part I, question [10.3](#); any claim of a complete transseries must account for them.

23.13 Certified computation, sampling and formalization (06, 10)

Make the constants in the contour, moving-window and L^1 remainders and in the tail bounds explicit and computable, and implement the positive geometric tails with interval arithmetic. This would give certified total-variation tolerances for sampling only the large parts and certified evaluations of the periodic mean and variance functions (06), and finite- n probability and integer-quantile decisions (10). An exact sampler for the whole partition must still handle the total-size constraint efficiently; an asymptotic law does not settle that algorithmic problem. A feasible formalization order is the finite Gaussian multi-index operator and the exact counting identities first, then the analytic localization and conditioning steps (10); compare Part I, questions [10.1](#) and [10.8](#).

24 Conclusion of Part II

Catalan-part partitions have two sharply different endpoint scales. The number of parts is governed by an infinite convolution of exponentials, with a sparse exact exponential representation, a natural boundary and a convergent inverse-tail expansion; the largest index has a phase-dependent integer law. The common size constraint disappears at leading order between the endpoints, yet leaves explicit order- $1/\log n$ dependence, both inside the upper edge ([corollary 16.3](#)) and between the endpoints ([theorems 16.6](#) and [18.6](#)). For the whole upper process the conditioning can be neglected exactly when the observed window contains $o(\log n)$ active sizes, with the explicit curve $d(a)$ otherwise; uniform tails and the all-threshold expansion

close the gap between a fixed-cutoff law and moment asymptotics; and in the bulk the mass fluctuates as a Brownian bridge in the normalized index. The Gaussian coefficient operator exposes the constraint without Gaussianizing the statistics themselves.

A A compact audit of the algebra

This appendix records the dependency chain for the constants most likely to be affected by an indexing or sign error.

Quantity	Formula or origin
Allowed unit part	$c_1 = 1$ only; no second c_0 factor
Catalan logarithmic drift	$\beta = 3/2, b_1 = -9/8, b_2 = 1/2$
Summatory logarithmic coefficient	$b_1 - \beta/2 = -15/8$
Barnes constant	$K = -\frac{3}{4} \log(2\pi) - \log G(3/2)$
Bulk $1/m$ coefficient	$27/16$ before the phase correction
First local phase correction	$-(3/2)\partial_\alpha \Psi_\alpha$
Saddle mean correction	$q_0 = -1/2 + \Psi'_\alpha/\alpha$
Logarithmic power of μ	$3\beta/2 - b_1 - 1/2 = 23/8$
Normalized constant	$C = -K - c - \frac{1}{2} \log(2\pi)$
First Gaussian log correction	$-1/(12m)$

The cancellation producing D_α is the identity

$$\begin{aligned} \alpha b + (-A + \Psi'_\alpha + q'_0)a + P_1 + q_1 - b_1 - \frac{q_0}{2} - \frac{1 + q'_0}{2\alpha} - \frac{1}{12} \\ = \frac{1}{6} - \frac{1}{2\alpha} - \frac{3}{2}\partial_\alpha \Psi_\alpha + \frac{17}{8\alpha}\Psi'_\alpha - \frac{(\Psi'_\alpha)^2 + \Psi''_\alpha}{2\alpha^2}, \end{aligned}$$

with a, b, q_0, q_1 as in (5.3) and (5.9)–(5.10). The supplied symbolic audit treats the phase and its derivatives as independent symbols and verifies this rational identity exactly. This check isolates coefficient algebra from the separate analytic justification of the asymptotic operations.

B Provenance and scope of the search

The external check was performed on October 1, 2026. It included the OEIS entry and linked survey, searches for `A033552 asymptotic` and `partitions into Catalan numbers asymptotic`, the classical periodicity reference, and the ProveIt transseries documentation and repository code search. The phrase “Catalan partitions” alone is ambiguous in the literature and was not treated as identifying this sequence without checking the definition.

No claim of an exhaustive MathSciNet, zbMATH, or journal-archive search is made. The absence of a posted OEIS asymptotic is an observable entry-level gap, not evidence that no general theorem could already imply a leading term. Accordingly the main theorems are stated on their mathematical content, and the article does not claim a resolved named conjecture or verified historical priority. No repository file or OEIS entry was edited as part of this work.

C Provenance of the collection edition

[Write note, batch 7301, 1 October 2026.] This appendix was added in the write.

Source. The report is built from one manuscript, number 39 of batch 73O1 of ProveIt’s incoming-reports intake: the archive `Catalan_Partitions_Research.zip` (17 files, a 21-page PDF), which arrived in commit `f8c3a392a` and was placed in the collection by commit `9df4ba51a`. Its pinned repository commit is `e6190a94552f119e9a2140e5fea8450a045a0837` (1 October 2026); the delivered `SOURCE_NOTES.md` also names `1f1981f682b2878bde51a6ad40c22777f362fc05` and wrongly calls it a tree (see the note in [section 1](#)). The title-page author line is kept as delivered; the delivered PDF metadata read “Research report prepared with ChatGPT”, and the write replaced that field by the collection’s “AI-assisted research report prepared for Vladimir Reshetnikov”. The report is AI-assisted and unrefereed.

What the write changed. Every one of the 93 delivered labels received the prefix `ctp:`, and every reference was updated with it; this appendix adds one label. The write added the [Write note] paragraphs (after the principal-formula box, in [section 1](#) twice, after the proof of [lemma 4.1](#), in [section 9](#) and here), the bibliography entries [9] and [10], the `\writenote` macro, the PDF author field above, `cleveref` type hints (`\label[lemma]`, `\label[corollary]`, `\label[proposition]`) on the five labels of lemmas, corollaries and propositions, and `\crefalias{section}{appendix}` after `\appendix`. These environments share the theorem counter, and the delivered PDF printed every reference to them as “theorem”; only those printed names changed, no label name or number. No delivered sentence, statement, proof or table was changed or removed.

Delivered names and shipped paths. `article.tex`, `README.md` (rewritten as the collection’s report guide) and `SOURCE_NOTES.md` are at the report root; `code/` keeps its three Python files, and `build.sh` moved there from the package root; `data/` keeps its seven files, and `requirements.txt` moved there from the package root. All but the article and the README are byte-identical to the delivery. Not shipped: the delivered `article.pdf` (replaced by a build of this edition) and the checksum ledger `SHA256SUMS.txt` (verified 16/16 at intake, then retired). `data/build_and_quality_checks.json` describes the delivered 21-page PDF.

Relation to repository material. No repository report or formal development treats A033552. The Mellin computation of [lemma 4.1](#) reappears, for a different product, in the Fabius-tree compilation cited in the note after it [10]. The batch’s other partition reports, on square-root-restricted partitions (A097356) and on distinct-partition norms (A022629), study different products in different saddle regimes and share only the standard saddle, Edgeworth and Lambert- W toolkit; no proposition is proved in two of them.

[Write note, batch 98, 5 October 2026.] Part II, added in batch 98, records its own provenance in [section 12.1](#). No other report in the repository treats A033552; both batch-98 manuscripts became Part II of this one.

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- [8] The mpmath developers, *Zeta functions, L -series and polylogarithms*, mpmath documentation, including Euler–Maclaurin evaluation options. <https://mpmath.org/doc/1.1.0/functions/zeta.html>. The accompanying numerical run used mpmath 1.3.0; see the recorded version in data/verification.json.
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- [10] [Added in the write.] ProveIt, Fabius-tree compilation *Geometric Uniform Convolutions and New Frontiers*, Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/frontier-compilations/Geometric_Uniform_Convolutions_and_New_Frontiers/fabius-frontier-report-H.tex, section “Mellin analysis and the hidden log-periodic phase” (Lemma “Mellin kernel”).
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